Discrete feature growth: what it is, how it compares with gradient-trained networks, and what it learns

Claude (Anthropic)*
prompted by Kaarel Hänni

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Abstract

Discrete feature growth (DFG) is one greedy loop: fit the labels by least squares on the features found so far, and admit every Fourier character or signed-sum indicator whose correlation with the residual passes a union-bound test over its class. DFG v1.3, run alone here, is the one new learning algorithm in v3, the stacked portfolio of this study's previous part, and the source of its Boolean results.

On its held-out suite, 12 fresh Boolean targets, DFG needs clearly less data than the per-target best of three network variants on 10 targets and more on none: the mean $\log_2$ ratio of n^*, the training-set size at 10% test error, is -1.29. Over 126 in-sample targets it is -1.87. DFG's n^* grows with the input dimension d with exponents 0.01–0.50, the networks' with 0.91–2.40. The low-degree algorithm, with DFG's characters and test but no search, needs 3.1× more data and is level with the networks.

DFG's models are short lists of named features, often the target's exact structure. On parities a proved criterion predicts every fit with $n \geq 32$; on signed-sum targets the search, not the test, sets the data cost, and residue classes mod 3, with almost no weight at Fourier degrees 1–2, are never found. DFG admits characters in order of detectability, the networks in order of construction. A post-hoc core with five parts nearly matches DFG at a tenth of its FLOPs. DFG needs ± 1 inputs: real-valued data, where earlier adaptations lost to a regularized MLP, is not covered.

*Written by an AI system in a series of sessions prompted and steered by Kaarel Hänni. Not reviewed by a human. Code, result files, the protocol and the underlying records are on the `claude/nifty-mccarthy-j0lw2a` branch of `https://github.com/kaarelh/AI-works`, folder `gd-circuit-growth/`; how the analyses were checked is in Appendix C.

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1 Introduction

The question. From its sixth part on, this study has asked whether a learner without continuous feature parameters can do at least as well as gradient-trained networks, with the same training data and no more FLOPs. Its tenth part answered with v3, a stacked portfolio whose input type decides which of its members carry a fit [13]. Kaarel Hänni then asked for "a discrete method that is more like 'one thing', less an ensemble of different things", perhaps one of v3's components, the aim being "a specific interesting object worth studying", and for a paper like the Part 10 paper about that object. This is that paper.

The suggestion: DFG v1.3, run alone. The object is DFG (discrete feature growth) v1.3, v3's member for Boolean inputs, as Part 10 froze it, run by itself (Section 3.4). There are three reasons for choosing it.

- *It is the one new learning algorithm in v3.* v3's other members are LightGBM [22], ExtraTrees [19], nearest-neighbour votes and logistic regression, combined by a convex stacker [35]; its image members are LightGBM on fixed banks of pixel-subset sums.
- *It is the source of the portfolios' Boolean results.* Part 8's post-hoc ablations traced the portfolio's toy wins to DFG's character search [11], and on Boolean inputs DFG is the member that matters in v3 [13].
- *It is one thing:* one loop, one kind of test and one linear readout. Its fitted model is a short, explicit list of named features (Section 6).

Three alternatives were considered and set aside. v3's image boosters are gradient-boosted trees on a fixed, data-derived feature bank; what is new in them is the bank, and the learner is LightGBM. The portfolio itself is the ensemble the request asked to avoid. A single DFG for real-valued data would be a new design, with a dictionary derived from the data, not a frozen object that can be studied as it is; it is left for future work (Section 7.6). The choice has a price, its domain. DFG v1.3 takes ± 1 inputs and two classes, so it runs on the study's 11 Boolean suites, on 2 of its 26 tabular datasets, and on no images.

The object, in one paragraph. DFG is one greedy loop over an explicit, finite dictionary of Boolean features: Fourier characters $\chi_S(x) = \prod_{i \in S} x_i$, indicators of the values of signed sums $\sum_{i \in S} s_i x_i$ of inputs, and a few composites of these. Each round fits the ± 1 labels by least squares on the features found so far. It then admits every candidate whose correlation with the residual passes a test whose threshold is a union bound over the candidate's whole class, about $s\sqrt{2 \ln N / n}$ for N candidates, n rows and residual scale s. Characters are enumerated class by class, up to degree 4 on all rows and degrees 5–6 on a subsample. Signed-sum blocks are proposed by a search seeded from the residual's degree-1 and degree-2 statistics, and each is tested by a χ^2 bound over every block of its size. The loop stops when a round admits nothing or the fit has every training label's sign right, and DFG predicts with the sign of the readout. At $n \leq 512$ DFG decides by its own cross-validation whether to add a best guess when nothing passes. Section 2 gives every step, five exact statements about the loop (Section 2.5), and its relation to the low-degree algorithm, forward stepwise regression and statistical-query learning (Section 2.6). Appendix A lists every name the paper uses.

What was found. Table 1 gives the comparisons.

evaluation	reference	DFG v1.3 alone	limits
held out: `dlheld7`, 12 targets (Section 4.1)	`nets_best`: best of 3 variants (hindsight)	mean $\log_2 n^*$ ratio -1.29 over the 11 targets both learn (2.4× less data; median -1.16; s.d. 0.91, s.e. 0.27); clearly better on 10 of 12, worse on 0	one family; on 5 of the 10 wins `lin` or `hgb` also beats `nets_best`; v3 needs less data (+0.21; clearly on the 3 controls); no learner learns f7_05
pooled: 126 targets of 10 suites (Section 4.2)	`nets_best`: best of 2 to 5 variants	-1.87 over 123 (3.7×); clearly better on 117, worse on 2	in-sample (Table 5); against `nets_wd` the margin is smallest on the three suites held out for some version (`dlheld2`, `dlheld3`, `dlheld7`)
input dimension: $d = 15, 60, 120$, 7 targets (Section 4.5)	`nets`, run at every d	exponent of n^* in d 0.01–0.50 (parities of degree 2, 3, 4: 0.24, 0.30, 0.50), networks 0.91–2.40 (1.03, 1.96, 2.40); median factor 6.2× at $d = 15$, 42.5× at $d = 120$	7, 7, 4 development targets; 2 seeds at $d \neq 30$; the weakest variant; 3 or 4 points per slope
no search: `lowdeg`, 138 targets (Section 4.4)	DFG alone; `nets_best`	`lowdeg` against DFG alone +1.64 over 122 (3.1× more data; worse on 112, better on 16, never on 12 that DFG learns); against `nets_best` -0.16 (53 better, 62 worse)	in-sample; Part 6's check, unchanged; faster only where DFG alone needs ≤ 133 points
tabular: 2 datasets, 9 cells (Section 4.6)	`mlp_best` (hindsight); `hgb`; v3	cells better / tie / worse: 0/8/1 against `mlp_best`, 0/9/0 against `hgb`, 0/8/1 against v3	kr_vs_kp and mushroom, the only 2 of 26 datasets with ±1-codable inputs and two classes; no real-valued data
post hoc: the core, 138 targets (Section 5.6)	DFG alone, same seeds; `nets_best`	against DFG alone +0.07 over 133 (28 lower, 26 higher; loses par5); against `nets_best` -1.74 (111/9), DFG alone -1.88 (117/5); `dlheld7` -1.30; median FLOPs 0.094 of DFG's	seeds 0, 1; defined after every result had been read: nothing is held out for it

Table 1: The results at a glance [Numerical]. n^*: the training-set size at which the seed-mean test error reaches 10%. A ratio is $\log_2(n^*_{\text{DFG}}/n^*_{\text{ref}})$ (for `lowdeg` and the core, of theirs to the reference's), negative when the first needs less data; means are over the targets both learn. "Clearly": beyond ±0.25, or only one of the two reaches 10% error; on tabular data, errors that differ by more than 10% and 2 s.e. (Section 3.3). The s.d. and s.e. on `dlheld7` are over its 11 learned targets, as independent draws from the family. Only `dlheld7` is held out for DFG v1.3. Generated by `paper_dfg/make_tables.py` (`tab_headline`) from `study/part11/RESULTS.json` and, for the core's FLOPs, `study/part11/interp/ablations/ablations.json`.

- **Against the networks** [Numerical] (Sections 4.1, 4.2 and 4.5). On the held-out `dlheld7`, DFG alone needs clearly less data than the best network variant on 10 of 12 targets and more on none, about 2.4× less on average. On five of the ten wins, logistic regression or default gradient boosting is also clearly better than the networks; on the other five, all sum-structured, DFG alone needs about 4.3× less data than the networks and no learner without a feature search comes close. On seven targets run at $d = 15$–120, DFG's n^* grows as $d^{0.01}$ to $d^{0.50}$ and the networks' as $d^{0.91}$ to $d^{2.40}$. v3, the portfolio, needs less data than DFG alone (1.14× less on average, over the 134 targets both learn), mostly on small formulas and thresholds that it learns by $n = 128$ (Section 4.3).
- **What carries the results** [Numerical] (Sections 4.4 and 5). Not the dictionary and the test alone: the low-degree algorithm, which has both, is level with the networks. Removing the sum blocks costs most: $+0.91$ in mean $\log_2 n^*$ over the 59 of 65 ablation targets both learn, and 3 targets lost. The residual-scaled tests and the small-n machinery come next. Post hoc, a core of five parts keeps most of DFG's margin over the networks (-1.74 against -1.88 pooled, -1.30 against -1.27 on `dlheld7`, on the same seeds) at a median tenth of its FLOPs. It loses where a removed part has no substitute: parities of degree 4–5 without a lower-degree footprint, dense halfspaces, and sums with a heavy inner term (Section 5.6).
- **What it learns** [Numerical] (Section 6.2). The models hold each target in the form of its dictionary that is cheapest to detect. Parities are one character from the first fit on, and symmetric targets are their exact signed block; at $n = 16384$, 8 of 37 analysed models are exactly the target's Fourier expansion. Where DFG errs, features are missing, not weights.
- **Two regimes** [Proved, Numerical] (Sections 5.2, 6.4 and 6.5). On parities the test sets the data cost. A proved criterion predicts the outcome in all 250 parity cells with $n \geq 32$, and moving the test's level moves n^* by the predicted amount. On signed-sum targets the search sets it: a stricter test raises the true block's threshold by about a quarter of a doubling of n, but on the 7 targets whose n^* lies a doubling or more beyond the size at which the block would pass, it moves n^* by at most 0.05. The residue classes mod 3 are the extreme case. Their Fourier weight at degrees 1–2, where the block search starts, is $1.6 \cdot 10^{-6}$ to $2.1 \cdot 10^{-4}$, and their block never enters DFG's model, although it would pass the test from $n = 128$ if the search proposed it.
- **Against the networks' order** [Numerical] (Section 6.6). DFG admits a target's characters in order of detectability, and the Part 5 networks acquire them in order of construction. Inside DFG's first round at $n = 16384$, its test statistics are positively rank-correlated with the networks' order on all 11 targets (median Kendall τ 0.54). But the orders invert where a large coefficient sits at high degree: the largest term of the multiplexer is DFG's first character and the networks' last.

Scope.

- *What is held out.* For DFG v1.3, only `dlheld7`: 12 targets drawn from the blind family of signed-sum targets and fixed in a commit before v1.3 was designed (Section 3.4). DFG v1.3 was developed on two other draws from the same family, so `dlheld7` tests it on fresh targets from a distribution its designer had seen.
- *Everything else is in-sample.* The other 10 suites were used in developing some version of DFG, or were held out only for an earlier version, and several of DFG's parts were added after

failures on their targets (Table 5). Their numbers describe the frozen learner; they do not test it. The ablations, the core and the analyses describe it too.

- *The object was chosen post hoc.* It was chosen after Part 10's results had been read, including v3's on `dlheld7`, where DFG v1.3 is v3's Boolean member. DFG alone's own results there were new, apart from a disclosed timing check on two targets. The choice is post hoc; the code is not (Section 3.4).
- *The references* are one network recipe, not tuned per target, and `nets_best` is a hindsight minimum over the 2 to 5 variants run on each suite (Section 3.2).
- *The domain is Boolean.* Inputs are uniform on $\{\pm 1\}^d$, mostly with $d = 30$. Two tabular datasets have only binary or categorical columns and can be written as ± 1; nothing here is about real-valued data or images (Section 4.6).
- *Seeds.* n^* is a mean over 3 training sets per target and n (2 on `random` and at $d \neq 30$), with no confidence interval, and the ± 0.25 band is a convention, not a significance test (Section 3.3). The ablations use one or two seeds, the core two, and most analyses seed 0.

Claims are labelled [Proved] (a proof is given, here or in the theory notes, which an independent verifier checked), [Numerical] (measured here, with the source file named) or [Heuristic] (a reading or a back-of-envelope argument, not tested here).

Background: Parts 1–10. Parts 1–5 found that rich-regime networks on Boolean targets grow features greedily, cheapest first, and pay in samples: for a k-bit parity on d inputs their n^* grows like $d^{1.0}$, $d^{1.9}$ and $d^{2.4}$ for $k = 2$, 3 and 4. That is close to the leap and correlational-SQ rate d^{k-1} [1, 2] for $k \leq 3$, and far from the $\approx k \log d$ of an Occam learner; a discrete model with a noise-limited feature search matched the networks' sample sizes within a factor of about 1.3 on held-out functions [12]. Part 6 built DFG from that picture, with enumeration and union-bound tests in place of the noisy search and at most one network run of FLOPs per fit, and on a blind 7-target suite it needed less data than the networks on every target [11]. Part 7 adapted DFG to real-valued standard tasks, where it was about level with a plain MLP on tabular data, behind a regularized one, and clearly behind on images at large n. Parts 8 and 9 stacked it with classical learners (v1, v2), whose toy wins came from DFG, and gave it signed-sum blocks (v1.2) [11]. Part 10 added weighted blocks, forced blocks and a beam search (v1.3) and built v3, which on the held-out `dlheld7` needed clearly less data than the best network variant on 10 of 12 targets and more on none, and whose analysis placed DFG's data cost on signed-sum targets in its search rather than its test [13].

Contributions.

1. What DFG v1.3 is, step by step and with its constants; five exact statements about its tests, the sample size of a parity, its sum blocks and its compute; and how it relates to familiar learners (Section 2).
2. A pre-registered evaluation of DFG alone (Section 3): on the held-out `dlheld7` and on every toy suite against the networks, the calibration references and v3; against the low-degree algorithm; in the input dimension; on tabular data; and in compute (Section 4; every target in Appendix B.1).
3. What its parts do: 12 pre-registered ablations against a post-hoc paired null, and a post-hoc core DFG with nine parts removed (Section 5).

4. A descriptive, independently verified account of what DFG learns: its models across target families, its false admissions on noise, its failures, when structure enters relative to its test, and its order of admission against the networks' (Section 6; detail in Appendix B.2).

5. What kind of learner DFG turned out to be, why it needs less data than the networks here, where it fails, and what to study next (Section 7); how Part 11 was run (Appendix C).

2 What DFG is

This section describes the paper's object: DFG v1.3 as frozen in `study/part10/hyb_portfolio3/dfg13.py`, run alone as `DFG(readout="logit").fit` with every other constructor argument at its default (Section 3.4). Parameter values below are those defaults. Table 3 lists them, with the constants fixed in the code, and line numbers refer to the frozen file. Tables 2 and 3 are generated from the frozen file itself by `paper_dfg/make_tables.py`, which also checks them against the theory notes `study/part11/interp/theory/THEORY.md`.

2.1 In one paragraph

DFG, *discrete feature growth*, is one greedy loop over an explicit dictionary of Boolean features. It starts from the constant. In each round it fits the ± 1 labels by least squares on the features it has, and takes the residual $r = y - \hat{y}$ and its root mean square s. It then enumerates candidate features class by class: Fourier characters $\chi_S(x) = \prod_{i \in S} x_i$, and indicators of the values of signed sums $\sum_{i \in S} s_i x_i$. A candidate is admitted only if its correlation with r passes a test whose threshold is a union bound over the candidate's whole class, about $s\sqrt{2 \ln N/n}$ for a class of N candidates. (At $n \le 512$ a forced variant may add the best candidate when nothing passes; DFG decides by cross-validation whether to use it.) DFG refits the linear readout on the enlarged set of features and repeats, until a round admits nothing or the fit has every training label's sign right. It predicts with the sign of that readout, refined by a few logistic Newton steps, plus one Hebbian step on the final residual. The fitted model is therefore a short list of named features with real weights. For example, with seed 0, DFG's model of par4 $= x_1x_2x_3x_4$ at $n = 64$ is the constant and $\chi_{\{1,2,3,4\}}$. Its model of sym9 $= [\,|x_1 + \cdots + x_9| \le 1\,]$ at $n = 128$ is the constant and one folded block, three indicators of the value of $|x_1 + \cdots + x_9|$. Both have test error 0 (`study/part11/interp/tables/atlas.md`; Section 6). Figure 1 gives the loop in outline.

2.2 The hypothesis class

DFG predicts $\operatorname{sign} h(x)$, and -1 where $h(x) = 0$, with

$$h(x) = \sum_{S \in \mathcal{S}} w_S\, \chi_S(x) + \sum_{i \in \mathcal{L}} w_i\, x_i + \sum_{B \in \mathcal{B}} \sum_{v \in \mathcal{V}_B} w_{B,v}\, \chi_{P_B}(x)\, [t_B(x) \in G_{B,v}] + \sum_{u \in \mathcal{U}} w_u ((u \cdot x)^2 - 1) + g \cdot x. \quad (1)$$

The five sums are the five parts of the dictionary.

- *Characters.* $\mathcal{S}$ holds the constant, $\chi_\emptyset = 1$, and every character admitted by its own test or forced (Section 2.4). The global classes reach degree 6. The extension and local classes, which grow from the characters already present, reach higher degrees.

- *The linear block.* $\mathcal{L}$ is a set of inputs whose degree-1 characters were admitted together, by one test on all of them. In the code they are ordinary degree-1 characters (line 832). They are written apart here because they entered together, without tests of their own.

fit(X, y), with n rows and d inputs. If $n > 512$, return **grow**$(X, y;$ default$, 1)$. Otherwise run **grow** for both variants, default and forced, on each of F cross-validation folds ($F = 10$ at $n \le 64$, else 5), with budget scale $1/(4F)$ each. Let e_{10} (e_{01}) count the held-out points where only the default (only the forced) variant errs. Return **grow**$(X, y;$ forced$, \frac{1}{2})$ if $e_{10} - e_{01} > 2\sqrt{e_{10} + e_{01}}$, else **grow**$(X, y;$ default$, \frac{1}{2})$.

grow$(X, y;$ variant$,$ scale$)$. Start with the features $\Phi = \{1\}$. In rounds $0, 1, \ldots, 7$:

1. fit the weights w by least squares on Φ (with a small ridge); $r = y - \Phi w$, $s = \mathrm{rms}(r)$;
2. stop if 0.75 of the budget is spent, or, from round 1 on, if $\mathrm{sign}(\Phi w) = y$ on every training row;
3. characters: in each class $\mathcal{C}$ (global, extension, local; Table 2), find the χ_S with $|\mathrm{mean}(r\chi_S)| > s\sqrt{2\ln(4K|\mathcal{C}|/\delta)/n}$;
4. dense moves: the linear block (every remaining x_i, by one χ^2 test), and spectral features $(u \cdot x)^2 - 1$ for the eigenvectors u of the residual pair matrix above its noise edge;
5. sum blocks: search for a signed subset (S, s) on r; keep it if the projection of r on its group indicators passes a χ^2 test with a union bound over every (S, s) of its size, both on r and on r projected off this round's characters; likewise products, modulated and weighted blocks;
6. if nothing passed: stop (default variant), or add the best candidate anyway (forced variant, at most twice per fit);
7. add what passed to Φ, keeping $|\Phi| \le \min(256, \lfloor n/2 \rfloor)$.

Readout. If $n \ge 8|\Phi|$, refine w by 15 Newton steps of logistic regression. Predict $\mathrm{sign}\big(\Phi(x)w + g \cdot x\big)$, with $g = X^\top r/n$ on the final residual.

Figure 1: One call of DFG v1.3, in outline (`study/part10/hyb_portfolio3/dfg13.py`: `DFG.fit`, lines 548–571; `DFG._fit`, lines 580–877). Here $K = 4$ and $\delta = 0.1$. Sections 2.3 and 2.4 give every step exactly.

- *Sum blocks.* A block B has inputs S_B, signs $s_i \in \{\pm 1\}$, and the integer statistic

$$t_B(x) = \sum_{i \in S_B} s_i x_i + a_B\, \tau_B(x),$$

  or its absolute value if the block is *folded*. An unweighted block has $a_B = 0$. A *weighted* block (v1.3) has $a_B \in \{\pm 2, \pm 3\}$ and an inner term τ_B on inputs outside S_B: a literal x_j, a 2-parity $x_j x_l$, or an AND of two literals written as ± 1. The groups $G_{B,v}$ are contiguous runs of the values of t_B that occur in training, each holding at least $\max(3, \lfloor 0.005\, n \rfloor)$ training points. Rare values are merged into their neighbours (`_edges`, lines 86–97). $\mathcal{V}_B$ is every occupied group except the most populous one, whose indicator the constant replaces. A *modulated* block multiplies its indicators by a character χ_{P_B} that is already a feature, on inputs outside S_B; otherwise $\chi_{P_B} = 1$. A *product* block $B_1 \times B_2$ of two blocks with disjoint inputs uses the joint cells $G_{B_1,v_1} \times G_{B_2,v_2}$ of the pair $(t_{B_1}(x), t_{B_2}(x))$. When every value of t_B is its own group, every function of t_B is a combination of the constant and the block's indicators (Proposition 2.5). A function of a signed sum of k inputs then costs at most $k + 1$ weights, where its Fourier expansion can need all 2^k characters on S_B.

- *Spectral features.* Each $u \in \mathcal{U}$ is a unit eigenvector of the residual pair matrix (Equation (4)). The feature $(u \cdot x)^2 - 1 = 2\sum_{i<j} u_i u_j x_i x_j$ is a fixed combination of degree-2 characters.

- *The kernel background.* $g = \frac{1}{n}\sum_t r_t x_t$ is computed on the final least-squares residual: one Hebbian step over all d inputs. It is computed, not fitted.

The weights w are fitted on the $p \le \min(256, \lfloor n/2 \rfloor)$ admitted features, by least squares and then, when $n \ge 8p$, by logistic Newton steps (Section 2.3).

class	candidates	rows tested	N at $d = 30$	$\ln(4KN/\delta)$	smallest n that can pass
Characters χ_S, test (2)					
global, $k = 1$	every x_i	all	30	8.48	16.95
global, $k = 2$	every pair	all	435	11.15	22.30
global, $k = 3$	every triple	all	4 060	13.38	26.77
global, $k = 4$	every 4-set; round 0	all	27 405	15.29	30.59
global, $k = 5$	round 0, final fit	first 1 374 (687)	142 506	16.94	33.88
global, $k = 6$	round 0, final fit	first 147 (73)	593 775	18.37	36.74
extension, +1 input	$T \cup \{j\}$	all	$bd \le 960$	8.48–11.94	16.95–23.88
extension, +2 inputs	$T \cup \{j, l\}$	all	$b\binom{d}{2} \le 13\,920$	11.15–14.62	22.30–29.23
local	$S \subseteq V$, $\lvert V\rvert \le 12$	all	$\binom{\lvert V\rvert}{\lvert S\rvert} \le 924$	5.08–11.90	10.15–23.81
Dense moves					
linear block	every remaining x_i	all	χ^2_d threshold 51.2, (3)		62
spectral features	≤ 4 eigenvectors	all		edge (4)	120
Sum blocks on k inputs, test (5), $\mu = 0$; searched on the first 2 048 rows					
$k = 2$	(S, s), folded or not	all	4.2×10^4	15.71	40.4
$k = 3$	(S, s), folded or not	all	7.8×10^5	18.64	46.9
$k = 5$	(S, s), folded or not	all	1.1×10^8	23.59	57.9
$k = 9$	(S, s), folded or not	all	1.8×10^{11}	30.97	74.1
$k = 14$	(S, s), folded or not	all	5.7×10^{13}	36.75	86.6
$k = 24$	(S, s), folded or not	all	2.4×10^{14}	38.18	89.7
weighted	$+a\tau$, $a \in \{\pm 2, \pm 3\}$	all	$\times$ 17 640	+9.78	
modulated	by χ_P, $\mu \le 4$	all	$\times (1 + \mu)$	$\le +1.61$	
product	$A \times B$, disjoint	all	$N_A N_B$	$L_A + L_B$	

Table 2: DFG's dictionary at $d = 30$ [Proved, by the code]. Each row is a class: a set of candidates enumerated together and tested with one union bound. "Rows tested": a budgeted search runs on the first m rows, with m set by its share of the budget (Section 2.4). At $d = 30$ that is every row of the grid ($n \le 16384$) except for degrees 5 and 6, which see the first 1374 and 147 rows (687 and 73 under the cross-validation's half budget, $n \le 512$). N is the count the union bound pays for, and $\ln(4KN/\delta)$ the log term of the threshold, with $K = 4$ and $\delta = 0.1$. For the extensions, $b \le 32$ is the number of current characters extended; for the local class, V is the set of inputs of the current characters. "Smallest n that can pass": no member can pass its test on fewer rows, for any labels (Proposition 2.3). For characters this is $2c^2 \ln(4KN/\delta)$; for blocks it is $1 + 2\sqrt{L_B} + 2L_B$, the threshold of Equation (5) at $\nu = 1$. For the linear block it is the room condition $n \ge 2(p + k)$, which is 62 in round 0 with all 30 inputs; for the spectral features it is $n \ge 4d$. For sum blocks, N_B counts the supports, the sign patterns up to a global sign, the two fold variants, the $1 + \mu$ choices of modulator and the $k_{\max} = 24$ sizes. Weighted and modulated blocks multiply it by the factors shown, and a product pays for both factors. Source: `study/part10/hyb_portfolio3/dfg13.py` via `paper_dfg/make_tables.py` (`tab_dictionary`; values in `paper_dfg/tables/numbers.json`).

The dictionary. Table 2 lists the classes. The dictionary is explicit and finite. At $d = 30$ the global classes hold $\sum_{k \le 6} \binom{30}{k} = 768\,211$ characters. The unweighted blocks (S, s, fold) on 2 to 24 inputs number 2.0×10^{14}, close to $3^{30} = 2.1 \times 10^{14}$, the number of signed subsets of 30 inputs. A class's size N enters its test only through the log term $\ln(4KN/\delta)$. The global classes are fixed by d. The extension and local classes are built from the characters already present, so they are small and their thresholds are low. Features on inputs already in use are therefore cheaper to admit: the discrete counterpart of the composition effect that Parts 1–5 saw in the networks [12]. A block on k inputs pays $\ln \binom{d}{k} + k \ln 2 + \ln\big(4K(1 + \mu)k_{\max}/\delta\big)$. That is $k \ln 2 + \ln(24(1 + \mu))$ more than a character of degree k, although the block class of size k is 2^k times larger.

Which parts matter [Numerical; post hoc]. Not every part is needed for DFG's results. The protocol's addendum, written after every pre-registered result had been read (`study/part11/PROTOCOL.md`), defines a `core` run. It keeps the global characters of degree at most 3, the unweighted sum blocks (seeded by pairs and single inputs), the residual-scaled tests, the least-squares refit and the small-n cross-validation. It drops the linear block and the spectral features, the extension and local classes, the degree-4 to degree-6 searches, the weighted blocks, the eigenvector seeds, the logistic refinement and the kernel background. Paired with DFG alone on the same training sets, the core's mean $\log_2 n^*$ ratio is $+0.07$ (median -0.01) over the 133 of 138 targets both learn. It needs clearly less data on 28 and clearly more on 26, and it loses par5, which needs the degree-5 search (`study/part11/RESULTS.json`, `q5b_core`; Section 5). On these suites, then, the loop and its tests, with low-degree characters and signed-sum blocks, carry most of the results.

2.3 One round

A round has five steps: the residual, the character tests, the dense moves, the sum blocks and the refit. Each step uses only statistics of the current residual on the training rows.

Residual and noise level. The features are the columns of $\Phi \in \mathbb{R}^{n\times p}$, the first one constant. The weights minimize $\sum_t (y_t - (\Phi w)_t)^2 + 10^{-3}\, n \sum_{j\geq 2} w_j^2$ (line 589). The constant is not penalized, so the residual $r = y - \Phi w$ has mean 0 up to rounding. The noise level is $s = \sqrt{\max(\mathrm{mean}(r^2), 10^{-12})}$.

The character test. A candidate S belongs to a class of N candidates, tested on the first m rows ($m = n$ unless the class is subsampled). It passes if

$$z_S = \frac{|v_S|}{c\,\sigma_m \sqrt{2L_N/m}} > 1, \qquad v_S = \frac{1}{m}\sum_{t\leq m} r_t\, \chi_S(x_t), \qquad L_N = \ln\frac{4KN}{\delta}, \tag{2}$$

with $\sigma_m = \mathrm{rms}(r_1, \ldots, r_m)$ (so $\sigma_n = s$), $c = 1$, $\delta = 0.1$ and $K = 4$ in every round (lines 602–603). $K_{\text{later}} = 3$ caps only the global degree after round 0. The standard error of v_S is about $\sigma_m/\sqrt{m}$, so the test asks for $c\sqrt{2L_N}$ standard errors: a Bonferroni threshold over the class's N candidates. The classes are:

- *global*: every χ_S with $|S| = k$, for $k \leq K = 4$ in round 0 and $k \leq K_{\text{later}} = 3$ after; $N = \binom{d}{k}$. Degrees 1 and 2 come from the degree-1 coefficients $g = X^\top r/n$ and the residual pair matrix $M = \frac{1}{n}\sum_t r_t x_t x_t^\top$. Degrees 3 and 4 come from the same matrix with the rows weighted by x_a or $x_a x_b$ (lines 616–648). In round 0 of the final fit, degrees 5 and 6 ($K_{\text{hi}} = 6$) come from $\frac{1}{m}\sum_t r_t \chi_{T_1}(x_t)\chi_{T_2}(x_t)$ over disjoint T_1, T_2 of degrees $\lceil k/2\rceil$ and $\lfloor k/2\rfloor$, one product of two character tables (lines 651–663);
- *extension*: for each of the up to 32 current characters χ_T with the largest $|w_T|$, the sets $T \cup \{j\}$ ($N = bd$ for b such bases) and $T \cup \{j, l\}$ ($N = b\binom{d}{2}$), from the degree-1 and pair statistics of $r\chi_T$ (lines 665–681);
- *local*: if the current characters use $2 \leq |V| \leq 12$ inputs, every $S \subseteq V$, from one Walsh–Hadamard transform of r on V; $N = \binom{|V|}{|S|}$ (lines 683–692).

A candidate that passes in several classes keeps its largest z_S. The round admits the up to 64 candidates with the largest z_S, within the room $\min(256, \lfloor n/2\rfloor) - p$.

Dense moves. Two moves catch signal spread over many small coefficients, none of which would pass a test of its own (lines 694–709). Let lin be the inputs whose character is neither a feature nor a candidate this round, and $k = |\text{lin}|$. The *linear block* admits all of them at once if

$$T_{\text{lin}} = \frac{n}{s^2} \sum_{i \in \text{lin}} g_i^2 > k + 2\sqrt{k \ln(1/\delta)} + 2\ln(1/\delta) \quad \text{and} \quad n \geq 2(p+k). \tag{3}$$

This is Laurent and Massart's tail bound for χ^2_k at level δ [26]. The block carries halfspaces and majorities. The *spectral features* take the eigenvectors u of $\tilde{M}$, the symmetrized M with zero diagonal and with the entries of pairs that are features or candidates set to 0, whose eigenvalues satisfy

$$|\lambda| > 2s\sqrt{d/n}\left(1 + \sqrt{2\ln(1/\delta)/d}\right), \tag{4}$$

at most four of them, and only if $n \geq 4d$. The first factor is the edge of the semicircle law for a $d \times d$ Wigner matrix with entries of size $s/\sqrt{n}$, about the size of $\tilde{M}$'s entries under noise. Each such u adds the feature $(u \cdot x)^2 - 1$.

The sum-block test. Take a block B with occupied groups v, each of n_v training points, and a vector ρ: the residual r, or $r\chi_P$ for a block modulated by χ_P. The block's statistic is the squared norm of the projection $\Pi_B \rho$ of ρ on the block's group indicators, in units of s^2. The block passes if

$$T_B(\rho) = \frac{1}{s^2} \sum_v n_v \bar{\rho}_v^2 = \frac{|\Pi_B \rho|^2}{s^2} \;>\; \nu + 2\sqrt{\nu L_B} + 2L_B, \tag{5}$$

$$L_B = \ln \frac{4KN_B}{\delta}, \qquad N_B = \tbinom{d}{k}\, 2^{k-1} \cdot 2 \cdot (1+\mu) \cdot k_{\max},$$

where $\bar{\rho}_v$ is the mean of ρ over group v, $\nu = \max(\text{occupied groups} - 1, 1)$, $k = |S_B|$, $\mu \leq 4$ is the number of current characters used as modulators, and $k_{\max} = 24$ (line 719). N_B counts the supports, the 2^{k-1} sign patterns up to a global sign, the two fold variants, the $1 + \mu$ modulators and the 24 sizes. By Laurent and Massart, $\Pr(\chi^2_\nu \geq \nu + 2\sqrt{\nu L} + 2L) \leq e^{-L}$. With Gaussian noise, a fixed candidate would therefore pass with probability at most e^{-L_B}. The union over every $(S, s, \text{fold}, \text{modulator}, \text{size})$ would then be at most $\delta/(4K)$, whichever candidates the search visits. That calibration is Gaussian; it is not proved for DFG's residuals, whose groups depend on the data (Section 2.5). A weighted block adds $\ln(2(4d + 20\binom{d}{2})) = 9.78$ nats at $d = 30$ for its inner term (line 136). The count is: $a \in \{\pm 2, \pm 3\}$ times a literal on any input, or a 2-parity or an AND of two literals on any pair, times 2 for a separate share of δ. A product pays $L_{B_1} + L_{B_2}$. A block is searched on the round's residual, but admitted only if it also passes the same threshold on $\tilde{r}$, the residual projected off the constant and this round's passing characters (up to 64), with $\text{mean}(\tilde{r}^2)$ in place of s^2 (lines 725–738). So it must add something beyond those characters. If $\text{mean}(\tilde{r}^2) \leq 10^{-4} s^2$, the characters already explain the residual, and no block is searched.

The block search. The test does not say which (S, s) to test. `_block_search` (lines 179–500) proposes candidates from the residual alone. It scores them by their margin, the statistic minus its threshold, on the first $\min(n, 2048)$ rows. Its starting points are:

- prefixes of the inputs ordered by $|u_i|$, for the top two eigenvectors u of $\tilde{M}$, with signs $\text{sign}(u_i)$; every prefix length $k \geq 2$, folded and unfolded. If f is a function of $t = \sum_{i \in S} s_i x_i$, then $\mathbb{E}[f\, x_i x_j] = a_2 s_i s_j$ for distinct $i, j \in S$ and 0 for every other pair (Proposition 2.5), so $\tilde{M} \approx a_2(ss^\top - I)$ on S, and its top eigenvector gives both the support and the signs;

- prefixes of the inputs ordered by $|g_i|$, with signs $\text{sign}(g_i)$;
- greedy growth from the 3 strongest pairs of $\tilde{M}$ (both relative signs) and the 2 strongest single inputs. Each step adds the input and sign that maximize T_B, until 24 inputs or 5 steps without a better margin;
- at $n \leq 128$, in a round's first additive search, a beam of width 32 over (S, s), started from the best pairs and scored on the first $\min(n, 1024)$ rows;
- in the additive search, v1.3's weighted candidates. These are eigenvector prefixes whose first input has weight 2 or 3; an inner term $\tau \in \{x_j x_l, \text{AND}(\pm x_j, \pm x_l)\}$ on the strongest pair of $\tilde{M}$, with $a \in \{\pm 2, \pm 3\}$; and extensions $t + a\tau$ of the two best unweighted candidates. The extensions use a literal outside S, an input of S re-weighted, or a 2-parity or AND on one of the 24 pairs outside S with the largest $|\tilde{M}_{jl}|$.

The three best candidates are refined by single moves (add an input with either sign, remove one, flip a sign) while the margin improves, at most 72 moves. Weighted candidates are refined in the same way, with their inner term fixed. The best weighted candidate replaces the best unweighted one if its margin is larger. The winner's groups are then formed on all n rows, and the winner is tested by Equation (5) on all n rows.

Block moves in a round. At $n \geq 32$ (lines 715–815):

- up to two *additive* blocks. The second is searched after the first candidate's group indicators are projected out of the residual, whether or not the first passed;
- *products*. When the round has two additive candidates, a block B is searched over the joint cells with the first one, A, starting also from the second. When A has at least two inputs of each sign, a *split* is also tried: B is searched over the joint cells with the block of A's positive inputs, starting from its negative inputs. The better product is admitted only if its margin on $\tilde{r}$ exceeds A's. Otherwise the product of the round's two additive candidates is tested, if their inputs are disjoint;
- one *modulated* block for each of the up to 4 current characters χ_P with the largest $|w_P|$, searched on $r\chi_P$ with P's inputs excluded;
- the products of pairs of additive blocks, admitted in earlier rounds, that have disjoint inputs.

Refit and stopping. The round adds the admitted characters, the linear block's inputs, each admitted block's indicators (if they fit in the room) and the spectral features, and refits the weights. DFG stops (lines 596–598, 824):

- when a round admits nothing (in the default variant; Section 2.4 for the forced one);
- when no room is left, $p \geq \min(256, \lfloor n/2 \rfloor)$;
- from round 1 on, when the fit has every training label's sign right;
- at the start of a round, if more than 0.75 of the call's budget is spent;
- after 8 rounds.

The code also stops if $s < 10^{-6}$. That cannot happen, since $s \geq 10^{-6}$ by its definition (`study/part11/interp/calib/VERIFY_CALIB.md`).

Readout. After the last round, if $n \geq 8p$, the least-squares weights are refined by 15 Newton steps on the mean logistic loss plus $\frac{1}{2} \cdot 10^{-4}|w|^2$, started from $4w$. The result is rescaled so that its output on the training rows has the norm of the least-squares output (lines 849–855). Part 6 added this step after the least-squares readout plateaued at 5–7% test error on a dense halfspace (thr30) where the networks reached 1% [11]. The prediction adds the kernel background $g \cdot x$ (line 856).

2.4 Small samples and compute

The forced variant. With few points the union bound passes nothing. In round 0, where only the global classes run, no character can pass its test at $d = 30$ below $n = 17$, the degree-1 floor $2c^2 \ln(4Kd/\delta) = 16.95$, and no degree-4 character below $n = 31$ (Table 2). Later rounds have smaller classes and lower floors. In a round where nothing passes, the forced variant adds a guess anyway, at most twice per fit (lines 817–822). If $n \geq 48$ and the round's first additive block candidate has T_B/threshold above the square of the best character's z, it forces that block; both ratios are on a χ^2 scale. Otherwise it adds the character with the largest z among those offered. These are every degree-1 character, the largest pair entry, the largest entry of each row block of degrees 3 and 4, and, from round 1 on, the local subsets above the smallest local threshold (`study/part11/interp/theory/THEORY.md`, §0). A forced guess passes no test. Forcing a block is on by default (`sum_force=True`); the docstring's "off by default" (line 73) is out of date.

The paired cross-validation. At $n \leq$ `cv_n` $= 512$, `DFG.fit` chooses between the default and the forced variant by cross-validation (lines 548–571). There are $F = 10$ folds at $n \leq 64$ and 5 above, assigned by a random permutation of the rows. Each variant is fitted on each fold's training rows, without the degree-5/6 search, and predicts the held-out fold. Let e_{10} be the number of points where the default errs and the forced variant does not, and e_{01} the reverse. The forced variant is chosen if $e_{10} - e_{01} > 2\sqrt{e_{10} + e_{01}}$: McNemar's test for paired errors [30], one-sided at two standard errors. The chosen variant is then fitted on all n rows, and the fold fits' FLOPs count toward the fit. On the 37 targets of Section 6 (seed 0), the cross-validation chose the default variant in 204 of the 222 cells with $n \leq 512$ and the forced variant in 18 (`study/part11/interp/ATLAS.md`, §2.3) [Numerical]. No level is proved for this choice, because the folds' held-out errors are dependent (Section 2.5).

The kernel background. The output adds $g \cdot x$, with $g = X^\top r/n$ on the final residual. With no feature but the constant, $r = y - w_0$, and the predictor is $\mathrm{sign}(w_0 + g \cdot x)$: the centred Hebbian classifier, one step on all d inputs, the counterpart of the networks' near-kernel component. Once the features fit the labels, $r \approx 0$ and $g \approx 0$. The background has no fitted parameter and costs $2nd$ FLOPs. Least squares, likewise, is kept to $p \leq n/2$ features, and the linear block is admitted only if $n \geq 2(p + k)$, so the fit never sits at its interpolation peak.

The budget. The unit is one training run of a toy network at the same d, $\mathrm{GD}(d) = 394\,PT$ FLOPs for $P = 128d + 16768$ parameters and T steps: 1.62×10^{11} at $d = 30$ (Section 3.3). A fit call gets GD · scale, with scale 1 at $n > 512$. At $n \leq 512$ the final fit gets $\frac{1}{2}$ and each of the $2F$ fold fits $1/(4F)$, so the shares add up to one run. Within a call:

- the exhaustive searches run on the first m rows, with m set so that each costs at most a fixed share of the call's budget. The shares are 15% for degree 4 (round 0 only), 3% per round for degree 3, 3% per round for the two-input extensions, and 3% each for degrees 5 and 6 (round 0 of the final fit only). The search uses $m = \min(n, \lfloor \text{share} \cdot \mathrm{GD} \cdot \text{scale}/c_{\text{row}} \rfloor)$ rows, with c_{row} the code's per-row cost estimate (Proposition 2.6), and its threshold uses m (Equation (2)).

Tests		
`K`	4	global degree in round 0
`K_later`	3	global degree, rounds ≥ 1
`K_hi`	6	top degree, round 0
`c`	1	threshold multiplier
`delta`	0.1	level δ of every test
`noise`	`residual`	tests scaled by $s = \mathrm{rms}(r)$
Rounds and classes		
`rounds`	8	rounds at most
`max_new`	64	characters admitted per round
`max_feats`	256	features (also $\leq n/2$)
`max_ext_bases`	32	characters extended
`local_max`	12	largest $\lvert V\rvert$, local class
`blocks`	True	linear and spectral moves
Small n and budget		
`small_n`	True	forced variant and its test
`cv_n`	512	the test runs at $n \leq$ this
`background`	True	kernel background $g \cdot x$
`budget`	(0.15, 0.03, 0.03)	shares: degree 4; 3; ext. by 2
`budget_hi`	0.03	share: degrees 5, 6 each
Readout		
`readout`	`ls`	run as `logit`
`ridge`	0.001	ridge of least squares ($\times n$)
`logit_lam`	0.0001	L2 penalty, logistic steps

Sum blocks		
`sums`	True	sum blocks
`sum_nmin`	32	smallest n for the block search
`sum_kmax`	24	largest block; the $k_{\max}$ of N_B
`sum_mod`	4	modulators μ at most
`sum_add`	2	additive blocks per round
`sum_gcap`	2048	rows for the search's scores
`sum_mc`	0.005	smallest group over n (at least 3)
`sum_prod`	True	product and split blocks
`sum_seeds`	(2, 3, 2)	eigenvector, pair, single seeds
`sum_inner`	True	weighted blocks $t + a\tau$
`sum_ntau`	1	pairs tried as inner terms
`sum_beam`	32	beam width
`sum_beam_nmax`	128	beam only at $n \leq$ this
`sum_force`	True	forced variant may force a block
`sum_force_nmin`	48	smallest n for a forced block
`sum_force_r2`	0	smallest T_B/n of a forced block

Fixed in the code		line
15	logistic Newton steps	851
$n \geq 8p$	when they run	849
$4w$	where they start	850
0.75	budget share that ends the rounds	598
10 / 5	folds at $n \leq 64$ / above	553
$2\sqrt{e_{10} + e_{01}}$	margin of the paired test	567
$\frac{1}{4F}$, $\frac{1}{2}$	budget scale: fold fit, final fit	564, 569
32 / 64	fewest rows: degrees 3, 4 / 5, 6	630, 655
4; $4d$	spectral features; smallest n	707
2	forced guesses per fit	817
10^{-4}	least $\mathrm{mean}(\tilde{r}^2)/s^2$ for blocks	734
3	fewest points per group	721
$\pm 2, \pm 3$	weights a of inner terms	273
5	growth steps without gain	179
24	pairs for inner-term extensions	180
1024	rows of the beam	180
$3k_{\max}$	refinement moves at most	376

Table 3: DFG v1.3's constructor defaults, as run (`DFG(readout="logit")`, every other argument at its default), and constants fixed in its code, with their lines [Proved, by the code]. Source: `study/part10/hyb_portfolio3/dfg13.py`, read by `paper_dfg/make_tables.py` (`tab_params`). The defaults are read from the constructor's signature, and each fixed constant is checked to occur at the line given.

A search whose m falls below 32 rows (64 for degrees 5 and 6) is skipped. At $d = 30$ only degrees 5 and 6 are subsampled, to 1374 and 147 rows (687 and 73 at $n \leq 512$). At $d = 60$ the degree-4 search sees at most 2212 rows, and at $d = 120$ at most 364, with that d's 40000-step run as the unit;

- a round starts only while the call has spent less than 0.75 of its budget. The other parts are not budgeted: degrees 1 and 2, the one-input extensions, the local class, the eigendecomposition, the sum blocks, least squares, the Newton steps and the background. Only this cap bounds them, so a call can exceed 0.75 by the cost of its last round.

The budget in practice [Numerical]. The cap stopped none of the 6786 fit calls recorded by the false-admission and compute analysis; the largest spent 0.471 of its budget (`study/part11/interp/calib/CALIB.md`, §3.4, confirmed by `study/part11/interp/calib/VERIFY_CALIB.md`). Over the 4730 fits in the result files, the largest uses 0.519 of one network run (Section 4.7).

2.5 Simple guarantees

Five statements about DFG can be made exactly. They are P1–P5 of `study/part11/interp/theory/THEORY.md`, restated here for the code as it is. They concern round 0, single tests, the block dictionary and the code's arithmetic; none bounds the test error of a fitted model. Each statement is followed by what it does not cover, and by checks computed by `study/part11/interp/theory/check_theory.py` and stored in `study/part11/interp/theory/check_theory.json` under the keys given. "By the code" means a direct reading of the cited lines. Throughout, the rows are iid uniform on $\{\pm1\}^d$, as on the toy suites; the column permutation of `harness.Permuted` preserves this. We write $L_k = \ln(4K\binom{d}{k}/\delta)$ for the log term of the global degree-k class.

Proposition 2.1 (Round 0's character and linear tests under the null; P1)**.** *[Proved]. Let the labels be independent of the rows, with any joint law. In round 0 of any fit call (a final or a fold fit, either variant, any budget scale) the following hold.*

(a) For each degree k the call searches, and every $u > 0$,

$$\Pr\Big(\max_{|S|=k} z_S \ge u\Big) \le 2\tbinom{d}{k}\Big(\frac{\delta}{4K\binom{d}{k}}\Big)^{u^2c^2}.$$

At $u = c = 1$ this is $\delta/(2K) = 0.0125$ per degree, for every d and n.

(b) The linear block's statistic over all d inputs satisfies $\Pr(T_{\mathrm{lin}} > d + 2\sqrt{dx} + 2x) \le e^{-x}$ for every $x > 0$. Hence a linear block is admitted with probability at most $\delta + \delta/(2K)$.

(c) If round 0 admits nothing, the default variant's model is the constant plus the kernel background. The default variant's model therefore holds a feature other than the constant with probability at most $3\delta/4 + \delta = 0.175$, plus the probabilities that round 0 admits a spectral feature or a sum block.

Proof. (a) Condition on y. Round 0's fit is on the constant alone, so $r = y - w_0$ with $w_0 = \sum_t y_t/(n + 10^{-9})$, and r, s and σ_m are functions of y. For a fixed nonempty S the signs $\varepsilon_t = \chi_S(x_t)$ are iid uniform and independent of y. The statistic is $\frac{1}{m}\sum_{t\le m} r_t\varepsilon_t$ and its threshold $c\sigma\sqrt{2L_k/m}$, where $\sigma^2 \ge \frac{1}{m}\sum_{t\le m} r_t^2$: $\sigma = \sigma_m$ if $m < n$, and $\sigma = s \ge \mathrm{rms}(r)$ if $m = n$. Hoeffding's inequality [21], $\Pr(|\sum_t a_t\varepsilon_t| \ge \lambda) \le 2\exp(-\lambda^2/(2\sum_t a_t^2))$, with $a_t = r_t$ and $\lambda = uc\sigma\sqrt{2mL_k}$, gives $\Pr(z_S \ge u \mid y) \le 2e^{-u^2c^2L_k}$. A union bound over the $\binom{d}{k}$ sets and the expectation over y finish the proof. The degree-5/6 search offers $\frac{1}{m}\sum_t r_t\chi_{T_1}(x_t)\chi_{T_2}(x_t)$ only for disjoint T_1, T_2 (line 662). That is the same statistic for $S = T_1 \cup T_2$, and a set reached by several splits is one event.

(b) Condition on y and let $a = r/|r|$. Since $ns^2 \ge |r|^2$, $T_{\mathrm{lin}} \le \sum_i Z_i^2$ with $Z_i = \sum_t a_t x_{t,i}$. The Z_i are independent Rademacher sums with $\sum_t a_t^2 = 1$. For $0 \le \lambda < \frac{1}{2}$ and $\xi \sim N(0,1)$, $\mathbb{E}e^{\lambda Z_i^2} = \mathbb{E}_\xi \prod_t \cosh(\sqrt{2\lambda}\,\xi a_t) \le \mathbb{E}_\xi e^{\lambda\xi^2} = (1-2\lambda)^{-1/2}$. Hence $\ln \mathbb{E}e^{\lambda(\sum_i Z_i^2 - d)} \le d\lambda^2/(1-2\lambda)$. This bound on the log-Laplace transform is all that the Chernoff argument in the proof of Lemma 1 of [26] uses, and that argument gives the tail. At the code's $x = \ln(1/\delta)$ the bound is δ. The block's inputs are all d unless a degree-1 character passed, which (a) bounds by $\delta/(2K)$.

(c) By the code, a round that admits nothing ends the fit (line 824). In round 0 there are no current characters or blocks, so the extension, local and modulated classes and the products of admitted blocks are empty. Add (a) over degrees 1–6 and (b); the $\delta/(2K)$ term of (b) is (a)'s degree 1. □

Not covered. The statements are for exact arithmetic; rounding matters only within rounding error of a threshold. Several parts of a fit are not covered.

- The spectral test: its edge (4) is the asymptotic semicircle edge with a Gaussian correction, not a bound.
- The sum-block tests: Equation (5) is calibrated for Gaussian noise, but the groups are a data-dependent partition of the rows. The union bound counts every candidate, but the tail of a fixed candidate is not proved to be at most e^{-L_B}.
- The forced variant, which adds features by design.
- The cross-validation's choice between the variants: a z-test at 2 on dependent held-out errors.
- Rounds ≥ 1: they run only after an admission, and their residual then depends on the rows.

Other input laws [Proved]. The verifier of THEORY.md adds that (a) does not need uniform rows for the searches on all n rows. Round 0's residual sums to zero, up to the 10^{-9} term, so $\sum_t r_t \chi_S(x_t) = \sum_t r_t(\chi_S(x_t) - \mu_S)$ with $\mu_S = \mathbb{E}\chi_S$: a sum of independent centred terms of range $2|r_t|$, to which Hoeffding's bound applies unchanged for iid rows from any law (`study/part11/interp/theory/VERIFY_THEORY.md`). This covers the character tests on the dependent one-hot columns of the tabular data (Section 4.6). It does not cover subsampled searches, whose rows' residuals need not sum to zero, or (b).

Checks [Numerical].

- *The character tests.* 16 null configurations of round 0 were simulated, 300 data sets each: at $d = 30$ with $n = 32$–2048 and three label laws, at $d = 15$ with degrees 1–6, and at $d = 60$ in a fold fit whose degree-4 search is subsampled. All 280 checks of (a), at levels $u = 0.7$–1, held (`P1.sims`). The bound is loose by about the Gaussian Mills ratio: at $d = 30$ the Gaussian approximation gives 0.87–1.15×10^{-3} per degree, against the bound's 12.5×10^{-3} (`P1.bound_over_gauss_d30`).
- *The linear block.* 43 of the 4800 null data sets passed it (0.9%), against the bound $\delta = 10\%$; the χ^2_d tail at the threshold is 0.92% at $d = 30$ (`P1.linear_chi2_level`).
- *Full fits on null labels* ($d = 30$). With fair labels, 78 of 80 final models are the constant plus the background (`P1.full_fits`). One holds a pair at $z = 1.06$, the event (a) bounds; the other is a forced-variant model chosen by the cross-validation. With $\Pr(y = 1) = 0.85$ at $n = 128$, 7 of 30 hold spurious spectral or sum-block features (`P1.nonconstant_fits`).
- *The spectral test on biased labels.* Near $n = 4d$ its null pass rate grows with d. For $\Pr(y = 1) = 0.8$ it is about 5%, 47% and 99% at $d = 30$, 60 and 120 (`study/part11/interp/calib/CALIB.md` with the corrections of `study/part11/interp/calib/VERIFY_CALIB.md`).

Section 6.3 reports DFG's false admissions in full.

Proposition 2.2 (The sample size of a parity; P2)**.** *[Proved]. Let $y = \chi_S(x)$ with $|S| = k$. Take round 0 of a fit call on n rows in which the degree-k search runs on all of them: $k \leq 2$ always, and $k = 3, 4$ when $m_k = n \geq 32$. Then* $\mathrm{mean}(r\chi_S) = 1 - w_0\bar{y}$ *and* $\mathrm{mean}(r^2) = 1 - 2w_0\bar{y} + w_0^2$. *With $w_0 = \bar{y}$ (dropping the 10^{-9}), χ_S passes its test if and only if*

$$n\,(1 - \bar{y}^2) > n_{\mathrm{thr}}(k, d) = 2c^2 \ln\!\big(4K\tbinom{d}{k}/\delta\big) = 2c^2\big(k \ln d - \ln k! + \ln(4K/\delta)\big) + O(k^2/d). \qquad (6)$$

Since $\sum_t y_t$ is a sum of n independent fair signs, exactly $\Pr(\mathit{pass}) = \Pr\big(|\sum_t y_t| < \sqrt{n(n - n_{\mathrm{thr}})}\big)$.

Proof. $y\chi_S = \chi_S^2 = 1$ and $y^2 = 1$ give the two means. With $w_0 = \bar{y}$, $z_S > 1$ reads $1 - \bar{y}^2 > c\sqrt{1 - \bar{y}^2}\sqrt{2L_k/n}$. Divide by $\sqrt{1 - \bar{y}^2}$ and square; if $\bar{y}^2 = 1$, both sides of the equivalence fail. Finally, $\ln \binom{d}{k} = k \ln d - \ln k! + O(k^2/d)$. □

Passing round 0's test is not the same as being in the final model. Admission also needs a place among the round's 64 largest z and room for the feature, which is not proved but never bound in the parity cells. The forced variant can admit χ_S without passing, and a later round could admit it after another admission. Whether the final model holds χ_S is therefore the measured statement below. The constant $2c^2\ln(4K/\delta) - 2c^2\ln k!$ is not small: n_{thr} is $2k\ln d$ plus 8.8, 6.6 and 3.8 for $k = 2$, 3 and 4, so it grows like $2k\ln d$, while $2k\ln d$ alone is 9–44% below it for $d = 15$–120 (`study/part11/interp/theory/VERIFY_THEORY.md`).

Proposition 2.3 (A floor for every target; P2′)**.** *[Proved]. For any labels, in any round, no character passes the test of a class of N characters tested on m rows unless $m > 2c^2\ln(4KN/\delta)$. No block passes its test unless $n > \nu + 2\sqrt{\nu L_B} + 2L_B \geq 1 + 2\sqrt{L_B} + 2L_B$.*

Proof. By Cauchy–Schwarz, $|\frac{1}{m}\sum_{t\leq m} r_t\chi_S(x_t)| \leq \sigma_m \leq \sigma$, so $z_S > 1$ needs $c\sqrt{2\ln(4KN/\delta)/m} < 1$. For a block, $\sum_v n_v\bar{\rho}_v^2 \leq \sum_t \rho_t^2 \leq ns^2$: a projection does not increase the norm, $|\rho_t| = |r_t|$, and $s^2 \geq \text{mean}(r^2)$. So $T_B \leq n$. The same holds for $\tilde{r}$ in its own units, for products, and for the deflated residual of the second additive search. □

The character half is THEORY.md's P2′; the block half is added here. Both bound passing a test, not admission: the forced variant's guesses ($n \leq 512$) and the linear block's test lie outside these floors. A k-parity with $\bar{y} = 0$ attains the character floor, so $n_{\text{thr}}(k, d)$ is exactly a parity's sample size. The last column of Table 2 gives the floors at $d = 30$.

Checks [Numerical].

- *Every parity cell with $n \geq 32$.* Proposition 2.2's criterion, with each training set's own $\bar{y}$, predicts the outcome in all 250 parity cells with $n \geq 32$: par2–par4 at $d = 15$–120, every seed (`P2.agree`). The test error is 0 where the criterion holds, and 0.488–0.506 elsewhere.
- $n = 16$. There $n_{\text{thr}} > 16$ at every d, and nothing passes. The 7 error-free par2 cells at $d \leq 60$ come from the forced variant, chosen by the cross-validation. With that, the criterion reproduces all 11 measured n^* (16–56) to within 0.2 (`P2.pred`).
- *par4 at $d = 30$.* Its n^* is 52, not 28, because 2 of its 3 seeds have $|\sum_t y_t| > 6$ at $n = 32$, where $n_{\text{thr}} = 30.59$. Under Proposition 2.2's exact law that event has probability 0.12.

Against the networks [Numerical; Heuristic beyond $d = 120$]. n_{thr}, which grows like $2k\ln d$, is 19.5–42.0 for $k = 2$–4 and $d = 15$–120 (`P2.n_thr`). DFG's n^* on the parities moves by at most one grid step over that range (Figure 3). The networks' n^* grows as $d^{1.03}$, $d^{1.96}$ and $d^{2.40}$ for $k = 2$, 3 and 4 (Table 15), about d^{k-1} for $k \leq 3$. DFG pays in compute instead: an exhaustive degree-k search costs about $n\binom{d}{k}$ FLOPs times a constant (Proposition 2.6). Under DFG's fixed budget, at $n > 512$, the global search can pass no degree-3 character from $d = 630$ on and no degree-4 character from $d = 179$ on, because there $m_k < 32$ or $m_k \leq n_{\text{thr}}$ (Proposition 2.3; `P5.d_parity_unreachable`). Characters of other targets can still enter through the extension and local classes, which run on all rows. At $d = 1000$ the test alone would pass the parities of degree 2–4 at $n \approx 56$–57. DFG v1.3 skips its degree-3 and degree-4 searches there ($m_3 = 17$, $m_4 = 0$), and an exhaustive degree-4 search on 64 rows would cost 55.9 network runs (`P2.d1000`, `P5.exhaustive`).

Proposition 2.4 (When a character passes in round 0; P3)**.** *(a) [Proved]. In round 0, for a set S searched on all n rows, $v_S = \hat{c}_S - w_0\bar{\chi}_S$ with $\hat{c}_S = \text{mean}(y\chi_S)$ and $\bar{\chi}_S = \text{mean}(\chi_S)$, and $s^2 = 1 - 2w_0\bar{y} + w_0^2$. Moreover $z_S > 1$ iff $n > 2c^2L_{|S|}\,s^2/v_S^2$. For iid uniform rows and nonempty S, $\mathbb{E}v_S = \hat{f}(S)\,(1 - 1/(n + 10^{-9}))$.*

(b) [Heuristic]. Replacing v_S by $\hat{f}(S)$ and s^2 by $1-\hat{f}(\emptyset)^2$, the round-0 test admits S once

$$n > n_{\text{glob}}(S) = 2c^2 \ln\!\left(4K\tbinom{d}{|S|}/\delta\right) \frac{1-\hat{f}(\emptyset)^2}{\hat{f}(S)^2}. \tag{7}$$

(c) [Proved]. In round j, S passes the test of a class of N_{cls} candidates tested on all rows iff $n > n_{\text{cls}} = 2c^2 \ln(4KN_{\text{cls}}/\delta)\, s_j^2/v_{S,j}^2$ (on a subsample, m and σ_m replace n and s_j). Hence

$$\frac{n_{\text{glob}}}{n_{\text{cls}}} = \frac{L_{|S|}}{\ln(4KN_{\text{cls}}/\delta)} \cdot \frac{1-\hat{f}(\emptyset)^2}{s_j^2} \cdot \frac{v_{S,j}^2}{\hat{f}(S)^2} :$$

a smaller union bound, a smaller residual, and the residual's coefficient on S.

Proof. (a) is the algebra of Proposition 2.2 with $\hat{c}_S$ in place of 1. For the expectation, $\mathbb{E}[w_0 \bar{\chi}_S] = \frac{1}{n(n+10^{-9})} \sum_{t,u} \mathbb{E}[y_t \chi_S(x_u)]$. Its terms with $u = t$ are $\hat{f}(S)$, and those with $u \neq t$ vanish because $\mathbb{E}\chi_S = 0$. (c) is the same algebra in round j. □

Checks [Numerical]. The checks use the 41 targets of `designed` and `random`, seed 0, at every n of the grid.

- *The round-0 decision.* Over 8681 (character, n) pairs, $n > n_{\text{glob}}$ agrees with DFG's round-0 decision in 8409 (96.9%; `P3.test0`). The agreement comes mostly from pairs far from n_{glob}: within one doubling of it, the two agree in 773 of 1012 pairs (76%), and more than two doublings away in 99.9–100% (`study/part11/interp/theory/VERIFY_THEORY.md`). The pass rate is 6% one doubling below n_{glob} and 94% one doubling above.
- *Entry through the round-0 test.* The 249 characters that first enter through it do so at a median $2^{0.26}\, n_{\text{glob}}$ (`P3.first`).
- *Entry through later-round classes.* Characters that enter through the local, extension or later global classes do so a median 1.4–3.7 doublings below n_{glob}, by class (`P3.entry_vs_nglob`). In Proposition 2.4(c)'s factorization the residual factor is the largest: its class medians are 3.2–19, and per character it ranges from 1.2 to 59.5. The union-bound factor has class medians 1.2–1.7, but single local-class characters gain up to 3.3 from it (`P3.factors`; per-character ranges from `study/part11/interp/theory/VERIFY_THEORY.md`).

Sections 6.5 and 6.6 use n_{glob} for when, and in what order, DFG admits a target's characters.

Proposition 2.5 (Sum blocks; P4)**.** *[Proved]. Let $f(x) = \psi(\beta(x))$, a function of $\beta = \#\{i \in S : s_i x_i = 1\} = (t+k)/2$, where $t = \sum_{i\in S} s_i x_i$ is a signed sum of k inputs.*

(a) If every value of t that occurs in training is its own group, the span of the block's indicators and the constant contains f exactly at every x whose value of t occurs in training, with at most $k+1$ weights. (DFG's small ridge shrinks the fitted weights slightly.)

(b) Every value is its own group iff every value occurs at least $\max(3, \lfloor 0.005n \rfloor)$ times. In the large-n limit this holds iff $k \leq 7$.

(c) A sign readout that uses one block is constant on its groups. It therefore errs on at least $\sum_{\text{groups}} \min(\Pr(f=1, \text{group}), \Pr(f=-1, \text{group}))$ of the inputs.

(d) $\hat{f}(T) = 0$ unless $T \subseteq S$, and $\hat{f}(T) = (\prod_{i\in T} s_i)\, a_{|T|}$ with $a_j = 2^{-k} \sum_\beta \psi(\beta) \mathcal{K}_j(\beta)$, where $\mathcal{K}_j(\beta) = \sum_l (-1)^{j-l} \binom{j}{l} \binom{k-j}{\beta-l}$. The Fourier weight at degree j is $W_j = \binom{k}{j} a_j^2$.

(e) For a residue class, $\psi(\beta) = \pm(1 - 2[\beta \equiv \kappa \bmod 3])$, and $j \geq 1$: $|a_j| \leq \frac{4}{3}\, 3^{j/2}\, 2^{-k}$, so $W_j \leq \frac{16}{9}\binom{k}{j} 3^j 4^{-k}$.

(f) Every starting point of the block search is computed from statistics of one or two inputs, or is an earlier candidate grown from such a starting point. In round 0, $\mathbb{E}\tilde{M}_{ij} = \hat{f}(\{i,j\})(1 - 1/(n + 10^{-9}))$ for $i \neq j$, and $\mathbb{E}g_i = \hat{f}(\{i\})(1 - 1/(n + 10^{-9}))$.

Proof sketch. (a) A function η of the group equals $\eta(v_{\max}) + \sum_{v \neq v_{\max}} (\eta(v) - \eta(v_{\max}))[\text{group} = v]$ on every occupied group.

(b) By the code (`_edges`): scanning the values upward, a group closes once it holds $\max(3, \lfloor 0.005n \rfloor)$ points, and a short last group is merged into the one before it. The rarest values, $t = \pm k$, have probability 2^{-k}, which is at least 0.005 iff $k \leq 7$.

(c) Any combination of the block's features and the constant is constant on each group.

(d) Inputs outside S are independent of f. On S, substitute $z_i = s_i x_i$; by symmetry the coefficient depends only on $|T|$, and $\mathcal{K}_j(\beta) = \sum_{z:\, \#\{z_i = 1\} = \beta} z_1 \cdots z_j$.

(e) Write $[\beta \equiv \kappa] = \frac{1}{3}\sum_{m=0}^{2} \omega^{m(\beta-\kappa)}$ with $\omega = e^{2\pi i/3}$. Since $\beta = \sum_{i \in S}(1 + z_i)/2$ with independent z_i,

$$\mathbb{E}\big[\omega^{m\beta} z_1 \cdots z_j\big] = \Big(\frac{\omega^m - 1}{2}\Big)^j \Big(\frac{1 + \omega^m}{2}\Big)^{k-j},$$

which vanishes at $m = 0$ for $j \geq 1$; the terms $m = 1$ and $m = 2$ are complex conjugates. Moreover $|\omega - 1|/2 = \sqrt{3}/2$ and $|1 + \omega|/2 = \frac{1}{2}$, so $|a_j| \leq 2 \cdot \frac{2}{3} \cdot 3^{j/2} 2^{-k}$.

(f) The starting points are the eigenvectors and the strongest entries of $\tilde{M}$; the coefficients g_i and single inputs; inner terms on $\tilde{M}$'s strongest pair; at $n \leq 128$, a beam started from two-input blocks; and, in the product and split searches, the round's earlier candidates. A two-input block's groups are functions of $x_i + sx_j$, and these span $\{1, x_i + sx_j, x_i x_j\}$. The expectations follow as in Proposition 2.4(a). □

What this says about the residue classes [Numerical; Heuristic where marked]. Take the three residue-class targets f7_05, h6_00 and fd_10, on $k = 14$, 13 and 10 inputs, which DFG alone never learns (Section 4.2).

- *Fourier weight.* Every one of the $2^k - 1$ characters on S is nonzero, while $W_1 + W_2 = 1.6 \times 10^{-6}$, 5.4×10^{-6} and 2.1×10^{-4}. These are exact rationals (`P4.targets`).
- *The representation is not what fails.* One block with one group per value represents each target with at most $k+1$ weights, and least squares on such a block reproduces seed 0's training labels at $n = 16384$ exactly. DFG's own grouping merges the rarest values, which costs 0.1–0.2% error (`P4.grouping`). Part 10 found the true block passing the test from $n = 128$ [13].
- *The search is* (Section 6.4). By (f), its starting points see these targets only through a_1 and a_2. [Heuristic]: by the BBP threshold for a rank-one perturbation of a random matrix [3, 6], the eigenvector seed needs $n \approx s^2 d/(a_2^2(k-1)^2)$: 1.1×10^7, 3.1×10^6 and 8.6×10^4 rows for the three targets, far above the grid's 16384.

Proposition 2.6 (Compute; P5)**.** *[Proved, by the code]. (a) The global searches of a round are charged, with a multiply-add counting 2 (lines 617–658):*

- *$2nd$ FLOPs for degree 1 and $2nd^2 + nd$ for degree 2;*
- *$(d-2)(2m_3d^2 + 2m_3d)$ for degree 3 and $\binom{d-2}{2}(2m_4d^2 + 3m_4d)$ for degree 4;*
- *$m_k(2N_1N_2 + N_1k_1 + N_2k_2)$ for degrees $k = 5, 6$, with $N_i = \binom{d}{k_i}$, $k_1 = \lceil k/2 \rceil$ and $k_2 = \lfloor k/2 \rfloor$.*

The exhaustive degree-3 and degree-4 searches therefore cost about $2nd^3 \approx 12n\binom{d}{3}$ *and* $nd^4 \approx 24n\binom{d}{4}$. *The matrix products also compute redundant entries, so one character costs* $12n$ *or* $24n$ *FLOPs, not* $2n$.

(b) Each budgeted search runs on $m = \min(n, \lfloor \mathit{share} \cdot \mathrm{GD} \cdot \mathit{scale}/c_{\mathrm{row}} \rfloor)$ *rows, where* c_{row} *is the code's per-row cost estimate. It is* $d(2d^2 + 2d)$ *for degree 3;* $\binom{d}{2}(2d^2 + 3d)$ *for degree 4;* $b(2d^2 + 3d)$ *for the two-input extensions of* b *bases; and* $2N_1N_2 + N_1k_1 + N_2k_2$ *for degrees 5 and 6. These estimates are at least the charged per-row costs, so each budgeted search costs at most its share of the call's budget. The shares of the* $2F + 1$ *calls at* $n \le 512$ *add up to one run,* $\frac{1}{2} + 2F \cdot \frac{1}{4F} = 1$. *The unbudgeted parts are bounded only by the round cap, up to the cost of the last round.*

Checks [Numerical].

- *An analytic FLOP model* of the "parity path" (round 0 admits χ_S alone, round 1 stops on the sign rule) reproduces the recorded FLOP count to 10^{-9} in 228 of the 275 parity cells. It reproduces 4 more once a false linear block is added, which refits confirm; the other 43 cells are at $n \le 32$ (`P5.check`).
- *The budget bound.* Over $d = 5$–399 and the four budget scales, no budgeted search's charged cost exceeds its share; the largest ratio is $1 - 1.4 \times 10^{-8}$ (`P5.max_share_used_over_share`).
- *Against one network run.* $\mathrm{GD}(d)$ is linear in d, while an exhaustive degree-4 search on 16384 rows costs 0.072, 1.04 and 6.5 network runs at $d = 30$, 60 and 120 (`P5.exhaustive`). The budget's subsampling is what keeps DFG inside one run.

What these statements do not give. None of them bounds the test error of a fitted model. Proposition 2.1 covers round 0 only. Proposition 2.2 is for noiseless parities, and Proposition 2.4(b) is first order. Proposition 2.5 says what a block can represent and what the search's starting points see, not when the search succeeds. Proposition 2.6 counts FLOPs, not CPU time (Section 4.7).

2.6 How DFG relates to familiar learners

[Heuristic], except where a statement restates the code or Section 2.5.

- *Forward stepwise regression and orthogonal matching pursuit.* DFG's loop is forward selection: score candidates by their correlation with the residual, admit, and refit all coefficients jointly by least squares. The joint refit is what distinguishes orthogonal matching pursuit [32] from matching pursuit [29]. DFG differs in three ways.
  - Its dictionary is never stored as a matrix. It is enumerated class by class, with a fast transform where one exists (the Walsh–Hadamard transform of the local class).
  - Its entry rule is a significance test, not "take the best". If χ_S is orthogonal on the sample to the current features, it would remove nv_S^2 from the residual sum of squares. Then $z_S > 1$ iff that reduction, in units of s^2, exceeds $2c^2 \ln(4KN/\delta)$. With $c = 1$ this is the per-variable penalty $2 \ln p$ of the risk inflation criterion [17], and the universal threshold $\sqrt{2 \ln p}$ of wavelet shrinkage [15], with $p = 4KN/\delta$: a Bonferroni rule within each class.
  - A round admits every passing candidate, up to 64, not one at a time.
- *The low-degree algorithm and sparse Fourier learning.* The low-degree algorithm [27] estimates every Fourier coefficient up to degree k from random examples and predicts with the sign of the truncated expansion. `lowdeg` (Section 3.2) is its thresholded version. DFG is a greedy, budgeted, residual-based version of it.

- It estimates coefficients of the residual, not of the labels, and refits the weights instead of using the raw estimates.
- It adds classes that grow from what it has: the extensions and the local transform.
- It adds a second dictionary, the signed-sum blocks.

Kushilevitz and Mansour [25], building on Goldreich and Levin [20], find every large Fourier coefficient of any degree in time polynomial in d, but they need membership queries. DFG has only random examples, so it enumerates each class. A degree-k class costs about $n\binom{d}{k}$ FLOPs (Proposition 2.6), and characters above degree 6 are reached only through the extensions, the local class and the blocks.

- *Statistical queries.* Each character test estimates the correlation $\mathbb{E}[r\chi_S]$ of the residual with a fixed function of x: a correlational statistical query [9, 23], asked of the residual instead of the label. The union bound is the tolerance budget for all N queries of a class, answered from the same n rows; that is why the sample cost is logarithmic in N (Proposition 2.2). For parities, statistical-query learners need a number of queries, or an inverse tolerance, polynomial in the number of candidate parities [7]. DFG pays the query count in compute (the enumeration) and only its logarithm in samples. Analyses of gradient training on sparse parities find costs polynomial in d, with an exponent that grows with k [2, 4].
- *Occam's razor and minimum description length.* The log term $\ln(4KN/\delta)$ is, up to the constant $\ln(4K/\delta)$, the number of nats needed to name one candidate in its class. In those terms the character test reads: the residual sum of squares it removes, over $2s^2$ (the Gaussian code length it saves), exceeds $c^2 \ln(4KN/\delta)$. That is a two-part code in the sense of minimum description length [34], without the usual $\frac{1}{2}\ln n$ for the coefficient. The parity criterion $n > 2c^2 \ln(4K\binom{d}{k}/\delta)$ is the matching Occam sample size [8]: logarithmic in the size of the class. A fitted model's description length is the sum of its features' names; par4's $\chi_{\{1,2,3,4\}}$ costs $\ln(4K\binom{30}{4}/\delta) = 15.29$ nats at $d = 30$.
- *Boosting.* Like gradient boosting [18], and L2 boosting in particular [10], DFG fits the residual stagewise with base learners from a dictionary. Unlike boosting it has no shrinkage and no early stopping: a candidate enters only if it passes its test, and every weight is refitted jointly after each round, as in orthogonal matching pursuit. Its base learners are fixed Boolean features, not fitted trees. And its loop is short. At $n = 16384$, 2048 and 512, all the target characters that DFG carries enter in a single round in 32, 30 and 24 of the 33 fits on the 11 Part 5 targets (`study/part11/interp/order/ORDER.md`) [Numerical].
- *Gradient-trained networks.* Part 6 designed DFG from what Parts 1–5 had seen the study's networks do [11, 12]:
  - they grow features greedily, cheapest first;
  - a feature's cost is set by its correlation with the residual against the noise of the competing directions;
  - features on inputs already in use are cheaper;
  - a near-kernel linear component is present from the start;
  - on every plateau the network computes the sign of a least-squares fit on the features grown so far.

  DFG keeps this picture and replaces the search: enumeration and explicit tests take the place of noisy gradient steps in weight space. Inside DFG's first round at $n = 16384$, the

object	how it is chosen	continuous?
characters χ_S	enumeration in classes; the union-bound test (2)	no
sum blocks: (S, s), fold, inner term $a\tau$, modulator, product partner	the block search (eigenvector, pair and single-input seeds, greedy growth, beam, single moves); the χ^2 test (5)	no
a block's groups	contiguous runs of observed values with at least $\max(3, \lfloor 0.005n \rfloor)$ points	no
the linear block's inputs	one χ^2 test (3)	no
the variant (default or forced)	the paired test on held-out errors	no
the settings K, c, δ, budget shares, ...	constructor defaults, fixed across all data sets (Table 3)	no
spectral directions u	top eigenvectors of the residual pair matrix	borderline: real directions, computed in closed form
the kernel background g	one Hebbian step, $g = X^\top r/n$	yes, computed in closed form
readout weights w	least squares with a small ridge, then 15 Newton steps of logistic regression	yes: a linear readout

Table 4: What DFG v1.3 chooses discretely and what it fits continuously [Proved, by the code].

order of its test statistics is positively rank-correlated with the order in which the Part 5 networks acquire the same characters, on all 11 targets (Kendall τ 0.22–0.81, median 0.54; `study/part11/interp/order/ORDER.md`; Section 6.6) [Numerical]. The trade is samples for FLOPs. On a k-parity DFG needs $n_{\text{thr}}(k, d)$ samples, which grow like $2k \ln d$, and about $n\binom{d}{k}$ FLOPs of search; the networks need about d^{k-1} samples for $k \leq 3$, at a cost per step linear in d (Propositions 2.2 and 2.6; Section 4.5).

2.7 What is discrete

Table 4 lists what DFG fits.

- *Every feature but the spectral ones comes from a finite dictionary.* The candidates are enumerated, or proposed by a combinatorial search over signed subsets, and admitted by a test with an explicit threshold, or, in the forced variant, as the best-scoring candidate when nothing passes. A fitted model can therefore be printed as a list of named features (Section 6).
- *Only the readout is fitted continuously.* Its weights are fitted by least squares and then by a few Newton steps on a convex logistic loss. The background g is a closed-form statistic of the residual, not a fitted parameter. No feature is fitted by gradient descent, and the readout's 15 Newton steps are the only iterative continuous optimization in a fit.
- *One borderline case.* The spectral directions u are real vectors, computed in closed form as eigenvectors of the residual pair matrix rather than trained. Each feature $(u \cdot x)^2 - 1 = 2\sum_{i<j} u_i u_j x_i x_j$ is a fixed combination of degree-2 characters. The post hoc `core` run, which has no spectral features, keeps DFG's results (Section 2.2).

3 How it is evaluated

This section gives the tasks, the references, the metrics and budgets, and the Part 11 protocol, including what is held out for DFG v1.3. Appendix A collects the names of learners, references, sets, targets and terms.

3.1 Tasks

Toy Boolean targets. Inputs are uniform on $\{\pm 1\}^d$ with $d = 30$, and labels are the ± 1 values of a target f. There are 11 suites, 138 targets in all, defined in `study/boolfuncs.py` and listed with their status in Table 5. The data are the networks' own (`study/discrete_learner/harness.py`). For each target and seed, the training set of size n is the first n rows of one draw of 16384 points, so training sets are nested in n, and each target has one fixed test set of 4096 points.

- `designed` (21), the study's original targets (Parts 1–5): a dictator, AND and majority of 3 inputs, `maj9`, a halfspace with Gaussian weights on 10 inputs, parities of 2–4 inputs, a staircase, compositions of parities and majorities, a multiplexer, a DNF, a decision list, a sparse polynomial threshold and four random 6-gate formulas.
- `random` (20): random fan-in-2 AND/OR/XOR formulas on 4–8 inputs with 3–8 gates, written in Part 3.
- `dlheld`, `dlheld2`, `dlheld3` (12, 7, 10): dense halfspaces and majorities, compositions of majorities, a 5-bit parity, the 6- and 11-multiplexers, a staircase, an intersection of two halfspaces, a DNF and a CNF, a decision list, an integer-weight threshold, symmetric functions such as $[|x_1 + \cdots + x_9| \leq 1]$, parities mixed with gates, and random formulas.
- `dlsymdev` (8) and `dlheld4` (12): symmetric and signed-sum targets with threshold, band, tail and exact-count accept sets, some combined with a parity, a gate or a second sum; `dlheld4` adds 3 non-symmetric controls.
- `dlheld5` (12), drawn from a random family: 8 functions of a signed sum of 5–13 inputs with a random accept set, two of them combined with a gate; 2 thresholds of a signed sum plus twice a parity; and 2 random formulas as controls.
- `dlheld6`, `dlfamdev`, `dlheld7` (12 each): three draws, with seeds 52817, 61616 and 70707, from the blind family.

The blind family is a random generator of signed-sum targets. An agent that had not seen DFG's sum-block design wrote it in Part 9, from the pre-registered one-line description of a held-out suite, before drawing from it (`study/part9/heldout6/FAMILY.md`). Each draw has 12 slots.

- *Nine structured slots*, built from functions $g(s)$ of a signed sum $s = \sum_i \epsilon_i x_i$ of 4–14 inputs: $g(s)$ alone (3 slots); $g(s)$ times a parity (2); $g(s)$ with a gate, that is an AND or OR with one or two literals, a majority with two literals, or a multiplexer between two sums (2); and composite sums (2), either $A(s + a\tau)$ with an inner term $a\tau$, $a \in \{2, 3\}$, τ a literal, a 2-parity or an AND of two literals, or two sums joined by XOR, AND or OR.
- *Three non-symmetric controls*: a read-once formula, a DNF and a decision list.

Each g has an accept set drawn uniformly from five types: a threshold, a centred band, an interval, a residue class mod 3 or 4, and an arbitrary set. A is a threshold or a band. Targets have 5–14 relevant inputs, and the only rejection rule is $P(f = +1) \in [0.25, 0.75]$. DFG v1.3 was developed on

dlheld6 and dlfamdev (Section 3.4), so dlheld7 tests it on fresh targets from a distribution its designer had seen.

The *sum-structured targets* are the 54 targets of dlsymdev, dlheld4–dlheld7 and dlfamdev other than their 14 non-symmetric controls. Every suite puts a target's relevant inputs first. DFG alone and lowdeg therefore see the columns in a fresh random order in every fit, drawn from the fit's random stream (harness.Permuted), as v2 and v3 did, so that tie-breaking by column index cannot help them.

Input dimension. Q4 uses the designed targets on which the networks were trained at other input dimensions in Part 3: maj9, majpar, mux, par2, par3, par4 and stair5 at $d = 15$ and $d = 60$, and maj9, mux, par2 and par3 at $d = 120$ (7, 7 and 4 targets). Each is the same function of the first inputs as at $d = 30$, and its data are drawn in the same way.

Tabular data. DFG is a learner for ± 1 inputs and two classes. Of the study's 26 PMLB datasets [31], two have only binary or categorical columns and two classes, and DFG runs on these two only: kr_vs_kp (3196 rows; 35 binary columns and 1 categorical) and mushroom (8124 rows; 16 categorical columns and 6 binary). Binary columns become ± 1, and categorical ones a one-hot code over the values seen in training, written as ± 1. This is v3's view of such data (study/part11/dfg_real.py). It gives 38 inputs on kr_vs_kp, and 87–112 on mushroom, depending on the values in the training set. Each dataset has a fixed test set, a random 25% of its rows (799 and 2031 rows). For each of 3 seeds the remaining pool (2397 and 6093 rows) is permuted, and the training set of size n is its first n rows. The grid is $n \in \{100, 300, 1000, 3000\}$ up to the pool size, plus the whole pool. DFG runs neither on images nor on real-valued columns.

3.2 References

Networks on the toy targets. These are the companion paper's μP two-hidden-layer ReLU MLPs, d-128-128-1 [12, 36]: Adam [24] at learning rate 10^{-3}, batches of 64 sampled from the fixed training set, 20000 steps (40000 at $d = 120$), read at the final step. There are five variants:

- nets: square loss. It was run on every suite except dlheld7 and dlfamdev, and it is the only variant run at $d \neq 30$.
- nets_wd: square loss with decoupled weight decay 0.1 [28]; every suite except dlfamdev.
- nets_cewd: logistic loss with weight decay 0.1; dlheld3–dlheld6 and dlsymdev.
- nets_cewd0.3, nets_cewd1.0: logistic loss with weight decay 0.3 and 1.0; dlheld6 and dlheld7.

nets_best is, per target, the variant with the smallest n^* among those run on its suite. It is a minimum over 2 variants on designed, random, dlheld and dlheld2, over 3 on dlheld3–dlheld5 and dlsymdev, over 5 on dlheld6, and over the 3 that Part 10 pre-registered on dlheld7: a hindsight choice that favours the networks. dlfamdev has no network runs. The networks are one recipe, not tuned per target or suite. The comparison is therefore with a standard gradient-trained network, not with the best network for each target.

Calibration references. lin is L2-regularized logistic regression on the raw inputs, its penalty chosen by 5-fold cross-validation over 10 values. hgb is scikit-learn's histogram gradient boosting at default settings [33] (both in study/hybrid/refs.py). They search no features, so they show

which targets need none. Both were run in Parts 8–10, with the columns in their natural order. `hgb` has no runs on `designed`, `dlsymdev` and `dlfamdev`, and `lin` none on `dlfamdev`.

The dictionary without the search: `lowdeg`. `lowdeg` (study/discrete_learner/sparse_fourier.py) is Part 6's first feasibility check, run unchanged with the inputs permuted per fit. It is the low-degree algorithm [27] with a significance threshold:

1. compute the empirical Fourier coefficients of the labels, $\hat{c}_S = \frac{1}{n}\sum_t y_t \chi_S(x_t)$, for every S with $|S| \le 4$;
2. keep S if $|\hat{c}_S| > \sqrt{2\ln(\binom{d}{|S|}/\delta)/n}$ with $\delta = 0.1$, a union bound over the characters of that degree;
3. compose: the products $\chi_{S_1}\chi_{S_2} = \chi_{S_1 \triangle S_2}$ of two kept characters, where they have degree above 4, are tested in the same way, with the number of such candidates in place of $\binom{d}{|S|}$, and this is repeated, at most 4 times, while something new is kept;
4. predict $\operatorname{sign}(\hat{c}_\emptyset + \sum_{S \text{ kept}} \hat{c}_S \chi_S)$.

It shares DFG's global characters up to degree 4 and the form of its test, and nothing else. It tests the labels, not a residual, so its noise scale is 1 in place of s, and its union bound has no factor $4K$. It has no greedy rounds, extension or local classes, sum blocks or block moves, and no refitted readout, kernel background or small-n variant. Its FLOPs are counted in the same way as DFG's.

Portfolios that contain DFG. v3 (`hyb_portfolio3`, Part 10 [13]) is the stacked portfolio whose Boolean member is DFG v1.3. DFG's default and forced variants are two of its members, next to LightGBM, ExtraTrees, nearest-neighbour votes and logistic regression on the raw columns. A convex stacker combines them, and DFG has a share of the budget. v2 (`hyb_portfolio2`, Part 9) and v1 (`hyb_portfolio`, Part 8) are the earlier portfolios, with DFG v1.2 and v1.1 [11]. On toy inputs v3's results are those of `p10_sum`, which is v3's code there (checked identical when v3 was assembled). v2's are its own or, on `designed` and `dlsymdev`, those of `p9_sum`, the same code on toy inputs. Both ran on every toy suite, with the inputs permuted per fit. DFG alone against v3 measures what the rest of the portfolio adds on Boolean inputs. v1 is a reference on tabular data only.

Earlier versions of DFG. Part 6 ran DFG v1 at $d = 15$ and 30, and v1.1 at $d = 60$ and 120, on the Q4 targets in the natural column order (study/discrete_learner/results/). They are context for Q4, not a paired comparison. On real-valued data the record is Part 7's three adaptations of DFG, frozen there [11]: Boolean-bit characters (`dfg_bits`), thresholded data-derived directions (`dfg_dirs`) and threshold and AND gates (`dfg_gates`). Their results on kr_vs_kp and mushroom are context for Q6.

References on tabular data. The MLP references are those of Parts 7–10, ReLU MLPs with softmax cross-entropy:

- `mlp`: 256-256, Adam, learning rate and stopping point chosen on validation;
- `mlp_reg`: the same with an L2 penalty chosen on validation;
- `mlp_strong`: quantile-transformed inputs, AdamW with cosine decay, weight decay chosen on validation, dropout 0.1, an ensemble of 3. It was run on mushroom, not on kr_vs_kp.

`mlp_best` is the per-cell minimum of the seed-mean test error over the MLP variants run on that cell: `mlp` and `mlp_reg` on kr_vs_kp; on mushroom also `mlp_strong`, its single network `mlp_strong1`, and `mlp_plus`, a variant with a wider learning-rate grid ($n \leq 3000$). It is a hindsight choice that favours the MLPs. `mlp_best+` adds Part 9's wider ensembles, none of which ran on these two datasets, so it equals `mlp_best` here. `hgb`, v3, v2 and v1 complete the references.

3.3 Metrics and budgets

Toy targets. n^* is the training-set size at which the seed-mean test error reaches 0.1, interpolated in $\log n$ between the grid points $n = 16, 32, \ldots, 16384$. It is "never" if the error does not reach 0.1 by $n = 16384$. DFG alone runs on the networks' seeds and training sets: 3 seeds per target, 2 on `random`.

- A comparison of learners A and B on a target is the ratio $\log_2(n_A^*/n_B^*)$, negative when A needs less data.
- A is *clearly better* when the ratio is below -0.25 or only A reaches 10% error, and *clearly worse* in the mirror cases.
- A suite mean is the mean ratio over the targets both learn. Pooled summaries pool the per-target ratios over suites, rather than averaging suite means.

The verdict has no noise term. n^* is a mean over 2 or 3 training sets per (target, n), with no confidence interval attached, so the ± 0.25 band is a convention, not a significance test. These are the definitions of Parts 8–10 (`study/hybrid/compare_hybrid.py`), and `study/part11/compare_dfg.py` computes every comparison from the result files.

Input dimension. At $d = 15$, 60 and 120, DFG alone and the networks have seeds 0 and 1; at $d = 30$ both have 3. At $d = 60$ and 120 the networks were trained on grids of their own, which reach up to 65536 points, and their n^* is read on those grids. The growth of n^* with d is summarized by the least-squares slope of $\log n^*$ on $\log d$. It is fitted for each learner over the same values of d: those at which both DFG v1.3 and the networks reach 10% error.

Tabular data. A cell is a (dataset, n) pair. Its verdict compares the seed-mean test errors of two learners. They are *better* or *worse* when the errors differ by more than 10% relative and by more than 2 standard errors, and otherwise a *tie*. The s.e. comes from the seed spread of both learners and is at least the test set's binomial noise. Mean $\log_2$ error ratios floor errors at 0.002.

Budgets. Compute is counted in FLOPs, a multiply-add counting 2. Everything a fit does is included, DFG's cross-validation fits too, and prediction is excluded.

- *The unit* is one training run of one toy network: $6P \cdot 64 \cdot T + 10P \cdot T$ FLOPs for $P = 128d + 16768$ parameters and T steps (`harness.gd_flops`). That is 1.62×10^{11} at $d = 30$ with $T = 20000$; 1.47 and 1.93×10^{11} at $d = 15$ and 60; and 5.06×10^{11} at $d = 120$, where the networks ran 40000 steps.
- *The networks.* Each variant spends exactly one unit per fit, and `nets_best` is a hindsight minimum over up to five such runs.
- *DFG alone* spends what its own budget allows (Section 2): fixed shares of one unit for its exhaustive searches, and no new round once 0.75 of the unit is spent.

- *On tabular data* DFG keeps that internal budget, in units of one toy-network run at its encoded width. Its FLOPs are reported against the budget that Parts 7–10 used for real data, three training runs of the baseline `mlp`, which is the baseline's own cost including its learning-rate search.

The references run their standard recipes uncapped.

CPU time is the process time of a fit. It is reported separately, because equal FLOPs are far from equal CPU time. DFG alone's was measured while four worker processes shared a four-CPU machine at low priority, and v3's in Part 10 under that run's own load, so CPU comparisons between them are rough. The networks' result files hold no timing, so the compute comparison with them is in FLOPs only.

3.4 The Part 11 protocol, and what is held out

Pre-registration. The Part 11 protocol, `study/part11/PROTOCOL.md`, was written on 2 November 2026, before any Part 11 result was produced except one timing check disclosed below. It was committed at 22:15 UTC together with the runners (`aca1f3c8`), before the first run, and its pre-registered part has not changed since; one post-hoc addendum, which defines the core run of Section 5.6, was added after every pre-registered run had finished (Appendix C.2). The first results were committed at 22:23. It fixed:

- the object and how it is run;
- seven questions, each with its runs, suites and seeds;
- that grids, seeds, test sets, n^*, the verdict rule and the suite means are those of Parts 8–10, and that the results of the networks, `lin`, `hgb`, v1, v2 and v3 are the existing files, with nothing re-run for them;
- that a run that crashes may get harness-level fixes only (paths, sharding, resumption), never a change to the learner, and that the crash is reported;
- what is held out.

The object. DFG v1.3 is the file `study/part10/hyb_portfolio3/dfg13.py`, frozen by Part 10's toy builder at 19:44 UTC on 27 October 2026, with md5 `69cabe5c50345cb24079198ed52b562f`. It is identical to `study/part10/p10_sum/dfg13.py`.

- *How it is run.* Part 11 runs it alone, as `DFG(readout="logit").fit(X, y, rng)` with every other constructor argument at its default, inside `harness.Permuted`. The run is called `dfg13-perm`. The runners, `study/part11/eval_dfg.py` for the toy suites and `study/part11/dfg_real.py` for tabular data, check the md5 when they load the file. No line of it changes in Part 11, and nothing is tuned.
- *Alone and inside v3.* Alone, DFG chooses between its default and forced variants itself, by its paired cross-validation test at $n \leq 512$; it adds its own kernel background and stays within its own budget. Inside v3 the same code ran as two stacked members, the default and the forced variant, with a share of v3's budget, and the stacker chose between them.
- *Why this object.* It was chosen after Part 10's results were read. It is the one component of v3 that is a new learning algorithm; the others are LightGBM, ExtraTrees, nearest-neighbour votes, logistic regression and a stacker, or LightGBM on a fixed feature bank. Part 8's ablations

found that the portfolios' toy results come from it. And its fitted model is an explicit, short list of features.

Questions and runs. The protocol fixed seven questions.

- *Q1, held out.* On `dlheld7`: DFG alone against `nets_best`, each network variant, `lin`, `hgb`, v3 and v2, target by target and as suite means with clearly better and clearly worse counts.
- *Q2, every suite.* The same on all 11 toy suites (138 targets), each with its status, and DFG alone against v3: does the rest of the portfolio add anything on Boolean targets?
- *Q3, the dictionary without the search.* `lowdeg` on all 11 suites.
- *Q4, input dimension.* DFG alone at $d = 15$, 60 and 120 on the 7, 7 and 4 targets above: n^* against d, against the networks.
- *Q5, what the parts do.* 12 ablations, each DFG alone with one constructor argument changed, on `designed`, `dlsymdev`, `dlfamdev`, `dlheld6` and `dlheld7` (65 targets). Six run on seeds 0 and 1 and six on seed 0. Each is compared with DFG alone on the same seeds, so on the same training sets and column permutations (Section 5).
- *Q6, tabular.* DFG alone on kr_vs_kp and mushroom, on the standard learning curves with 3 seeds, against the MLP references, `hgb` and v3; v2 and v1 are also shown.
- *Q7, what it learns.* Descriptive analyses, on seed 0 unless stated and checked by independent verifiers (Section 6): what DFG's fitted models contain across target families, against each target's own representation in DFG's dictionary; when a target's structure first appears in the model, against when its test would admit it; the order in which DFG admits a target's characters, against the order in which the Part 5 networks acquire them; false admissions on labels independent of the inputs; and where the compute goes.

Section 4 reports Q1–Q4 and Q6, Section 5 Q5, and Section 6 Q7.

What is held out. For DFG v1.3, only `dlheld7` is held out (Table 5). It was generated in Part 10 from the blind family and fixed in a commit (`dc70266c`) before any Part 10 design work. Part 10's builders were not allowed to load, generate or inspect it; the toy builder, which built v1.3, recorded that it saw only the one-line signature of the suite's generator function. No result of a Part 10 learner on it was read before v3 was frozen [13].

- *Everything else is in-sample.* Every other suite was used in developing some version of DFG, or was held out only for an earlier version. Several of DFG's parts were added after failures on such targets: the degree-5/6 search with `par5` known to be in `dlheld`; the logistic readout after `thr30` (`dlheld`); sum blocks after `sym9` (`dlheld3`); weighted blocks after h6_10 (`dlheld6`). Those suites still measure the learner, but they do not test it, and some selection in its favour is expected there [Heuristic].
- *The other sets.* The Q4 targets are `designed` targets, and kr_vs_kp is a development set. Mushroom was not used to develop any version of DFG, but v1–v3 ran DFG on it, and their results were known.
- *The ablations and the analyses* (Q5, Q7) describe the frozen learner. They are not tests of a design.

set	size	origin, and its use in developing DFG	status for DFG v1.3	references run
`designed`	21	Parts 1–5. Development for v1 and v1.1 (Part 6 chose DFG's settings on its test error), and for v1.2 and v1.3 (the Part 9 and Part 10 development evaluations)	development	`nets`, `nets_wd`
`random`	20	Part 3. A development check for v1 and v1.1 in Part 6, run and read with intermediate versions	development	`nets`, `nets_wd` (2 seeds)
`dlheld`	12	Part 6, held out for v1; its degree-5/6 search was added knowing `par5` was in the suite. v1.1 was made after v1's run on it	development	`nets`, `nets_wd`
`dlheld2`	7	Part 6, written after v1.1 was committed: blind for v1.1	held out for v1.1 only	`nets`, `nets_wd`
`dlheld3`	10	Part 8, held out for v1 (DFG v1.1). v1's failure on `sym9` motivated v1.2's sum blocks; a regression check in Part 9	held out for v1.1 only	`nets`, `nets_wd`, `nets_cewd`
`dlsymdev`	8	Part 9. Development for v1.2 and v1.3	development	`nets`, `nets_wd`, `nets_cewd`
`dlheld4`	12	Part 9, meant to be held out for v1.2; its builder saw the suite's source	exposed to v1.2	`nets`, `nets_wd`, `nets_cewd`
`dlheld5`	12	Part 9, drawn after v1.2 was frozen, from a family specified after reading v1.2's design	exposed to v1.2	`nets`, `nets_wd`, `nets_cewd`
`dlheld6`	12	Part 9, the blind family's first draw; held out for v1.2. Development for v1.3, whose weighted blocks were designed after v2's failure on h6_10	development	all five variants
`dlfamdev`	12	Part 10, a development draw from the blind family. Development for v1.3	development	none
`dlheld7`	12	Part 10, a draw from the blind family fixed in a commit (`dc70266c`) before v1.3 was designed; never used in its development	**held out**	`nets_wd`, `nets_cewd0.3`, `nets_cewd1.0`
`designed` at $d = 15, 60, 120$	7, 7, 4	Networks trained in Part 3. Part 6 ran v1 at $d = 15$ and v1.1 at $d = 60$ and 120	development	`nets` (2 seeds)
kr_vs_kp	3196 rows	Part 7, a development set of the standard tasks; the Part 10 toy builder's tabular check	development	`mlp`, `mlp_reg`
mushroom	8124 rows	Part 7, held out there. Never used to develop DFG; v1–v3 ran DFG on it, and their results were read before Part 11	seen through v1–v3	`mlp`, `mlp_reg`, `mlp_strong`

Table 5: Every set and its status for DFG v1.3. Sources: `study/part11/PROTOCOL.md` ("Held out") and the protocols and records of Parts 6, 8, 9 and 10. "Development": used in developing some version of DFG. "Exposed": the Part 9 sum-block design and the suite were not independent. "Held out for v1.1 only": held out for that version (inside v1 for `dlheld3`), and seen since. "References run": the network variants run on the suite, over which `nets_best` is the per-target minimum, or the MLP variants on the dataset. Networks have 3 seeds except where stated. `lin` and `hgb` are listed in Section 3.2; v3, v2 and `lowdeg` ran on every toy suite.

Disclosures.

- *A timing check on held-out targets.* Before the protocol was written, DFG alone was fitted on two `dlheld7` targets, f7_03 and f7_08 (seed 0; $n = 64$, 512, 4096 and 16384), to time it, and the 8 test errors were printed. The protocol records them. The queue's runs reproduce all 8 to the three decimals recorded (`study/part11/RESULTS.md`, "Consistency checks"). The code had been frozen in Part 10 and nothing is tuned in Part 11, so no setting could respond to them.

- *Smoke tests* of the ablations and of `lowdeg` used three `designed` targets: `maj9`, `stair5` and `par4`.
- *A replaced ablation setting.* The first choice for the lenient ablation, $\delta = 10$, crashed: the linear block's test takes the square root of a multiple of $\ln(1/\delta)$, which is negative for $\delta > 1$. It was replaced by $\delta = 0.9$ before any run, and the protocol records both.
- *The object was chosen post hoc.* It was chosen knowing v3's results on every suite, including `dlheld7`, where DFG is v3's toy member. DFG alone's results there were new, the timing check aside, but they are not independent of v3's, which were known. The choice is post hoc; the code is not.

4 How well it does

This section compares DFG v1.3, run alone, with the networks and with every other reference. It starts with the one held-out suite (Section 4.1) and all 11 toy suites (Section 4.2), then compares DFG alone with the portfolio v3 that contained it (Section 4.3) and with the low-degree algorithm, which has its dictionary but not its search (Section 4.4). It ends with the scaling in the input dimension (Section 4.5), the two tabular datasets DFG can run on (Section 4.6) and compute (Section 4.7).

Numbers come from `study/part11/RESULTS.json`, which `study/part11/compare_dfg.py` computes from the result files, unless another file is named. The tables and figures are generated from it by `paper_dfg/make_tables.py` and `paper_dfg/make_figures.py`. Derived numbers that the text quotes and no table shows are in `paper_dfg/tables/numbers.json` and in the figures' `.json` files. The rules are those of Section 3.3. n^* is the training-set size at 10% test error, and a comparison is the ratio $\log_2(n^*_{\mathrm{DFG}}/n^*_{\mathrm{ref}})$, negative when DFG needs less data. A target is *clearly* better or worse when the ratio is beyond ± 0.25, or when only one of the two learners reaches 10% error; means are taken over the targets both learn. DFG alone, v3, v2 and `lowdeg` see the input columns in a fresh random order in every fit; `lin` and `hgb` were run in the natural order.

4.1 The held-out suite `dlheld7`

`dlheld7` is the only toy suite held out for DFG v1.3. Its 12 targets were drawn in Part 10 from the blind family of signed-sum targets and fixed in commit `dc70266c`, before DFG v1.3 was designed, and DFG's development never used them (Section 3.4, Table 5). What is *not* held out is the choice of the object. DFG v1.3 was chosen as this paper's subject after Part 10's results had been read, including v3's on `dlheld7`, where DFG v1.3 is v3's toy member. The choice is post hoc; the code is not. Table 6 gives every target's definition and n^*, and Table 7 the summaries against each reference.

Against the networks [Numerical].

- *Against `nets_best`:* mean $\log_2 n^*$ ratio -1.29 over the 11 targets both learn, about $2.4\times$ less data (median -1.16). DFG alone is clearly better on 10 of the 12 targets and clearly worse on none. The 11 ratios have s.d. 0.91; if the targets are treated as independent draws from the family, the mean has a standard error of about 0.27.
- *Against each variant:* -1.62 against `nets_wd` (clearly better on 11 of 12), -1.38 against `nets_cewd0.3` and -1.36 against `nets_cewd1.0` (10 each). No variant is clearly better than DFG alone on any target.

	discrete, inputs permuted			calibration		networks				$\log_2$ ratio of DFG alone to	
target	DFG alone	v3	lowdeg	lin	hgb	nets_wd	nets_cewd0.3	nets_cewd1.0	nets_best	nets_best	v3
f7_00	211	**203**	7061	never	6208	2418	1545	1565	1545	−2.87	+0.06
f7_01	119	**106**	595	111	684	205	121	119	119	−0.00	+0.17
f7_02	190	**106**	435	never	120	337	341	348	337	−0.83	+0.85
f7_03	**352**	352	4205	never	2199	1710	1665	1599	1599	−2.18	−0.00
f7_04	97	**70**	121	never	84	164	175	181	164	−0.76	+0.47
f7_05	never	never	never	never	never	never	never	never	never	n/a	n/a
f7_06	82	**44**	179	127	59	108	124	124	108	−0.39	+0.90
f7_07	**167**	174	1123	never	1373	445	443	415	415	−1.32	−0.06
f7_08	644	**634**	never	never	never	5843	3239	2932	2932	−2.19	+0.02
f7_09	222	**215**	591	never	never	1000	883	928	883	−1.99	+0.05
f7_10	84	**82**	336	87	114	210	120	119	119	−0.50	+0.03
f7_11	51	55	**29**	119	63	114	144	141	114	−1.16	−0.12

target	definition
f7_00	OR([|s1| ≤ 0], [s2 == −1]), s1 = x1+x2−x3−x4, s2 = −x5−x6−x7+x8+x9+x10+x11
f7_01	[s ≤ 1], s = x1+...+x13
f7_02	OR(AND(x2,−x6,−x8), AND(x1,−x2,−x4,−x5), AND(x3,−x5,−x7))
f7_03	−x11 ? [((s1+6)/2) mod 4 != 2] : [|s2| > 2], s1 = x1+...+x6, s2 = x7+...+x10
f7_04	if AND(−x6,−x9) → +1; elif −x8 → −1; elif −x4 → +1; elif −x2 → −1; elif x7 → −1; elif x5 → −1; elif AND(x3,x1) → −1; else +1
f7_05	[((s+14)/2) mod 3 != 0], s = x1+...+x14
f7_06	OR(AND(AND(−x1,x2),AND(AND(−x3,x4),−x5)),AND(OR(−x6,x7),x8))
f7_07	[−2 ≤ s ≤ 8] * x13, s = −x1+x2−x3+x4+x5+x6+x7+x8−x9+x10−x11+x12
f7_08	[s in {−11,−9,−3,1,9,11}], s = x1+...+x11
f7_09	[s ≤ 0] * x11 * x12, s = x1−x2+x3−x4+x5−x6+x7+x8+x9+x10
f7_10	[u ≤ −1], u = s + 2*(x6), s = x1+...+x5
f7_11	−x11 ? [s1 in {−4,4}] : [|s2| ≤ 4], s1 = x1+x2−x3+x4, s2 = −x5−x6+x7−x8−x9+x10

Table 6: The held-out toy suite `dlheld7` [Numerical]. Top: n^* per target (mean of 3 seeds, interpolated in $\log n$; "never": 10% test error not reached by $n = 16384$) for DFG alone, v3, `lowdeg`, the calibration references `lin` and `hgb`, the three network variants run on the suite and their per-target minimum `nets_best`. The last two columns give the $\log_2$ ratio of DFG alone's n^* to that of `nets_best` and of v3 (n/a: neither learns the target). Bold: the smallest n^* in the row. Bottom: the definitions. $[\cdot]$ is +1 when the condition holds and −1 otherwise; `*` multiplies ±1 values; `c ? a : b` is a when $c = +1$; a literal `x5` or `-x5` inside AND/OR or a decision list stands for $x_5 = +1$ or $x_5 = -1$. f7_02, f7_04 and f7_06 are the family's non-symmetric controls. Source: `study/part11/RESULTS.json` (`q1_dlheld7`); definitions: `study/boolfuncs.py`.

- *Largest margins:* f7_00, an OR of conditions on two signed sums: 211 against 1545 (−2.87); f7_08, an arbitrary accept set of an 11-input count: 644 against 2932; f7_03, a multiplexer between a residue class mod 4 of a 6-input count and a condition on a 4-input count: 352 against 1599; f7_09, a threshold of a signed 10-input sum times $x_{11}x_{12}$: 222 against 883.

Against v3, `lowdeg` and the calibration references [Numerical].

- *v3* needs less data than DFG alone: the mean ratio is +0.21 (1.16× more data for DFG alone) and the median +0.05. DFG alone is clearly worse on 3 targets, the three non-symmetric controls: f7_02, a DNF (190 against 106), f7_04, a decision list (97 against 70), and f7_06, a read-once formula (82 against 44). It is clearly better on none. Against v2 the summary is

	DFG alone against the reference						the reference against `nets_best`	
reference	both reach	mean $\log_2$ ratio	median	clearly better	clearly worse	only the reference / only DFG reaches	mean (better/worse) [both reach]	median
v3	11	+0.21	+0.05	0	3	0 / 0	−1.50 (10/0) [11]	−1.29
v2	11	+0.21	+0.04	0	3	0 / 0	−1.50 (10/0) [11]	−1.33
`lowdeg`	10	−1.90	−1.71	10	1	0 / 1	+0.70 (3/8) [10]	+1.07
`lin`	4	−0.45	−0.35	9	0	0 / 7	−0.06 (1/7) [4]	−0.02
`hgb`	9	−1.39	−0.44	8	2	0 / 2	+0.28 (4/6) [9]	−0.06
`nets_wd`	11	−1.62	−1.33	11	0	0 / 0	–	–
`nets_cewd0.3`	11	−1.38	−1.41	10	0	0 / 0	–	–
`nets_cewd1.0`	11	−1.36	−1.32	10	0	0 / 0	–	–
`nets_best`	11	−1.29	−1.16	10	0	0 / 0	–	–

Table 7: DFG alone against each reference on `dlheld7` (12 targets) [Numerical]. Left: the number of targets that both reach 10% test error, the mean and median $\log_2 n^*$ ratio of DFG alone to the reference over them, the targets on which DFG alone is clearly better and clearly worse, and the targets that only the reference or only DFG alone reaches. Right, for calibration: the reference itself against `nets_best`, as mean (clearly better / clearly worse) [both reach] and median. Source: study/part11/RESULTS.json (`q1_dlheld7.summary`, `.context`).

the same (+0.21; 0 better, 3 worse).

- *`lowdeg`:* −1.90 over the 10 targets both learn, clearly better on 10 and clearly worse on 1 (f7_11: 51 against 29). `lowdeg` never learns f7_08.
- *Calibration:* `lin` reaches 10% error on 4 targets and `hgb` on 9. DFG alone's mean is −0.45 against `lin` over those 4 and −1.39 against `hgb` over those 9. It is clearly better than `lin` on 9 targets and than `hgb` on 8, and clearly worse than `hgb` on 2, f7_02 and f7_06.

Five of the ten wins need no feature search [Numerical]. On five targets a learner that searches no features is already clearly better than `nets_best`: `hgb` on the three controls and on the multiplexer f7_11, and `lin` on the halfspace f7_10, by $\log_2$ ratios of −1.49 to −0.44. DFG alone beats `nets_best` on these too (−1.16 to −0.39, mean −0.73). Against the better of the two calibration references, however, it is clearly worse on f7_02 (+0.65) and f7_06 (+0.47), tied on f7_04 (+0.20) and f7_10 (−0.06), and clearly better only on f7_11 (−0.31). v3, which stacks DFG with LightGBM and logistic regression, is ahead of both calibration references on all five (−0.43 to −0.09).

On the other seven targets neither calibration reference beats the networks. `hgb` needs 684–6208 samples or never reaches 10% error, and `lin` learns only f7_01. There DFG alone is clearly better than `nets_best` on five targets, all sum-structured (f7_00, f7_03, f7_07, f7_08, f7_09), with a mean of −2.11, about 4.3× less data. It ties on f7_01 and does not learn f7_05.

Reading [Heuristic]. The five calibration wins say that the networks are slow on small formulas and on this halfspace, not that DFG's search is needed there. On the controls a tree ensemble is the better learner, and v3 inherits that from its tree members. The evidence that the search earns its cost on fresh targets is the other five, the sum-structured ones, where DFG alone needs about 4× less data than the best network variant and no learner without feature search comes close.

The tie and the target no learner learns [Numerical].

- f7_01 = $[x_1 + \cdots + x_{13} \leq 1]$ is a halfspace. DFG alone (119), `nets_best` (119, `nets_cewd1.0`), `lin` (111) and v3 (106) are all within the tie band of one another: a logistic regression on the

suite	status for v1.3	targets	DFG never	nets_best	single variants: min / max mean	v3	v2	lin	hgb	lowdeg
designed	development	21	0	−2.17 (20/0) [21]	−2.82 / −2.17 (2 variants)	+0.25 (1/9) [21]	+0.22 (3/9) [21]	−0.19 (17/1) [7]	–	−0.92 (16/3) [21]
random	development	20	0	−2.04 (20/0) [20]	−2.70 / −2.04 (2 variants)	+0.22 (2/7) [20]	+0.20 (2/5) [20]	−0.62 (20/0) [5]	−1.54 (18/0) [19]	−0.71 (15/4) [20]
dlheld	development	12	0	−2.22 (11/0) [12]	−2.71 / −2.22 (2 variants)	−0.18 (1/4) [12]	−0.16 (1/3) [12]	−0.49 (10/0) [4]	−1.97 (9/1) [11]	−1.53 (9/2) [11]
dlheld2	held out for v1.1 only	7	0	−1.59 (7/0) [7]	−2.22 / −1.59 (2 variants)	+0.40 (0/2) [7]	+0.26 (1/2) [7]	−0.05 (5/0) [3]	−2.39 (5/2) [7]	−2.00 (6/0) [7]
dlheld3	held out for v1.1 only	10	0	−1.34 (7/1) [10]	−2.05 / −1.48 (3 variants)	+0.46 (0/5) [10]	+0.42 (0/5) [10]	−0.11 (7/2) [4]	−1.31 (6/2) [9]	−0.74 (6/3) [10]
dlsymdev	development	8	0	−2.14 (8/0) [8]	−2.99 / −2.15 (3 variants)	+0.08 (0/1) [8]	−0.10 (1/0) [8]	n/a (8/0) [0]	–	−3.98 (8/0) [7]
dlheld4	exposed to v1.2	12	0	−1.67 (12/0) [12]	−2.58 / −1.73 (3 variants)	+0.20 (1/3) [12]	+0.01 (3/2) [12]	−0.72 (12/0) [1]	−2.61 (10/2) [7]	−2.89 (12/0) [11]
dlheld5	exposed to v1.2	12	1	−2.49 (11/0) [11]	−3.20 / −2.51 (3 variants)	−0.06 (1/1) [11]	−0.09 (2/1) [11]	n/a (11/0) [0]	−4.19 (11/0) [6]	−2.05 (11/0) [6]
dlheld6	development	12	1	−1.27 (11/1) [11]	−2.15 / −1.35 (5 variants)	+0.22 (0/3) [11]	−0.02 (2/3) [11]	−0.22 (10/0) [2]	−1.86 (9/2) [7]	−2.29 (10/1) [9]
dlfamdev	development	12	1	–	–	+0.26 (0/4) [11]	+0.20 (1/5) [11]	–	–	−1.67 (9/2) [10]
dlheld7	**held out**	12	1	−1.29 (10/0) [11]	−1.62 / −1.36 (3 variants)	+0.21 (0/3) [11]	+0.21 (0/3) [11]	−0.45 (9/0) [4]	−1.39 (8/2) [9]	−1.90 (10/1) [10]

Table 8: DFG alone on every toy suite [Numerical]. Each entry is the mean $\log_2 n^*$ ratio of DFG alone to the reference over the targets both learn; below it, the numbers of targets on which DFG alone is clearly better / clearly worse, and in brackets the number both learn. "n/a": the reference learns no target of the suite; "–": not run there. `nets_best` is the per-target minimum over the network variants run on the suite (2 to 5; Section 3.2); "single variants" gives the smallest and largest of DFG alone's means against those variants one at a time. "DFG never": targets that DFG alone does not bring to 10% error by $n = 16384$. Status for DFG v1.3: as in Table 5. Source: `study/part11/RESULTS.json` (`q2_suites`).

raw inputs does as well.

- f7_05 $= [\frac{1}{2}(s + 14) \bmod 3 \neq 0]$, $s = x_1 + \cdots + x_{14}$, a residue class mod 3 of a 14-input count, is learned by none of the learners of Table 6, nor by v2, by $n = 16384$ (3 seeds each). Part 10 traced the same failure in v3's DFG member to the search, not the test: a residue class mod 3 has almost no Fourier weight at degrees 1–2, where the block search starts [13]. Section 6 shows what DFG alone's models on it contain.

4.2 Every suite

Table 8 summarizes all 11 suites and Table 9 pools them; Figure 2a shows every target with network runs, and Appendix B.1 lists every target's n^* for every learner. The status column gives each suite's status for DFG v1.3 (Table 5). Six suites were used in developing some version of DFG (`designed`, `random`, `dlheld`, `dlsymdev`, `dlheld6`, `dlfamdev`). `dlheld2` and `dlheld3` were held out for v1.1 only and have been seen since. `dlheld4` and `dlheld5` were exposed to v1.2: the suites and its sum-block design were not independent. `dlheld7` is held out. For DFG v1.3 every suite except `dlheld7` is therefore in-sample in some sense, and the protocol treats the results there as descriptions of the frozen learner, not as tests of its design.

Pooled [Numerical].

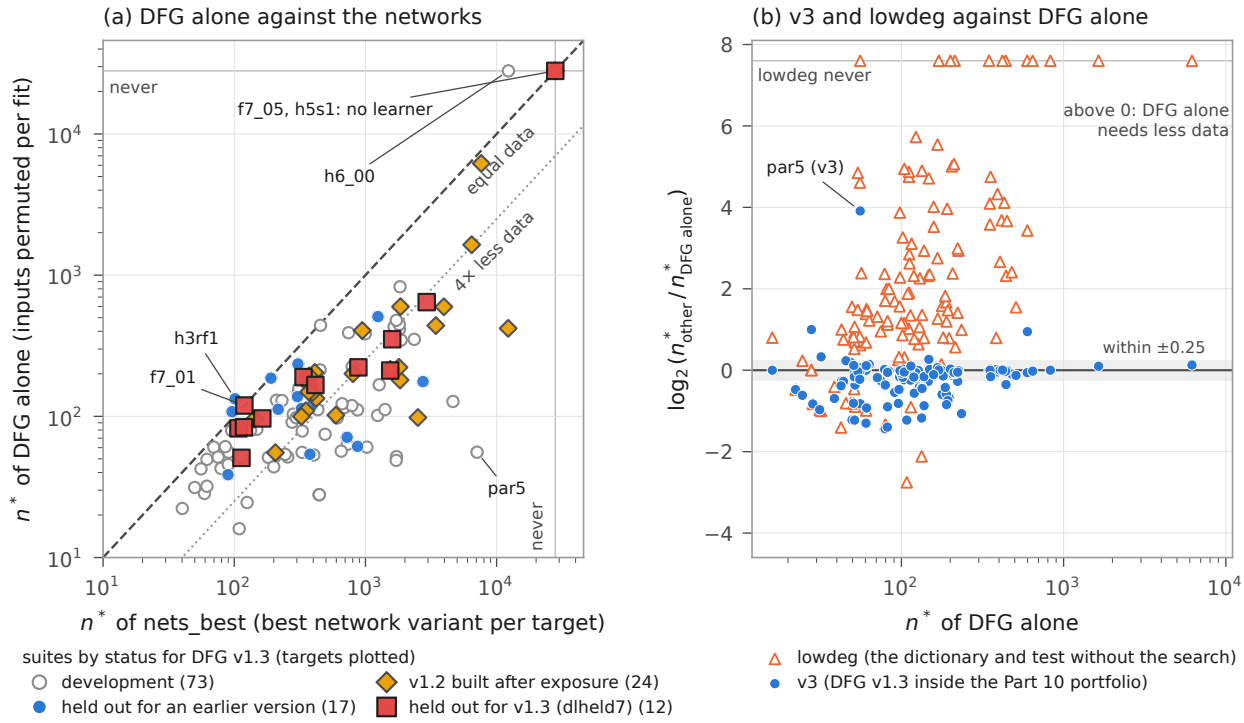


Figure 2: (a) n^* of DFG alone against `nets_best` for the 126 targets of the 10 toy suites with network runs [Numerical]. Colour and marker give the suite's status for DFG v1.3 (Table 8); the held-out `dlheld7` is drawn as large outlined squares. "never" (10% error not reached by $n = 16384$) is drawn at the edge. Points below the dashed line need less data than the best network variant, points below the dotted line at least 4× less. `nets_best` is a minimum over 2 to 5 variants, depending on the suite. (b) For the 134 targets of all 11 suites that DFG alone learns (`dlfamdev` included), the $\log_2$ ratio of v3's and of `lowdeg`'s n^* to DFG alone's, against DFG alone's n^*. Above 0, DFG alone needs less data; the grey band is the tie band ±0.25; `lowdeg`'s "never" is drawn at the top. Source: `study/part11/RESULTS.json` via `paper_dfg/make_figures.py`; the plotted points are in `paper_dfg/figures/fig_nstar.json`.

- *Against* ***`nets_best`****:* on the 126 targets of the 10 suites with network runs, DFG alone's mean is −1.87 over the 123 targets both learn, about 3.7× less data (median −1.65). It is clearly better on 117, clearly worse on 2, and needs at least 4× less data on 47 (Figure 2a).
- *Against single variants:* −2.04 against `nets_wd`, the one variant run on every suite with network runs (clearly better on 119 of 126, worse on 1), and −2.64 against `nets` (square loss; 9 suites, 114 targets).
- *Part 10's held-out set* (`dlheld`–`dlheld7`, 77 targets): DFG alone is at −1.71 and v3 at −1.87 against `nets_best` on the same targets.

By suite [Numerical]. The suite means against `nets_best` range from −2.49 (`dlheld5`) to −1.27 (`dlheld6`); `dlheld7` is at −1.29. They are not comparable across suites, because `nets_best` minimizes over 2 to 5 variants depending on the suite. Against `nets_wd` alone the means range from −2.72 (`dlheld5`) to −1.59 (`dlheld2`). The three smallest margins against `nets_wd` are those of the held-out suite and of the two suites held out for v1.1: `dlheld2` (−1.59), `dlheld3` (−1.60) and `dlheld7` (−1.62). The five development suites with network runs give −1.71 (`dlheld6`) to −2.45

reference	suites	targets	both reach	mean $\log_2$ ratio	median	clearly better	clearly worse	only the reference	only DFG	neither
DFG alone against										
`nets_best`	10	126	123	−1.87	−1.65	117	2	1	0	2
`nets`	9	114	112	−2.64	−2.65	110	0	0	0	2
`nets_wd`	10	126	123	−2.04	−1.90	119	1	0	0	3
`nets_cewd`	5	54	52	−1.88	−1.66	49	0	0	0	2
`nets_cewd0.3`	2	24	22	−1.37	−1.30	21	0	0	0	2
`nets_cewd1.0`	2	24	22	−1.37	−1.20	21	1	1	0	1
v3	11	138	134	+0.19	+0.07	6	42	0	0	4
v2	11	138	134	+0.11	+0.07	16	38	0	0	4
`lin`	10	126	30	−0.33	−0.25	109	3	0	93	3
`hgb`	8	97	75	−1.98	−1.95	76	11	0	19	3
`lowdeg`	11	138	122	−1.64	−1.35	112	16	0	12	4
against `nets_best` *on Part 10's held-out suites (*`dlheld`–`dlheld7`*)*										
DFG alone	7	77	74	−1.71	−1.40	69	2	1	0	2
v3	7	77	74	−1.87	−1.70	72	1	1	0	2

Table 9: DFG alone against each reference, with the per-target ratios pooled over the suites where both were run (not a mean of suite means) [Numerical]. Columns as in Table 7. "Only the reference": DFG alone never reaches 10% error and the reference does; "only DFG": the reverse. Bottom: DFG alone and v3 against `nets_best` on the same 77 targets of the seven suites that Part 10 pooled as its held-out set; for DFG v1.3, only `dlheld7` among them is held out. Source: `study/part11/RESULTS.json` (`q2_suites._pooled`, `._pooled_p10_heldout`).

(`dlsymdev`), and the two exposed suites −2.05 and −2.72.

Reading [Heuristic]. Some selection in DFG's favour on the suites it was developed on or exposed to is therefore likely. The suites also differ in kind, though. `designed` and `random` are small juntas and formulas, `dlsymdev` is symmetric, and `dlheld2`, `dlheld3` and the blind family mix sums, formulas and halfspaces. With 7 to 12 targets per suite, the data do not separate the two explanations. On the held-out suite the margin is about 3× (against `nets_wd`), not the 4× of the pooled numbers.

Where DFG alone is not better [Numerical]. Table 10 lists the 9 of the 126 targets with network runs on which DFG alone is not clearly better than `nets_best`.

- *Clearly worse (2).* h3rf1, a random 9-gate formula on 7 inputs: 133 against 101 (`nets_wd`). And h6_00, a residue class mod 3 of a signed 13-input count, which only the most regularized network learns (`nets_cewd1.0`, 12304).
- *Ties (5).* Three threshold-like targets: thrint, a weighted threshold (186 against 190), inter2, an AND of two halfspaces (441 against 454), and f7_01 (119 against 119). The random formula h3rf0 (108 against 96) and and3 = AND(x_1, x_2, x_3) (60 against 70).
- *Learned by no learner (2).* f7_05 and h5s1, a signed 9-input count with an arbitrary accept set, multiplied by the parity $x_{10}x_{11}$.
- *Never learned.* DFG alone does not learn 4 of the 138 targets: these two, h6_00 and fd_10 (`dlfamdev`, a third residue class mod 3, without network runs). v3 and `lowdeg` do not learn them either.

Reading [Heuristic]. On the two random formulas and and3, the networks' margin reflects DFG's slowness at small n more than a strength of the networks. v3 (59, 46 and 25) needs less data than both DFG alone and the networks, and so does `hgb` on the two formulas (63, 60; it was not run on

suite	target	definition	DFG alone	nets_best (variant)	$\log_2$ ratio	v3	lowdeg	lin	hgb
DFG alone clearly worse									
dlheld6	h6_00	[((s+13)/2) mod 3 != 1], s = x1+ x2+x3+x4+x5+x6+x7+x8−x9 −x10−x11−x12+x13	never	12304 (cewd1.0)	$+\infty$	never	never	never	never
dlheld3	h3rf1	random 9-gate formula on 7 inputs	133	101 (wd)	+0.39	59	31	101	63
ties ($\lvert\log_2$ ratio$\rvert \leq 0.25$)									
dlheld3	h3rf0	random 8-gate formula on 7 inputs	108	96 (wd)	+0.17	46	16	112	60
dlheld7	f7_01	[s ≤ 1], s = x1+...+x13	119	119 (cewd1.0)	−0.00	106	595	111	684
dlheld3	thrint	sign(3x1 + 2x2 + 2x3 + x4 + ... + x9)	186	190 (cewd)	−0.03	139	659	137	243
dlheld	inter2	AND of two Gaussian halfspaces on x1..x10	441	454 (wd)	−0.04	346	2204	never	416
designed	and3	AND(x1,x2,x3)	60	70 (wd)	−0.21	25	158	68	–
no learner reaches 10% error									
dlheld5	h5s1	[$\sum_{i\leq 9} s_i x_i$ in [−9, −5, −1, 3, 9]], s = [−1, 1, 1, 1, −1, 1, −1, −1, −1] xor chi(10, 11)	never	never	n/a	never	never	never	never
dlheld7	f7_05	[((s+14)/2) mod 3 != 0], s = x1+...+x14	never	never	n/a	never	never	never	never
DFG alone never learns it; no network runs									
dlfamdev	fd_10	[((s+10)/2) mod 3 == 0], s = x1+ x2+x3+x4−x5−x6+x7+x8+x9+ x10	never	–	–	never	never	–	–

Table 10: Every target on which DFG alone is not clearly better than `nets_best` (9 of the 126 with network runs), and the one other target it never learns (`dlfamdev` has no network runs) [Numerical]. n^* as in Table 6. In brackets, the variant attaining `nets_best`: wd, cewd and cewd1.0 stand for `nets_wd`, `nets_cewd` and `nets_cewd1.0`. Random formulas are named by their size; their gates are in `study/boolfuncs.py`. Source: `study/part11/RESULTS.json` (`q2_suites`, `q3_lowdeg`) via `paper_dfg/make_tables.py`.

`designed`). On h3rf1 and h3rf0 even `lowdeg` does (31, 16). On thrint and f7_01, logistic regression on the inputs is as fast or faster (`lin`: 137, 111). The residue classes are the search failure of Section 4.1.

4.3 What the portfolio added

Inside v3, DFG v1.3 ran as two members of a stack, its default and (at $n \leq 512$) its forced variant, next to LightGBM, ExtraTrees, k-nearest-neighbour votes and logistic regression. It had a share of the budget: 35% for the out-of-fold fits and at most 50% for the final ones. A convex stacker, fitted on the members' out-of-fold predictions (10-fold at $n \leq 500$), weighted the members (Section 3.2; `study/part10/hyb_portfolio3/learner.py`). Alone, DFG chooses between its two variants itself, by the paired test on its own cross-validation, and it has the whole budget (Section 2). Both see the same training sets in the same column order, so their results can be compared target by target. The comparison shows what the rest of the portfolio contributed on Boolean targets.

Where [Numerical].

- *Overall* (Table 11a). Over the 138 targets, DFG alone's n^* equals v3's on 5, is lower on 32 and higher on 101. Beyond ±0.25 it is lower on 6 and higher on 42. The mean ratio is +0.19, so

(a) per suite

suite	targets	DFG alone lower / higher (identical)	clearly lower / higher ($\lvert \log_2 \text{ratio} \rvert > 0.25$)	mean $\log_2$ ratio (both reach)
`designed`	21	4 / 16 (1)	1 / 9	+0.25
`random`	20	5 / 15 (0)	2 / 7	+0.22
`dlheld`	12	3 / 9 (0)	1 / 4	−0.18
`dlheld2`	7	1 / 6 (0)	0 / 2	+0.40
`dlheld3`	10	1 / 9 (0)	0 / 5	+0.46
`dlsymdev`	8	3 / 5 (0)	0 / 1	+0.08
`dlheld4`	12	2 / 10 (0)	1 / 3	+0.20
`dlheld5`	12	5 / 6 (1)	1 / 1	−0.06
`dlheld6`	12	2 / 9 (1)	0 / 3	+0.22
`dlfamdev`	12	3 / 8 (1)	0 / 4	+0.26
`dlheld7`	12	3 / 8 (1)	0 / 3	+0.21
all	138	32 / 101 (5)	6 / 42	+0.19

(b) by v3's n^*

v3's n^*	targets	DFG alone lower / higher	clearly lower / higher	mean (median)
≤ 64	45	9 / 35	3 / 24	+0.39 (+0.26)
64–256	67	12 / 55	1 / 17	+0.17 (+0.06)
256–1024	19	8 / 11	1 / 1	−0.16 (+0.00)
> 1024	3	3 / 0	1 / 0	−0.39 (−0.13)
never	4	0 / 0	0 / 0	–

Table 11: DFG alone against v3 on the 138 toy targets [Numerical]. (a) Per suite: the targets on which DFG alone's n^* is lower / higher than v3's (identical in brackets), those on which it is clearly lower / higher, and the mean $\log_2$ ratio over the targets both learn. (b) The same by v3's n^*; the 4 targets that neither learns are under "never". Source: `study/part11/RESULTS.json` (`q2_suites.dfg_vs_v3`, `.dfg_vs_v3_by_v3_nstar`).

DFG alone needs 1.14× v3's data, and the median is +0.07. A two-sided sign test over the 133 targets with unequal n^* gives $p = 1.6 \times 10^{-9}$. No target is learned by one and not the other.

- *By target size* (Table 11b). On the 45 targets that v3 learns by $n = 64$, DFG alone's mean is +0.39, clearly higher on 24. On the 67 that v3 learns at 64–256 it is +0.17 (17 clearly higher). On the 22 that need v3 more than 256 points, DFG alone is lower on average (−0.16 and −0.39).
- *By n* (Table 12a). v3's test error is lower by more than 0.02 on 63, 64 and 67 of the 138 targets at $n = 16$, 32 and 64, on 38 at $n = 128$, 17 at 256 and 5 at 512, and on none at $n \geq 1024$. DFG alone's error is lower by that much on 2–26 targets at every n up to 4096.
- *Which targets* (Table 13). v3 learns 39 of the 42 targets on which DFG alone is clearly worse by $n = 128$. Most are small Boolean formulas: random gate formulas, DNFs and a CNF, decision lists, read-once formulas, mux6, and3 and andxor. The others are six halfspace-like targets (dict, maj3, maj15, thr10, thrint, and inter2, an AND of two halfspaces) and four sum targets (far9, tail13, fd_08, h5s4). The largest gaps are h6_04 (79 against 29), h2rf1 (82 against 31), and3 (60 against 25) and dnf4x2 (235 against 112). `hgb` alone needs at most DFG alone's n^* on 15 of the 28 of these targets on which it was run.

Why [Numerical, then Heuristic]. Part 10's seed-0 analysis refits of v3 record v3's error, the error of its own DFG member alone, and the stacking weights (Table 12b). At $n = 16$, 32 and 64, v3's error is lower than its own DFG member's by more than 0.02 in 57, 68 and 53 of 136–138 fits, and higher in 25, 19 and 11. The stacker gives the DFG members no weight in 52, 47 and 30 fits, and their

n	16	32	64	128	256	512	1024	2048	4096	8192	16384
(a) DFG alone against v3: the result files, every seed, 138 targets											
v3 lower by > 0.02	63	64	67	38	17	5	0	0	0	0	0
DFG alone lower by > 0.02	26	22	8	4	3	2	2	2	3	0	0
mean difference (points)	+2.3	+2.2	+2.8	+1.7	+0.4	−0.2	−0.1	−0.1	−0.2	−0.0	+0.0
(b) inside v3: Part 10's analysis fits of v3, seed 0											
fits analysed	138	138	136	134	122	87	54	33	16	10	7
v3 lower than DFG member	57	68	53	32	16	8	1	0	0	0	0
DFG member lower than v3	25	19	11	5	0	0	0	0	0	0	0
DFG members' weight 0	52	47	30	17	8	2	4	3	4	3	2
median DFG weight share	0.24	0.22	0.30	0.56	0.73	0.79	0.81	0.64	0.51	0.19	0.15

Table 12: DFG alone and v3 by training-set size [Numerical]. (a) Over the 138 toy targets, with the same seeds, training sets and column orders for both: the targets on which v3's seed-mean test error is lower than DFG alone's by more than 0.02, the reverse, and the mean difference of the errors (DFG alone minus v3, in percentage points). Source: `study/part11/results/toy__dfg13-perm__<suite>.json` and `study/hybrid/results/toy__p10_sum-perm__<suite>.json`. (b) Inside v3: Part 10's analysis refits of v3, seed 0, one per target and n, whose test error equals v3's evaluation cell. Each target's grid stops after two consecutive n with error ≤ 0.005, so there are fewer fits at large n. Rows: the fits in which v3's error is lower than that of its own DFG member alone by more than 0.02; the reverse; the fits in which the stacker gave the DFG members (default and forced) no weight; and the DFG members' median share of the stacking weight. Where DFG has no weight, the member's error is that of its last out-of-fold fit. Source: `study/part10/interp/results/toy_final/v3/` (fields `err`, `err_dfg`, `w_dfg`; defined in `study/part10/interp/toy_structure.py`).

median share of the weight is 0.22–0.30. From $n = 256$ on, v3 is never worse than its DFG member by more than 0.02, and the DFG members' median share is 0.73–0.81 at $n = 256$–1024.

Reading [Heuristic]. At small n the portfolio's other members carry v3. On formulas and small thresholds, trees, nearest neighbours and logistic regression reach low error with fewer points than DFG needs before its tests admit features. The stacker can also lean on DFG's forced variant, which DFG alone keeps only when its paired test prefers it. Once DFG's features pass their tests, its members take most of the weight and the two learners agree. Section 6 compares DFG alone's fitted models with v3's DFG member.

The reverse: what only the full-data fit finds [Numerical, then Heuristic]. On three of the six targets where DFG alone needs clearly less data, the files show what happens: DFG's fit on all n points finds a structure that fits on fewer points miss.

- *par3 and rf11.* rf11 is a random formula that reduces to the 3-parity $-x_2x_3x_4$. DFG alone has test error 0 at $n = 32$ in every seed, against v3's 0.47 and 0.50. In v3's seed-0 fits at $n = 32$ the stacker gave DFG no weight, and its last out-of-fold DFG fit (29 rows) had test error 0.51 on par3 and 0.50 on rf11.
- *par5.* DFG alone has error 0 from $n = 64$ on. v3's seed-mean errors at $n = 64$–512 are 0.36, 0.17, 0.17 and 0.35, and it reaches 10% only at 839. In v3's seed-0 fit at $n = 64$ the DFG member alone had error 0, but it held 0.30 of the weight and v3's error was 0.076.
- *The other three* are smaller or unexplained here: h5s2 (598 against 1154), rf00 (32 against 40) and ssym8 (148 against 177).

Reading [Heuristic]. Two rules of the frozen code (`study/part10/hyb_portfolio3/dfg13.py`) make the difference.

suite	target	DFG alone	v3	$\log_2$ ratio	hgb	lin	suite	target	DFG alone	v3	$\log_2$ ratio	hgb	lin
dlheld6	h6_04	79	29	+1.44	59	96	random	rf03	91	62	+0.54	139	never
dlheld2	h2rf1	82	31	+1.40	58	103	designed	dict	22	16	+0.48	–	27
designed	and3	60	25	+1.30	–	68	dlheld7	f7_04	97	70	+0.47	84	never
random	rf15	51	22	+1.23	53	67	dlsymdev	far9	98	71	+0.47	–	never
dlheld3	h3rf0	108	46	+1.22	60	112	dlfamdev	fd_07	134	98	+0.45	–	–
random	rf08	50	21	+1.22	53	69	dlheld3	thrint	186	139	+0.42	243	137
dlheld3	h3rf1	133	59	+1.17	63	101	dlheld4	tail13	134	100	+0.42	never	never
dlheld2	dnf4x2	235	112	+1.07	98	never	dlheld6	h6_03	129	98	+0.40	103	never
random	rf06	31	16	+0.97	58	37	dlheld6	h6_05	42	33	+0.38	59	48
designed	dlist5	61	32	+0.93	–	85	designed	andxor	52	40	+0.37	–	never
dlheld7	f7_06	82	44	+0.90	59	127	dlheld	hrf0	80	62	+0.36	63	191
dlfamdev	fd_00	138	75	+0.87	–	–	dlheld	inter2	441	346	+0.35	416	never
dlheld7	f7_02	190	106	+0.85	120	never	dlheld	maj15	130	105	+0.31	1278	110
designed	rand2	28	16	+0.83	–	50	random	rf16	44	36	+0.27	214	never
designed	maj3	56	31	+0.82	–	56	dlheld5	h5s4	222	184	+0.27	1465	never
dlheld4	h4rf1	96	54	+0.82	61	158	designed	thr10	146	121	+0.27	–	113
dlfamdev	fd_08	51	29	+0.81	–	–	random	rf10	43	36	+0.26	53	66
dlheld4	dnf3x3	193	115	+0.74	148	never	random	rf05	75	62	+0.26	250	never
dlheld3	dlist8	39	24	+0.69	61	91	dlheld4	ssym8	148	177	−0.27	8072	never
dlheld	mux6	197	123	+0.67	188	never	random	rf00	32	40	−0.33	60	83
designed	rand0	25	16	+0.62	–	never	dlheld5	h5s2	598	1154	−0.95	10860	never
designed	dnf3	178	118	+0.60	–	never	designed	par3	28	55	−0.99	–	never
dlheld3	cnf3	189	126	+0.58	164	never	random	rf11	28	56	−1.00	416	never
dlfamdev	fd_03	120	81	+0.56	–	–	dlheld	par5	56	839	−3.91	never	never

Table 13: The 48 targets on which DFG alone's and v3's n^* differ by more than 0.25 in $\log_2$, sorted by the ratio of DFG alone to v3 (positive: v3 needs less data) [Numerical]. `hgb` and `lin` (natural column order) are shown where they were run; "–": not run on the suite. Source: `study/part11/RESULTS.json` (`q2_suites.dfg_vs_v3`; `hgb` and `lin` from the per-target ratios there, checked against `study/hybrid/results/`).

- *Minimum rows.* DFG's degree-3 and degree-4 searches run only on at least 32 rows, and its degree-5 and degree-6 searches only on at least 64. At $n = 32$ the full fit therefore searches degree 3, while v3's 10-fold out-of-fold fits, on 28–29 rows, do not.
- *Budget share.* At $n \leq 512$ each of v3's out-of-fold DFG fits gets 1.75% or 3.5% of the budget. At $d = 30$ this leaves 24 or 48 rows for the degree-5 search, fewer than the 64 it needs, so those fits skip it. v3's final DFG fits and DFG alone's have 343–687 rows for it.

DFG alone uses its full-data fit whatever its folds show, whereas a stacker weights a member by its out-of-fold predictions. So a structure that only the full-data fit can find counts for little in v3. The parities are the cleanest case, and the effect is small overall: 6 of 138 targets. (The subsample sizes are computed from the code by `paper_dfg/make_tables.py` and recorded in `paper_dfg/tables/numbers.json`.)

4.4 The dictionary without the search

`lowdeg` is the thresholded low-degree algorithm [27] of Part 6's feasibility check, run unchanged (Section 3.2). It estimates the label's Fourier coefficient $\hat{c}_S$ on every character of degree $k \leq 4$, keeps S when $|\hat{c}_S| > \sqrt{2\ln(\binom{d}{k}/\delta)/n}$ with $\delta = 0.1$, tests products of kept characters in the same way, and predicts $\operatorname{sign}(\hat{c}_\emptyset + \sum_{\text{kept}} \hat{c}_S \chi_S)$. It therefore has DFG's character dictionary up to degree 4 and the

suite	targets	`lowdeg` vs DFG alone	`lowdeg` vs `nets_best`	`lowdeg` never	targets `lowdeg` never learns (DFG alone's n^*)
`designed`	21	+0.92 (3/16) [21]	−1.25 (13/6) [21]	0	–
`random`	20	+0.71 (4/15) [20]	−1.33 (16/2) [20]	0	–
`dlheld`	12	+1.53 (2/9) [11]	−0.26 (5/6) [11]	1	par5 (56)
`dlheld2`	7	+2.00 (0/6) [7]	+0.40 (2/5) [7]	0	–
`dlheld3`	10	+0.74 (3/6) [10]	−0.60 (5/4) [10]	0	–
`dlsymdev`	8	+3.98 (0/8) [7]	+1.69 (1/7) [7]	1	band11 (828)
`dlheld4`	12	+2.89 (0/12) [11]	+1.34 (3/9) [11]	1	band10 (440)
`dlheld5`	12	+2.05 (0/11) [6]	−0.54 (3/6) [6]	6	h5s0 (201), h5s1 (never), h5s3 (1642), h5s5 (420), h5s6 (597), h5s7 (6186)
`dlheld6`	12	+2.29 (1/10) [9]	+1.13 (2/9) [9]	3	h6_00 (never), h6_06 (347), h6_10 (213)
`dlfamdev`	12	+1.67 (2/9) [10]	–	2	fd_04 (170), fd_10 (never)
`dlheld7`	12	+1.90 (1/10) [10]	+0.70 (3/8) [10]	2	f7_05 (never), f7_08 (644)
pooled	138	+1.64 (16/112) [122]	−0.16 (53/62) [112]	16	

Table 14: The low-degree algorithm `lowdeg` on every suite [Numerical]. Entries as in Table 8, but the ratios are those of `lowdeg`'s n^* to the reference's (positive: `lowdeg` needs more data), so "better" and "worse" are `lowdeg`'s. Last column: the targets that `lowdeg` never brings to 10% test error by $n = 16384$, with DFG alone's n^* in brackets. Source: `study/part11/RESULTS.json` (`q3_lowdeg`); `lowdeg`'s runs: `study/part11/results/toy__lowdeg-perm__<suite>.json`.

same form of union-bound test. The differences in the test are small: DFG's log term is $\ln(4KN/\delta)$ with $K = 4$, about 2.8 nats more, and DFG tests the residual against the residual's s.d. instead of the labels. `lowdeg` has none of the rest of DFG:

- no residual and no greedy rounds;
- no sum blocks, no linear or spectral blocks, no extensions, no local search and no search of degrees 5–6;
- no refitted readout (least squares with logistic refinement), no kernel background and no small-n cross-validation.

Table 14 compares it with DFG alone and with `nets_best`; Figure 2b shows it target by target.

Results [Numerical].

- *Against DFG alone:* pooled over the 138 targets, `lowdeg` needs more data. Its mean ratio is +1.64 over the 122 targets both learn, about 3.1× more data (median +1.35). It is clearly worse on 112 targets and clearly better on 16.
- *Against the networks* `lowdeg` is about even: −0.16 against `nets_best` over the 112 of 126 targets both learn, clearly better on 53 and clearly worse on 62.

- *Never learned.* `lowdeg` never learns 16 targets: the four that DFG alone misses and 12 that DFG alone learns. Of those 12, par5 has degree 5 and is outside `lowdeg`'s dictionary. The other 11 are functions of a signed sum: the symmetric band11, band10 and f7_08; five signed counts with arbitrary accept sets (h5s0, h5s3, h5s5, h5s6, h5s7); two symmetric functions times a parity (h6_06, fd_04); and h6_10, a condition on $|s - 3x_{13}|$ with $s = x_1 + \cdots + x_{12}$.
- *Where `lowdeg` is faster.* All 16 such targets are learned at small n: DFG alone needs 22–133 points, `lowdeg` 16–61. They include the formulas h3rf0 (16 against 108) and h3rf1 (31 against 133), on which DFG alone is tied with or worse than the networks, and par4 (28 against 52).

Reading [Heuristic]. DFG alone's margin over the networks does not come from its dictionary or its test. With the dictionary and the test but none of the search, `lowdeg` is at par with the networks, and DFG alone needs about 3× less data than `lowdeg`. The margin comes from what `lowdeg` lacks. The sum blocks make the signed-sum targets learnable at all: a function of a k-input signed sum spreads its Fourier weight over many characters of several degrees, each too weak to pass its own test, while DFG tests the whole block with one χ^2 statistic. The residual rounds let weaker features pass once the stronger ones are fit, and the readout is refitted on the grown features. On small formulas the comparison runs the other way. At $n \leq 128$, `lowdeg`'s slightly more lenient test, applied to every degree at once on the labels, admits characters at sizes where DFG still keeps few or none. Section 5 removes DFG's parts one at a time.

4.5 Input dimension

The networks were trained at $d = 15$ and 60 on 7 `designed` targets and at $d = 120$ on 4 of them (maj9, mux, par2, par3). DFG v1.3 alone was run on the same targets at $d = 15$, 60 and 120 with seeds 0 and 1, as Part 6's DFG v1.1 runs were, and at $d = 30$ with seeds 0, 1 and 2. The networks here are `nets` (square loss), the only variant run at every d, with seeds 0 and 1 at $d \neq 30$ and 0, 1 and 2 at $d = 30$. They were trained for 20000 steps, except 40000 at $d = 120$. One network run, and with it DFG's budget, is therefore twice as large there. Table 15 gives n^* and the fitted exponents, and Figure 3 plots them.

Results [Numerical].

- *Exponents.* The least-squares slope of $\log n^*$ on $\log d$ is 0.01 (stair5) to 0.50 (par4) for DFG v1.3 alone, and 0.91 (maj9) to 2.40 (par4) for the networks, fitted per target on the same d.
- *The gap widens with d.* The median $\log_2$ ratio of DFG v1.3 to the networks is -2.64 at $d = 15$ (6.2× less data), -4.19 at $d = 30$ (18.3×), -4.99 at $d = 60$ (31.7×) and -5.41 at $d = 120$ (42.5×, 4 targets).
- *Parities.* DFG's n^* follows the union-bound line (Figure 3). n_{thr} grows from 19.5 to 27.9 for par2 and from 22.4 to 35.2 for par3 between $d = 15$ and 120, and from 24.6 to 36.3 for par4 between $d = 15$ and 60. On the grid, DFG's n^* moves by at most one grid step over the range.
- *Same as Part 6's DFG.* On these targets DFG v1.3 gives the n^* of Part 6's DFG (v1 at $d \leq 30$, v1.1 at $d \geq 60$): the median $\log_2$ difference is 0.00 at every d. The largest difference is maj9 at $d = 120$, 104 against 439.
- *Compute.* DFG's largest fit on these targets uses 0.07, 0.15, 0.18 and 0.18 of one network run at $d = 15$, 30, 60 and 120.

Reading [Heuristic].

	DFG v1.3 alone: n^* at $d =$				networks (`nets`): n^* at $d =$				
target	15	30	60	120	15	30	60	120	exponent DFG / networks
maj9	79	81	167	104	123	263	625	752	0.22 / 0.91
majpar	52	61	63	–	306	1380	3466	–	0.14 / 1.75*
mux	58	63	149	102	364	1156	2848	6073	0.36 / 1.35
par2	16	16	16	28	100	200	427	842	0.24 / 1.03
par3	28	28	28	56	241	843	3410	14029	0.30 / 1.96
par4	28	52	56	–	508	2728	14255	–	0.50 / 2.40*
stair5	109	111	110	–	433	1514	3487	–	0.01 / 1.50*

	DFG v1.3 against the networks: median $\log_2$ ratio (factor)	DFG v1.3 against Part 6's DFG: median	max FLOPs per fit / one network run	seeds (DFG; nets)
$d = 15$ (7 targets)	-2.64 (6.2×)	$+0.00$	0.07	2; 2
$d = 30$ (7 targets)	-4.19 (18.3×)	-0.01	0.15	3; 3
$d = 60$ (7 targets)	-4.99 (31.7×)	$+0.00$	0.18	2; 2
$d = 120$ (4 targets)	-5.41 (42.5×)	-0.00	0.18	2; 2

Table 15: n^* against the input dimension d on the `designed` targets that the networks were trained on at $d = 15$, 60 and 120 [Numerical]. DFG v1.3 alone (inputs permuted per fit) and the networks `nets` (square loss, the only variant run at every d); "–": not run. Seeds 0, 1 at $d \neq 30$ and 0, 1, 2 at $d = 30$ for both. The networks were trained for 20000 steps, 40000 at $d = 120$. Their grid extends to 65536 at $d = 60$ and is 64–8192 (maj9, par2), 256–32768 (mux) and 512–65536 (par3) at $d = 120$; n^* on the full grid equals the 16–16384 value everywhere. Exponent: the least-squares slope of $\log n^*$ on $\log d$ over the d at which both reach 10% error (*: $d = 15$–60 only). Bottom, per d: the median $\log_2$ ratio of DFG v1.3 to the networks (factor $2^{-\text{median}}$), the median $\log_2$ ratio of DFG v1.3 to Part 6's DFG (v1 at $d \leq 30$, v1.1 at $d \geq 60$, natural column order), DFG v1.3's largest FLOPs per fit on these targets, and the number of seeds. Source: `study/part11/RESULTS.json` (`q4_dscale`); DFG's runs: `study/part11/results/toy__dfg13-perm__designed_d<d>.json` and `study/part11/results/toy__dfg13-perm__designed.json`.

- *DFG against gradient descent.* DFG's data cost on a k-parity is the union bound's, $2\ln(4K\binom{d}{k}/\delta)$, which grows like $2k\ln d$: logarithmically in d. The networks' exponents grow with the parity's degree, 1.03, 1.96 and 2.40 for $k = 2$, 3 and 4 (par4 fitted on $d = 15$–60), as analyses of gradient training on sparse parities lead one to expect: a cost polynomial in d, with a degree that grows with k [2, 4]. On these targets the source of DFG's advantage is thus explicit. Exhaustive enumeration pays a logarithm of the dictionary's size, while gradient descent pays a power of d.
- *The size of the factors.* The comparison is with `nets`, the weakest variant at $d = 30$: on `designed`, DFG alone is at -2.82 against `nets` and -2.17 against `nets_wd` (Table 8). The factors are therefore larger than they would be against the best variant. The exponents are the robust part of the result.

4.6 Tabular data

DFG v1.3 needs ±1 inputs and two classes. Of the study's 26 tabular datasets only two qualify, kr_vs_kp and mushroom. Their binary and categorical columns are one-hot encoded as ±1, as in v3's DFG view (Section 3.1; `study/part11/dfg_real.py`), and DFG runs unchanged with its own budget. kr_vs_kp is a development set. Mushroom was held out in Part 7 and never used to develop DFG, but v1–v3 ran DFG on it and their results were known (Table 5). Table 16 gives the cells, verdicts and compute.

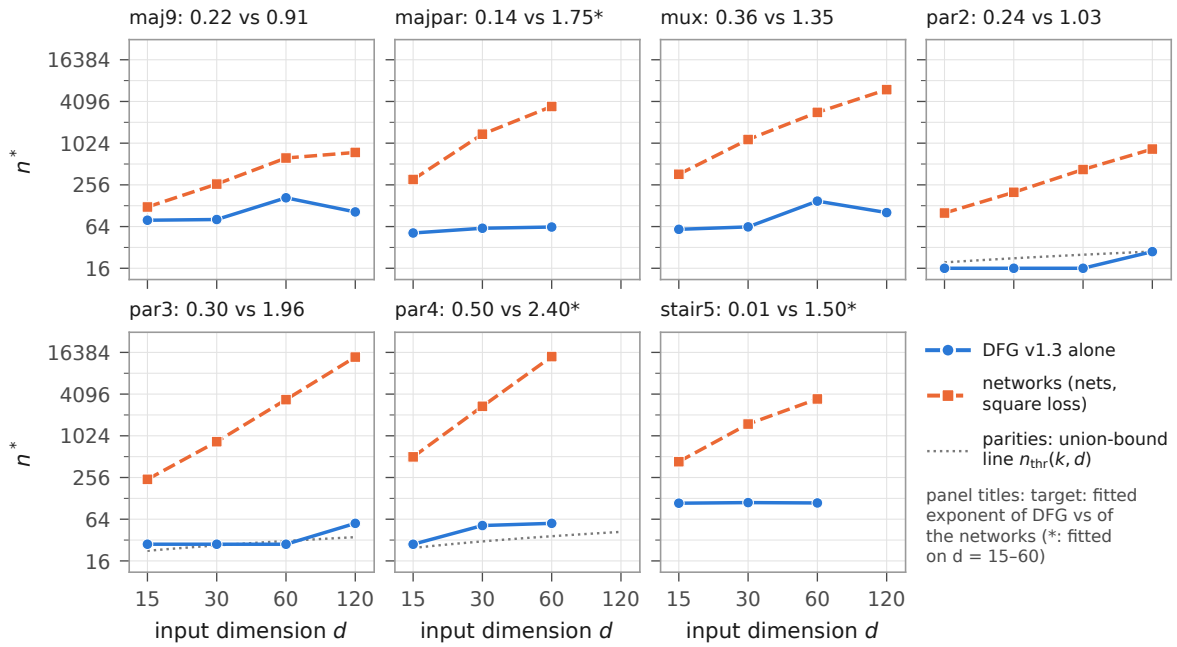


Figure 3: n^* against d (log–log) on the `designed` targets with network runs at $d = 15$, 60 and 120 (Table 15) [Numerical]. Panel titles give the fitted exponent of DFG v1.3 alone and of the networks (*: fitted on $d = 15$–60). Dotted, for the k-parities: the union-bound line $n_{\text{thr}}(k, d) = 2\ln(4K\binom{d}{k}/\delta)$ with DFG's $K = 4$, $\delta = 0.1$ and $c = 1$. On the first round's residual $r = y - \bar{y}$, a parity's statistic $|\text{mean}(r\chi_S)|/\text{sd}(r)$ is $\sqrt{1-\bar{y}^2} \approx 1$, so DFG's test admits it once $n(1-\bar{y}^2) > n_{\text{thr}}$. On the n-grid the test therefore passes at the first grid point above the line: 32 for par2 at every d, for par3 up to $d = 60$ and for par4 up to $d = 30$, and 64 for par3 at $d = 120$ and for par4 at $d = 60$. Refits of the stored fits show two exceptions (`paper_dfg/check_parity_variant.json`, refitted test errors identical to the result files). par2 is learned at $n = 16$, below the line, at $d = 15$, 30 and 60, because DFG's cross-validation kept its forced variant in all 7 of those fits; at $d = 120$ it did not. And par4 at $d = 30$ ($n_{\text{thr}} = 30.6$) passed at $n = 32$ in 1 of 3 seeds. Seeds: 2 at $d \neq 30$, 3 at $d = 30$; the $d = 120$ networks were trained for 40000 steps. Source: `study/part11/RESULTS.json` (`q4_dscale`) via `paper_dfg/make_figures.py`; the plotted values and n_{thr} are in `paper_dfg/figures/fig_dscale.json`.

Results [Numerical].

- *Against the best MLP variant per cell:* no cell is better, 8 of 9 are ties and 1 is worse: kr_vs_kp at $n = 100$, 14.06% against `mlp_reg`'s 7.80%. The mean $\log_2$ error ratio is -0.278, from cells where both errors are small and the cell rule sees no difference.
- *Against single references:* against the baseline `mlp`, 3 cells are better (kr_vs_kp at $n = 300$ and 1000, mushroom at 300), 5 are ties and 1 is worse ($n = 100$ on kr_vs_kp), with a mean of -0.482. Against `hgb` all 9 cells are ties (mean -0.022).
- *Against v3:* 8 ties and 1 worse cell (kr_vs_kp at $n = 100$: 14.06% against 6.88%), with a mean of $+0.213$: $+0.515$ on kr_vs_kp and -0.030 on mushroom.
- *Compute:* no fit uses more than 0.144 of the standard-task budget. The seed-mean CPU time per fit is at most 13.4 seconds per cell, 0.19–0.66 of v3's on the same cell.

On these two Boolean tables, then, DFG alone is level with gradient boosting and with the regularized MLPs, worse than v3 at the smallest n of kr_vs_kp, and nowhere clearly better than the best MLP.

		test error (%)							DFG alone		
dataset	n	DFG alone	`mlp`	`mlp_reg`	`mlp_strong`	`mlp_best`	`hgb`	v3	FLOPs / budget	CPU s per fit	v3 CPU s
kr_vs_kp	100	14.06	9.30*	7.80*	–	7.80*	10.76	6.88*	0.013	5.7	14.6
	300	3.71	6.13†	5.72	–	5.72	4.26	3.13	0.038	10.2	22.9
	1000	1.29	3.13†	1.67	–	1.67	1.21	0.75	0.030	3.6	15.8
	2397	0.29	0.50	0.46	–	0.46	0.13	0.29	0.062	7.8	40.6
mushroom	100	1.97	2.99	3.20	3.20	2.99	3.79	2.12	0.024	0.8	2.4
	300	0.53	1.35†	1.23	1.23	1.13	0.56	0.54	0.050	2.2	4.2
	1000	0.15	0.23	0.16	0.16	0.16	0.16	0.16	0.087	4.1	6.3
	3000	0.00	0.16	0.00	0.00	0.00	0.05	0.00	0.144	11.1	27.3
	6093	0.00	0.07	0.00	0.00	0.00	0.00	0.00	0.141	13.4	30.1

	kr_vs_kp		mushroom		both	
DFG alone against	cells (better / tie / worse)	mean $\log_2$ error ratio	cells (better / tie / worse)	mean $\log_2$ error ratio	cells (better / tie / worse)	mean $\log_2$ error ratio
`mlp`	4 (2/1/1)	−0.545	5 (1/4/0)	−0.432	9 (3/5/1)	−0.482
`mlp_reg`	4 (0/3/1)	−0.198	5 (0/5/0)	−0.386	9 (0/8/1)	−0.302
`mlp_strong`	–	–	5 (0/5/0)	−0.386	5 (0/5/0)	−0.386
`mlp_best`	4 (0/3/1)	−0.198	5 (0/5/0)	−0.342	9 (0/8/1)	−0.278
`mlp_best+`	4 (0/3/1)	−0.198	5 (0/5/0)	−0.342	9 (0/8/1)	−0.278
`hgb`	4 (0/4/0)	+0.208	5 (0/5/0)	−0.206	9 (0/9/0)	−0.022
v3	4 (0/3/1)	+0.515	5 (0/5/0)	−0.030	9 (0/8/1)	+0.213
v2	4 (0/3/1)	+0.368	5 (0/5/0)	−0.030	9 (0/8/1)	+0.147
v1	4 (0/3/1)	+0.540	5 (0/5/0)	−0.030	9 (0/8/1)	+0.223

Table 16: DFG alone on the two tabular datasets it can run on [Numerical]. Top: test error (%, seed mean of 3) per cell, against the MLP references and `hgb` and v3. The references are `mlp`; `mlp_reg`, L2-regularized; `mlp_strong`, run on mushroom only; and `mlp_best`, the per-cell best MLP variant by test error, chosen with hindsight. A reference's entry is marked * where DFG alone is clearly worse than it by the cell rule (errors differing by more than 10% relative and by more than 2 s.e.) and † where DFG alone is clearly better. Right: DFG alone's FLOPs as a fraction of the standard-task budget (three MLP training runs; maximum over seeds) and its CPU seconds per fit, and v3's CPU seconds on the same cells (seed means). Bottom: per reference, the cells (better / tie / worse for DFG alone) and the mean $\log_2$ ratio of DFG alone's error to the reference's (errors floored at 0.002), per dataset and over the cells of both. Source: `study/part11/RESULTS.json` (`q6_tabular`); DFG alone's runs: `study/standard_tasks/results/dfg13__kr_vs_kp.json` and `study/standard_tasks/results/dfg13__mushroom.json`.

Nine cells, with errors near 0 from $n = 1000$ on, resolve little.

Real-valued data: Part 7 [Numerical, then Heuristic]. On real-valued, multi-class data DFG has been run only through three adaptations of DFG v1.1, written in Part 7 [11]. dfg_bits uses thermometer bits and their characters, dfg_gates ANDs of threshold literals, and dfg_dirs thresholded directions. Each was frozen on 5 development datasets before 13 held-out ones were run (`study/standard_tasks/STANDARD-TASKS.md`; tables in `study/standard_tasks/RESULTS.md`).

- *Tabular, held out (11 datasets, 59 cells):* their mean $\log_2$ error ratio to the baseline MLP was +0.04 to +0.10, with 6–9 cells better and 10–15 worse. Against an L2-regularized MLP at $n \leq 3000$ (44 cells) it was +0.16 to +0.23, with 0–4 cells better and 11 worse. On the same cells default gradient boosting had −0.07 against the baseline MLP and +0.11 against the regularized one.

- *Images:* on MNIST and Fashion-MNIST (14 cells) they were at +0.26 to +0.41 against the baseline MLP, and worse in 7 of the 14 cells each.

These are adaptations of v1.1, frozen before the sum blocks (v1.2) and the weighted blocks (v1.3). Each replaced DFG's ±1 dictionary with its own, and two also loosened the test constant. They are evidence about DFG's recipe, not about DFG v1.3, which has not been run on real-valued data. Part 7's reading [Heuristic] was that the recipe fails there because the useful features of dense real-valued signal are not in any small enumerable dictionary, and that the tests can stop growth early on many weak, correlated effects.

4.7 Compute

The budget rule. Every DFG fit has the budget of one network training run at its d (`harness.gd_flops`: 1.62×10^{11} FLOPs at $d = 30$; at $d = 120$ the run has 40000 steps). Compute is DFG's own count of FLOPs, with a multiply-add counted as 2. The rule, from DFG's docstring (Section 2), has three parts.

- *Subsampled searches.* The exhaustive searches run on the first m training points, with m set so that each stays within a fixed share of the budget: 15% for degree 4, 3% per round for degree 3 and for the two-input extensions, and 3% each for degrees 5 and 6 in the first round. The test thresholds use m. A search with fewer than 32 rows (64 for degrees 5–6) is skipped.
- *Rounds.* A round starts only while the fit has spent less than 0.75 of its budget.
- *Cross-validation.* At $n \leq 512$ the final fit gets half of the budget, and the $2F$ fold fits of the cross-validation share the other half.

Results [Numerical] (Table 17).

- *FLOPs.* Over all 4730 fits (4334 at $d = 30$, 396 at $d \neq 30$), the largest uses 0.519 of one network run (h5s5 on `dlheld5` at $n = 16384$), and none exceeds 0.75. The largest fit per suite uses 0.154 (`random`) to 0.519 (`dlheld5`) of a run, always at $n = 16384$; at $d = 15$, 60 and 120 the largest use 0.067, 0.177 and 0.179. The median fit uses less than 0.001 of a run at $n \leq 32$, 0.025–0.055 at $n = 64$–512 and 0.149 at $n = 16384$.
- *CPU time.* The median fit takes 1.1–2.7 seconds per suite, and the longest takes 17.4 seconds (`dlheld3`). By n the median is largest at $n = 64$–512 (4.2–7.3 seconds) and falls to 1.2–2.4 seconds at $n \geq 1024$.
- *Against v3* on the same fits, the median of the per-fit CPU ratio is 0.06–0.92 at every n. On these rough measurements DFG alone is faster than the portfolio it was part of.
- `lowdeg` uses at most 0.095 of a network run per fit and a median of 0.02–0.03 seconds.

Reading [Heuristic]. FLOPs grow with n, but CPU time follows the cross-validation. At $n = 64$–512 DFG fits $2F$ fold models in addition to the final one ($F = 10$ at $n \leq 64$, 5 above), on folds large enough for the degree-3 and degree-4 searches, and these many small fits cost more time per FLOP. At $n \leq 32$ the folds are too small for those searches, and above $n = 512$ DFG fits only the final model. Each network fit spends one run by definition, and `nets_best` is a hindsight choice among 2 to 5 such runs per target. "No more FLOPs" therefore holds with a wide margin: the largest DFG fit uses about half of one network run, and the median fit at the largest n about a seventh.

(a) per suite: DFG alone

suite	d	fits	max FLOPs per fit / one network run (target, n)	fits above 0.75 run	CPU s per fit: median / max
`designed`	30	693	0.372 (thr10, 16384)	0	1.53 / 15.5
`random`	30	440	0.154 (rf06, 16384)	0	1.12 / 13.6
`dlheld`	30	396	0.420 (inter2, 16384)	0	2.43 / 13.2
`dlheld2`	30	231	0.307 (thr20, 16384)	0	2.07 / 12.0
`dlheld3`	30	330	0.275 (dlist8, 16384)	0	2.33 / 17.4
`dlsymdev`	30	264	0.163 (twosym, 16384)	0	2.12 / 11.6
`dlheld4`	30	396	0.162 (symand, 16384)	0	2.03 / 14.7
`dlheld5`	30	396	0.519 (h5s5, 16384)	0	2.70 / 15.5
`dlheld6`	30	396	0.420 (h6_09, 16384)	0	2.66 / 13.5
`dlfamdev`	30	396	0.471 (fd_08, 16384)	0	2.40 / 14.7
`dlheld7`	30	396	0.380 (f7_04, 16384)	0	2.36 / 16.6
`designed`, $d = 15$	15	154	0.067 (maj9, 16384)	0	0.14 / 4.5
`designed`, $d = 60$	60	154	0.177 (maj9, 16384)	0	1.51 / 13.1
`designed`, $d = 120$	120	88	0.179 (maj9, 16384)	0	3.19 / 16.2
`lowdeg`, 11 suites	30	4334	0.095	0	0.02–0.03 / 1.7

n	16	32	64	128	256	512	1024	2048	4096	8192	16384
DFG alone: FLOPs per fit / one network run											
median	0.000	0.000	0.025	0.029	0.036	0.055	0.058	0.072	0.082	0.104	0.149
max	0.000	0.001	0.033	0.039	0.050	0.096	0.077	0.137	0.200	0.312	0.519
CPU s per fit: median (max)											
DFG alone	0.02	0.16	7.29	5.27	4.24	4.26	1.20	1.57	1.68	1.85	2.41
	(0.04)	(0.91)	(16.6)	(11.8)	(10.9)	(17.4)	(5.1)	(9.6)	(11.3)	(13.0)	(15.5)
v3	0.31	0.58	8.22	10.71	8.44	4.59	4.16	6.06	3.67	5.04	7.64
	(1.9)	(1.6)	(19.4)	(30.0)	(28.8)	(18.6)	(23.9)	(45.8)	(18.2)	(30.4)	(34.5)
median of the paired ratio DFG alone / v3	0.06	0.32	0.89	0.49	0.51	0.92	0.27	0.25	0.46	0.38	0.32

Table 17: Compute of DFG alone per fit [Numerical]. Top: per suite, FLOPs as a fraction of one network training run at that d (DFG's own count, multiply-add = 2; the largest fit, with its target and n), the number of fits above 0.75 of a run, and CPU seconds per fit (process time of the fit only). Last row: `lowdeg` over its 11 suites (range of the suites' medians). Bottom: by n, over DFG alone's 4334 fits at $d = 30$ (394 per n), with v3's CPU seconds on the same fits and the median of the per-fit ratio. The CPU times were measured with 4 worker processes running in parallel at nice 10 on a 4-CPU machine, and v3's in Part 10 under that run's own load, so the CPU comparison is rough. Source: `study/part11/RESULTS.json` (`compute`), from the `flops` and `cpu_seconds` fields of the result files.

5 What its parts do

DFG v1.3 has many parts (Section 2, Table 3). This section asks which of them carry its results. It reports the protocol's 12 ablations (Q5), each DFG alone with one constructor argument changed, and then a post-hoc *core* DFG with nine parts removed at once (Section 5.6). Every run uses the frozen file; only constructor arguments change.

How the runs are compared. An ablation runs on DFG alone's training sets, column permutations and cross-validation folds, which all come from the fit's random stream. Its n^* is compared with DFG alone's n^* computed from the same seeds of DFG alone's own runs. Six ablations (and the core) ran on seeds 0 and 1, six on seed 0 only (Section 3.4). The ratio is $\log_2(n^*_{\text{run}}/n^*_{\text{DFG alone}})$, negative when the run needs less data. A target moves *lower* or *higher* when the ratio is beyond

∓ 0.25; means are over the targets both reach. The five ablation suites hold 65 targets (`designed`, `dlsymdev`, `dlfamdev`, `dlheld6`, `dlheld7`); only `dlheld7` is held out.

Two nulls. The protocol's null compares DFG alone's seed 1 with its seed 0. Those are different training sets, so it overstates the noise of a paired comparison: it moves 23 of the 65 targets beyond ± 0.25. A paired comparison needs a paired null, which was added post hoc: DFG alone refitted on the same training and test sets with another fit stream, so with another column order and other folds and nothing else (`study/part11/interp/ablations/perm_null.py`). The column order is irrelevant to the targets, and no test statistic depends on it, but it can matter through tie-breaking, the block search's orders and the folds. A change that leaves DFG's decisions alone therefore still perturbs its path, and the paired null measures by how much.

Families. Each target is assigned by hand, from its definition, to one of the nine groups of Section 6.2 and to three classes (post hoc; `study/part11/interp/ablations/families.py`). *Character targets* are parities, staircases, multiplexers, compositions of parities and majorities and thresholds of a few characters, DNFs, CNFs, decision lists, read-once and random formulas: 71 of the 138 targets, 25 of the 65. *Halfspaces* are majorities and thresholds of a weighted sum, with the dictator, AND(x_1, x_2, x_3) and an AND of two halfspaces (13 and 7). *Symmetric and mixture targets* are functions of one signed sum, alone or with a parity, a gate, a second sum or an inner term (54 and 33). This class differs from the 54 sum-structured targets of Section 3.1 by two targets each way: it includes sym8 and sym9, and it counts f7_01 and fd_01, thresholds of an unsigned sum, as halfspaces.

Sources. The summaries are in `study/part11/RESULTS.json` (`q5_ablations`, `q5b_core`), written by `study/part11/compare_dfg.py`. The groupings, the nulls, the predictions of Section 5.2 and the leave-one-in runs of Section 5.6 are in `study/part11/interp/ablations/ablations.json`, written by `study/part11/interp/ablations/ablations.py`; the analysis is `study/part11/interp/ablations/ABLATIONS.md`. Models are read from refits of single cells (`study/part11/interp/ablations/refit.py`). A refit uses the instrumented copy of Section 6.3, so that every feature carries the class that admitted it, and all 36 refits reproduce their run's result file exactly in test error and FLOP count (`study/part11/interp/ablations/results/refits.json`). In four cells the same fit through the evaluation's own loader of the frozen file gives the same model, term by term (`study/part11/interp/ablations/results/refit_crosscheck.json`). The analysis has not been checked by an independent verifier.

5.1 Twelve single changes

Tables 18 and 19 and Figure 4 give the results.

The paired null [Numerical; post hoc]. On seed 0 the re-permuted fits move 1 of the 65 targets beyond ± 0.25 (dlist5, $+0.39$), and 10 of the 65 ratios are not zero; the mean |ratio| is 0.012. On seeds 0 and 1 they move 3 (andxor -0.69, thrsum -0.66, h6_07 $+0.40$), with mean |ratio| 0.038. Over all 138 targets of the 11 suites they move 8 on seeds 0 and 1 (mean |ratio| 0.040). The unpaired null moves 23 to 31 of the 65 targets, depending on the pair of seeds.

What carries the results [Numerical].

run	change	seeds	all (65)	newly never	character targets (25)	halfspaces (7)	symmetric and mixtures (33)
`nosums`	no sum blocks	0, 1	+0.91 (2/31) [59]	band11, h6_01, f7_08	+0.16 (1/5) [25]	+0.05 (0/1) [7]	+1.82 (1/25) [27]
`strict`	stricter tests, $\delta = 10^{-4}$	0, 1	+0.06 (9/14) [62]	–	+0.17 (3/8) [25]	−0.23 (4/1) [7]	+0.04 (2/5) [30]
`lenient`	more lenient tests, $\delta = 0.9$	0, 1	+0.07 (5/13) [62]	–	+0.12 (2/8) [25]	−0.04 (2/1) [7]	+0.05 (1/4) [30]
`noeig`	no eigenvector seeds in the block search	0, 1	+0.03 (0/1) [62]	–	+0.00 (0/0) [25]	+0.00 (0/0) [7]	+0.05 (0/1) [30]
`noblocks`	no linear or spectral block	0, 1	−0.02 (6/4) [62]	–	−0.03 (2/1) [25]	−0.19 (3/0) [7]	+0.02 (1/3) [30]
`nocomp`	no extension or local classes	0, 1	+0.00 (2/2) [62]	–	+0.02 (1/2) [25]	+0.00 (0/0) [7]	−0.02 (1/0) [30]
`noweighted`	no weighted blocks	0	+0.02 (0/1) [62]	–	+0.01 (0/0) [25]	+0.00 (0/0) [7]	+0.03 (0/1) [30]
`nosmalln`	no forced variant, no cross-validation	0	+0.12 (0/10) [62]	–	+0.22 (0/8) [25]	−0.03 (0/0) [7]	+0.07 (0/2) [30]
`nobg`	no kernel background	0	−0.10 (7/3) [62]	–	−0.07 (3/2) [25]	−0.09 (1/0) [7]	−0.12 (3/1) [30]
`lsread`	least-squares readout	0	−0.01 (2/1) [62]	–	−0.01 (2/1) [25]	+0.00 (0/0) [7]	−0.02 (0/0) [30]
`K3`	global search to degree 3	0	+0.02 (2/1) [62]	–	+0.08 (1/1) [25]	−0.05 (1/0) [7]	−0.01 (0/0) [30]
`labelnoise`	tests scaled to the labels' noise	0	+0.14 (5/15) [62]	–	+0.23 (2/9) [25]	−0.17 (2/0) [7]	+0.14 (1/6) [30]
paired null	other column order and folds, same training sets	0	+0.00 (0/1) [62]	–	+0.00 (0/1) [25]	+0.00 (0/0) [7]	−0.00 (0/0) [30]
paired null	other column order and folds, same training sets	0, 1	−0.02 (2/1) [62]	–	−0.02 (1/0) [25]	+0.00 (0/0) [7]	−0.02 (1/1) [30]
unpaired null	seed 1 against seed 0: other training sets	1 vs 0	−0.04 (13/10) [62]	–	−0.02 (5/4) [25]	−0.17 (2/1) [7]	−0.04 (6/5) [30]

Table 18: The 12 ablations of Q5 on the 65 targets of the five ablation suites, against DFG alone on the same seeds and training sets [Numerical]. Each cell gives the mean $\log_2$ ratio of the run's n^* to DFG alone's over the targets both learn; below it, the targets on which the run needs clearly less / clearly more data (beyond ∓ 0.25, or only one reaches 10% error), and in brackets the number both learn. "Newly never": targets the run never brings to 10% error by $n = 16384$ while DFG alone does. Class sizes in the header (Section 5, "Families"). The three targets that DFG alone never learns (fd_10, h6_00, f7_05) are in no mean. Bottom: the paired null (post hoc), DFG alone refitted on the same training sets with another column order and other cross-validation folds, and the protocol's unpaired null. Sources: `study/part11/RESULTS.json` (`q5_ablations`; the pooled column is checked against it) and `study/part11/interp/ablations/ablations.json` via `paper_dfg/make_tables.py`.

- *Sum blocks.* Without them (`nosums`) the mean ratio is +0.91 over the 59 targets both learn, with 31 clearly higher and 2 lower, and DFG no longer learns band11, h6_01 and f7_08. The cost falls on the symmetric targets (+2.78, 9 of the 9 that DFG alone learns move, 3 of them to "never") and the mixtures (+1.54, 16 of 21 higher). Character targets lose little (+0.16; rand3, parmaj, h6_05, fd_07 and dnf3 move up). They are of the kinds the atlas found encoded in part by blocks before their characters are all detectable: parmaj by two

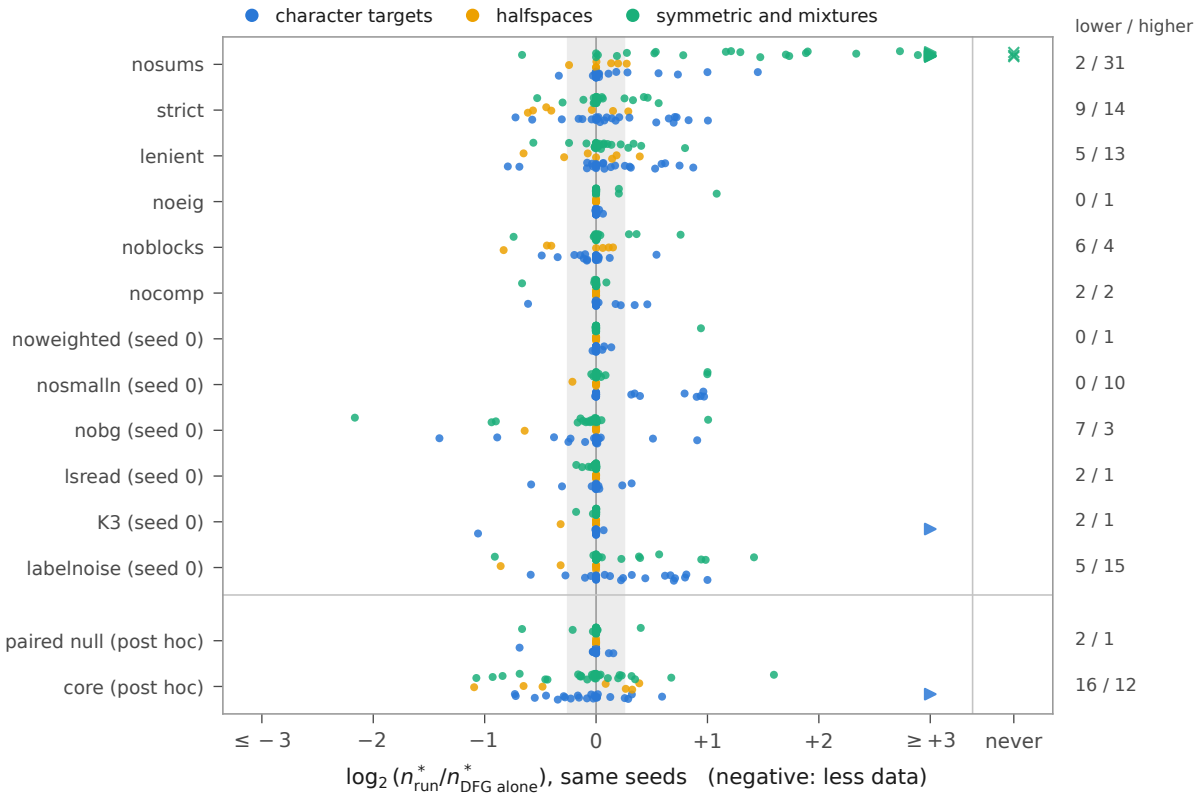


Figure 4: Every target of the five ablation suites, one row per run: the $\log_2$ ratio of the run's n^* to DFG alone's on the same seeds [Numerical]. Colour gives the target's class; the grey band is the tie band ± 0.25. Ratios beyond ± 3 are drawn at the edge (triangles), and "never" (the run never reaches 10% error where DFG alone does) in the right-hand column. Right: the numbers of targets clearly lower / clearly higher. Bottom: the paired null on seeds 0 and 1 (post hoc) and the core DFG of Section 5.6 on the same 65 targets (post hoc). Source: `study/part11/interp/ablations/ablations.json` via `paper_dfg/make_figures.py`; the plotted points are in `paper_dfg/figures/fig_ablations.json`.

folded blocks, formulas by the blocks of a clause or term (Section 6.2). The two seeds agree: 27 targets move up on both and none in opposite directions.

- *Tests scaled to the residual* (`labelnoise`, +0.14; 15 higher, 5 lower) and *the small-n machinery* (`nosmalln`, +0.12; 10 higher, none lower) come next, mostly on character targets (Sections 5.4 and 5.5).

What moves both ways [Numerical]. The two test levels, the kernel background and the linear and spectral blocks move 10 to 23 targets each, in both directions, with means between −0.10 and +0.07. Halfspaces gain from `strict` (−0.23), `noblocks` (−0.19) and `labelnoise` (−0.17), three changes that keep features out of the model at small n (Section 5.2). For `strict` and `lenient`, most movements are one grid step on one seed: 24 and 22 targets move on one seed only, 5 and 4 on both.

What can be removed [Numerical]. Five parts stay within the paired null, or close to it, on these 65 targets:

run	parities (3)	staircases (1)	half-spaces (7)	compositions (6)	formulas (11)	random formulas (4)	symmetric (12)	mixtures (21)
`nosums`	+0.00	+0.00	+0.05	+0.11	+0.17	+0.36	+2.78	+1.54
	(0/0)	(0/0)	(0/1)	(1/1)	(0/3)	(0/1)	(0/9)	(1/16)
`strict`	+0.40	+0.83	−0.23	+0.09	+0.01	+0.39	−0.01	+0.06
	(0/1)	(0/1)	(4/1)	(2/2)	(1/2)	(0/2)	(0/0)	(2/5)
`lenient`	−0.26	+0.00	−0.04	+0.36	+0.19	−0.11	+0.07	+0.05
	(1/0)	(0/0)	(2/1)	(0/3)	(0/5)	(1/0)	(0/1)	(1/3)
`noeig`	+0.00	+0.00	+0.00	+0.00	+0.01	−0.00	+0.02	+0.06
	(0/0)	(0/0)	(0/0)	(0/0)	(0/0)	(0/0)	(0/0)	(0/1)
`noblocks`	+0.00	+0.00	−0.19	−0.07	+0.01	−0.12	−0.00	+0.04
	(0/0)	(0/0)	(3/0)	(1/0)	(0/1)	(1/0)	(0/0)	(1/3)
`nocomp`	+0.00	+0.00	+0.00	−0.07	+0.08	+0.05	+0.00	−0.03
	(0/0)	(0/0)	(0/0)	(1/0)	(0/2)	(0/0)	(0/0)	(1/0)
`noweighted`	+0.00	+0.00	+0.00	+0.00	+0.02	+0.00	+0.00	+0.04
	(0/0)	(0/0)	(0/0)	(0/0)	(0/0)	(0/0)	(0/0)	(0/1)
`nosmalln`	+0.26	+0.00	−0.03	+0.38	+0.15	+0.22	+0.11	+0.05
	(0/1)	(0/0)	(0/0)	(0/3)	(0/3)	(0/1)	(0/1)	(0/1)
`nobg`	+0.00	+0.04	−0.09	−0.12	−0.10	+0.00	−0.04	−0.16
	(0/0)	(0/0)	(1/0)	(1/0)	(1/1)	(1/1)	(0/0)	(3/1)
`lsread`	+0.00	+0.00	+0.00	+0.06	−0.06	−0.01	−0.00	−0.02
	(0/0)	(0/0)	(0/0)	(0/1)	(2/0)	(0/0)	(0/0)	(0/0)
`K3`	+1.00	+0.07	−0.05	+0.00	−0.00	−0.26	+0.00	−0.01
	(0/1)	(0/0)	(1/0)	(0/0)	(0/0)	(1/0)	(0/0)	(0/0)
`labelnoise`	+0.00	+0.70	−0.17	+0.11	+0.33	+0.19	+0.10	+0.15
	(0/0)	(0/1)	(2/0)	(1/1)	(1/6)	(0/1)	(0/1)	(1/5)
paired null (0, 1)	−0.00	+0.00	+0.00	−0.11	+0.02	−0.00	−0.02	−0.01
	(0/0)	(0/0)	(0/0)	(1/0)	(0/0)	(0/0)	(0/0)	(1/1)
paired null (0)	−0.00	+0.00	+0.00	−0.02	+0.02	−0.00	+0.00	−0.00
	(0/0)	(0/0)	(0/0)	(0/0)	(0/1)	(0/0)	(0/0)	(0/0)
unpaired null	−0.33	+0.01	−0.17	+0.17	−0.01	−0.07	+0.01	−0.06
	(1/0)	(0/0)	(2/1)	(1/2)	(2/1)	(1/1)	(1/1)	(5/4)
core (post hoc)	+2.01	−0.08	−0.17	−0.15	−0.04	−0.27	−0.03	−0.05
	(0/1)	(0/0)	(3/3)	(3/1)	(2/2)	(2/1)	(0/0)	(6/4)

Table 19: The ablations by target family (the groups of Section 6.2; group sizes in the header) on the 65 targets of the five ablation suites [Numerical; families post hoc]. Each cell: mean $\log_2$ ratio over the targets both learn, and the targets clearly lower / clearly higher. The symmetric group holds the three targets no learner learns, so its means are over 9 targets (6 under `nosums`). Bottom: the paired and unpaired nulls, and the core DFG on the same 65 targets (post hoc). Source: `study/part11/interp/ablations/ablations.json`.

- the eigenvector seeds of the block search (`noeig`): 1 target moves, h6_10 (+1.08);
- the extension and local classes (`nocomp`): 4 move, among them andxor and thrsum, which the paired null also moves by as much;
- the weighted blocks (`noweighted`): 1 moves, h6_10 again (+0.94), the target they were designed after (Table 5);
- the logistic refinement (`lsread`): 3 move, by at most 0.58;
- the search of degrees 4–6 (`K3`): 3 move, of which only par4 by much (+3.01; Section 5.3).

Their mean |ratio| is 0.020–0.075, against 0.012 and 0.038 for the paired null on one and two seeds.

Reading [Heuristic]. On these suites DFG's results come from the sum blocks, with the residual scaling of the tests and the small-n machinery as smaller contributors. Five parts can each be removed without a change larger than an arbitrary perturbation of the path produces. Two qualifications apply. The suites contain few targets that need those parts: one pure 4-parity, no dense halfspace on more than 13 inputs, no degree-5 parity. And the parts are removed one at a time, so they may substitute for one another. Section 5.6 removes them together.

5.2 The test or the search

`strict` ($\delta = 10^{-4}$) and `lenient` ($\delta = 0.9$) move the log term of every test by $+6.91$ and -2.20 nats. If a target's data cost is set by a test, its n^* should move by a predictable amount; if it is set by the search, it should not move. Section 6.5 goes through these two runs target by target (Figure 6); this subsection summarizes them and adds the refits behind the asymmetry between the two.

Parities: the test, exactly [Numerical]. By Proposition 2.2, a k-parity passes round 0's test exactly when $n(1-\bar{y}^2) > n_{\text{thr}}(k,d)$, and n_{thr} moves by $+13.8$ and -4.4 rows with δ. At $d = 30$ it moves from 22.3, 26.8 and 30.6 ($k = 2, 3, 4$) to 36.1, 40.6 and 44.4 under `strict`, and to 17.9, 22.4 and 26.2 under `lenient`. Applied seed by seed to each training set's own $\bar{y}$, this predicts the measured changes of n^* to within 0.03: par3 $+0.98$ against $+1.00$ under `strict`, and 0 against 0 under `lenient`; par4 $+0.20$ against $+0.21$, and -0.79 against -0.79. par2 moves under neither, because the forced variant fits it at $n = 16$, below every n_{thr} here, and the cross-validation's choice does not depend on δ.

Other character targets [Numerical]. By Proposition 2.4(b), n_{glob} is proportional to the log term, so `strict` should raise it by $+0.54$ to $+0.86$ doublings (degree 4 to degree 1), and `lenient` lower it by -0.22 to -0.43. `strict` raises n^* beyond 0.25 on 8 of the 25 character targets (par3, stair5, rand3, rand2, majpar, sparsepoly, h6_03, h6_05), and their first fit moves one grid step. `lenient` is not the mirror image: it lowers n^* on 2 targets (par4, rand2) and raises it on 8 (andxor, majpar, fd_00, mux, fd_07, dlist5, h6_05, f7_02). Two things cause the asymmetry. Most of these targets are first fitted at $n = 32$–64, often by the forced variant. And the test most sensitive to δ is the linear block's, whose threshold on 30 inputs (Equation (3)) is 51.2 at $\delta = 0.1$, 33.8 at 0.9 and 81.7 at 10^{-4}. Refits show both.

- *majpar at $n = 64$* (seed 0). With $\delta = 0.1$ nothing passes in round 0. The forced variant adds $\chi_{\{1,2\}}$ and then $\chi_{\{3,4\}}$, and $\chi_{\{5,6\}}$ passes its own test in round 2: error 0.067. With $\delta = 0.9$, $\chi_{\{1,2\}}$ passes in round 0 and the linear block enters with 30 degree-1 characters, none of them in majpar's Fourier support; then nothing passes, and the error is 0.341. With $\delta = 10^{-4}$ the forced pairs are the same, but $\chi_{\{5,6\}}$ fails its test: error 0.269.
- *f7_02 at $n = 128$* (a DNF of three terms, seed 1), the other direction. With $\delta = 0.1$ the linear block enters, and the default variant is kept: error 0.202. With $\delta = 10^{-4}$ nothing passes. The forced variant guesses the blocks $(x_2 - x_6 - x_8)$ and $(-x_3 + x_5 + x_7)$, two of the DNF's terms; the third term's block then passes its test, and the cross-validation keeps the forced variant: error 0.

Halfspaces and sum targets [Numerical]. Halfspaces gain from a stricter test (-0.23; 4 of 7 lower). On f7_01 at $n = 128$, a degree-1 character that passes its own test first takes part of the exact block's signal in the block's second test, and with $\delta = 10^{-4}$ the block enters instead (Section 6.5). On the 33 symmetric and mixture targets the class means are $+0.04$ and $+0.05$ (30

learned; 5 and 4 higher, 2 and 1 lower). At the true block's detection point n_{detect} (Part 10's block statistics, re-thresholded for each δ), `strict` moves the block threshold by +0.20 to +0.30 doublings of n. On the 7 targets whose n^* lies a doubling or more beyond n_{detect}, it changes n^* by at most 0.05. On the other 23, n^* lies less than a doubling beyond n_{detect} (for 18 at or below it), and the grid cannot separate the test from the search.

Reading [Heuristic]. For character targets the round-0 test is binding when the first fit comes from it: exactly for the parities, within a grid step for staircases and small formulas. Where the first fit is the forced variant's, δ acts through the linear block and the cross-validation instead, and the sign of its effect does not follow from the test alone. For the sum targets the test is not binding: once the search proposes the block, the block passes far above its threshold (9 to 71 times for par4 in Section 5.3). This agrees with Part 10's reading that the search, not the test, sets their data cost [13].

5.3 Degree 4 without the degree-4 search

`K3` sets $K = K_{\text{hi}} = 3$: the global search stops at degree 3, and the degree-5/6 search is off. K also enters every class's union bound through $\ln(4KN/\delta)$ (Equation (2)), so every test becomes 0.29 nats more lenient. On the 65 targets `K3` moves 3 (Table 18).

par4 [Numerical; refits, seed 0]. Under `K3`, par4's test error is 0.49–0.50 at $n \leq 256$, 0 at 512, 0.50 at 1024, 0 at 2048 and 4096, and 0.49–0.50 at 8192 and 16384. In the three fitted cells the model is a folded block on the parity's four inputs, admitted in round 0 by an additive block search: $|-x_1 - x_2 - x_3 - x_4|$ at 512, $|x_1 - x_2 + x_3 + x_4|$ at 2048 and $|x_1 - x_2 - x_3 + x_4|$ at 4096, at 9.0, 34.5 and 70.9 times its threshold. A k-parity is a function of any signed sum of its k inputs, so one folded block represents it exactly (Proposition 2.5). In the four empty cells the round-0 block searches' best candidates stand at 0.13–0.45 of their thresholds, and the model is the constant and the background. With the degree-4 search, $\chi_{\{1,2,3,4\}}$ passes its round-0 test from $n = 64$ on (Proposition 2.2).

mux and andxor [Numerical; refits, seed 0]. These two targets have degree-4 coefficients of 0.75 and 0.5. Their test error under `K3` equals DFG alone's at every n, but the models differ.

- mux at $n = 256$: DFG alone's largest term is $\chi_{\{1,2,3,4\}}$ (share 0.51), admitted by the degree-4 test in round 0. `K3` holds no degree-4 character; it has the blocks $(-x_1 - x_2 - x_3 + x_4)$ and $(x_1 + x_2 + x_3 + 3x_4)$, and $\chi_{\{1,2,3\}}$ from the two-input extension of $\chi_{\{1\}}$.
- andxor at $n = 256$: DFG alone holds its exact expansion $\chi_{\{1,2\}}$, $\chi_{\{3,4\}}$, $\chi_{\{1,2,3,4\}}$. `K3` holds the two pair characters and the weighted block $(x_1 + x_2 - 3x_3x_4)$.

rand1, -1.06 [Numerical; refit, seed 0, $n = 64$]. With $K = 4$, round 0 admits the linear block (30 degree-1 characters, one of them relevant), and the forced variant then guesses $\chi_{\{4,12,17\}}$: error 0.412. With $K = 3$, $\chi_{\{1\}}$ passes its own, slightly more lenient test, so the linear block does not enter. The forced guess is then $\chi_{\{1,4,6\}}$, the local class adds $\chi_{\{4,6\}}$, and a block $(x_3 - x_5)$ follows: error 0. This difference comes from K in the union bound, not from the degree-4 search.

Reading [Heuristic]. The degree-4 search is redundant where a target's degree-4 characters share their inputs with lower-degree structure: blocks on those inputs, and the extension and local classes, reach them. It is indispensable for a degree-4 parity with no lower-degree footprint. The block test would admit the exact block from small n on, but the block search starts from degree-1 and degree-2 statistics (Proposition 2.5), which are zero for such a parity, so it finds the four inputs only by chance. The ablation suites contain one such target. Section 5.6 adds three more.

5.4 The small-n machinery and the background

The atlas computed two counterfactuals on its 37 targets (Section 6.2): the variant DFG's cross-validation did not choose, refitted on the same rows, and each fitted model without its kernel background. `nosmalln` and `nobg` run them as learners, and on the 21 targets the ablation suites share with the atlas the two agree.

No forced variant, no cross-validation [Numerical]. `nosmalln` fits the default variant on all n rows. At $d = 30$ it can differ from the atlas's "always default" refit only in the degree-5/6 subsample sizes (Proposition 2.6). Its seed-0 test error equals that refit's in all 126 cells with $n \leq 512$. Over the 65 targets, the mean test error rises by 0.012, 0.007, 0.022, 0.009 and 0.004 at $n = 16$, 32, 64, 128 and 256, and by nothing from 512 on. n^* rises on 10 targets and falls on none (mean +0.12). Since `nosmalln` is the default variant, every cell where DFG alone does better is one where DFG alone kept the forced variant. The first fits lost this way are par2 and rand2 at $n = 16$, majpar at 32 and 64, mux and sym7 at 64, fd_04 at 128 and f7_02 at 256. Each comes back one grid step later. Over the atlas's 222 cells, DFG's choice has mean test error 0.2257, always taking the default 0.2319, always taking the forced variant 0.2256, and the better of the two in each cell 0.2111 (Section 6.2).

No kernel background [Numerical]. The background is added after the weights are fitted. A fitted model without it therefore gives exactly DFG's prediction without it for the same variant. The ablation can differ from that only where the cross-validation chooses another variant, because the fold fits lose their background too. `nobg`'s seed-0 error equals the atlas's model-without-background error in 224 of 231 cells. The other 7 are all at $n \leq 128$, so in each of them the choice changed. One is refitted: f7_11 at $n = 16$. DFG alone keeps the default variant there (held-out errors 6 and 3), the constant plus the background, with test error 0.301. Without the background, the default variant's held-out errors rise to 13, the forced variant is kept, and its model $\chi_{\{11\}}$, $\chi_{\{8\}}$ has error 0.077. Over the 65 targets the mean error rises by 0.015 and 0.016 at $n = 16$ and 32, and falls by 0.009, 0.002, 0.003 and 0.002 at 64, 128, 256 and 512. n^* falls on 7 targets and rises on 3 (mean −0.10); f7_11 falls most (−2.17).

Reading [Heuristic]. With nothing else in the model, at $n = 16$–32, the background is a Hebbian classifier, and it helps targets with a strong degree-1 part. From $n = 64$ on it adds the noise of 30 residual correlations to a model that already fits. Through the cross-validation it also makes the default variant look better than it is, so the forced variant's right guesses are kept less often. Most n^* lie at 64–512, so the background costs data on balance on these suites.

5.5 Tests scaled to the residual

`labelnoise` replaces s by 1 in every threshold, so the tests are scaled to the labels' noise instead of the residual's. (The subsampled searches keep their subsample's root mean square, and a block's second test, on the projected residual, keeps that residual's variance.) In round 0 the residual is $y - \bar{y}$, with variance $1 - \bar{y}^2$, and after the first features fit it shrinks. By Proposition 2.4(c), characters admitted after round 0 enter 1.4–3.7 doublings below n_{glob} (class medians), mainly because the residual variance there is 3.2–19 times smaller (class medians; Proposition 2.4, "Checks").

Results [Numerical]. The mean ratio is +0.14 (15 higher, 5 lower). By family: formulas +0.33 (6 of 11 higher), the staircase +0.70, random formulas +0.19, mixtures +0.15 (5 of 21), compositions +0.11, symmetric targets +0.10, parities 0, halfspaces −0.17 (Table 19). The mean test error rises by 0.024 at $n = 128$ (18 targets higher by more than 0.02, 2 lower) and by 0.013 at 256, and by

at most 0.004 in absolute value elsewhere. Two refits at $n = 128$ (seed 0) show the two ways the scaling matters.

- *Later rounds*, rand3. DFG alone admits $\chi_{\{5,6\}}$ in round 0 ($s = 1.00$), $\chi_{\{4,5,6\}}$ by the one-input extension in round 1 ($s = 0.84$), two local characters in round 2 ($s = 0.77$) and modulated blocks in rounds 3 and 4: error 0.049. With $s = 1$ the first round is the same, but nothing passes in round 1, and the fit stops: error 0.317.
- *Biased labels*, far9 $= [|x_1 + \cdots + x_9| \geq 5]$, whose round-0 residual has $s = 0.71$. DFG alone admits the folded block $|x_1 + \cdots + x_9|$ and a spectral feature in round 0: error 0.006. With $s = 1$ the thresholds are 1.4 times higher, nothing passes, and the model is the constant and the background: error 0.181.

Reading [Heuristic]. The residual scaling is what makes the growth greedy in the sense of DFG's design: each round's tests are calibrated to the noise left after the previous round, so weaker features pass once stronger ones are in. Its cost on these targets is modest, because the features that bring most targets below 10% error enter in round 0 (Section 6.6). It is larger for formulas and staircases, whose later characters build on the first ones, and for targets with biased labels.

5.6 A core DFG (post hoc)

Everything in this subsection is post hoc. The protocol's addendum (`study/part11/PROTOCOL.md`) was written after every pre-registered result had been read, to ask whether the parts that are removable one at a time can be removed together. The `core` run keeps the global characters of degree at most 3; the unweighted signed-sum blocks, searched from pairs and single inputs, with their modulated, product, split, beam and forced moves; the union-bound tests scaled to the residual; least squares; and the small-n cross-validation. It removes nine parts: the linear and spectral blocks, the extension and local classes, the degree-4 to degree-6 searches, the weighted blocks, the eigenvector seeds, the logistic refinement and the kernel background. Through $K = 3$, every test is also 0.29 nats more lenient. The code is the frozen file, with the constructor arguments `readout="ls"`, `sum_inner=False`, `sum_seeds=(0,3,2)`, `blocks=False`, `max_ext_bases=0`, `local_max=0`, `K=3`, `K_hi=3` and `background=False`. It ran on all 11 suites, with seeds 0 and 1 and the standard grid. Its definition used the ablations' results on all five ablation suites, `dlheld7` included, so no suite is held out for it.

Against DFG alone [Numerical; post hoc]. Over the 138 targets the core's mean ratio is +0.07 (median −0.01) over the 133 both learn (Table 20). It needs clearly less data on 28 targets and clearly more on 26, and it loses par5. Its suite means range from −0.20 (`dlheld4`) to +0.46 (`dlheld2`). On the held-out `dlheld7` the mean is −0.03, with 2 targets lower and 2 higher. The core moves 54 of the 138 targets beyond ±0.25. The paired null on the same seeds and all 138 targets moves 8 (mean |ratio| 0.040; `study/part11/RESULTS.json`, `q5b_core.paired_null`). Most of the core's differences are therefore real, but they cancel on average. By family, the core loses on parities (+2.01 over the 3 it still learns, and par5) and halfspaces (+0.32; 7 higher, 3 lower), and it is level elsewhere (−0.08 to +0.21).

Against the networks [Numerical; post hoc]. The core's mean against `nets_best` is −1.74 over the 122 of 126 targets both learn, clearly better on 111 and clearly worse on 9. DFG alone on the same seeds is at −1.88, clearly better on 117 and worse on 5. On `dlheld7` the core is at −1.30 (9 better, 1 worse) and DFG alone at −1.27 (10 and 1). The core is clearly worse than `nets_best` on

suite	status for DFG v1.3	targets	core vs DFG alone	newly never	core vs `nets_best`	DFG alone vs `nets_best`
`designed`	development	21	+0.11	–	−2.03	−2.14
			(8/5) [21]		(20/1) [21]	(20/0) [21]
`random`	development	20	+0.26	–	−1.78	−2.04
			(3/3) [20]		(19/1) [20]	(20/0) [20]
`dlheld`	development	12	+0.15	par5	−1.57	−2.17
			(2/3) [11]		(8/2) [11]	(11/0) [12]
`dlheld2`	held out for v1.1 only	7	+0.46	–	−1.15	−1.62
			(0/3) [7]		(6/1) [7]	(7/0) [7]
`dlheld3`	held out for v1.1 only	10	−0.08	–	−1.30	−1.22
			(4/2) [10]		(8/1) [10]	(7/3) [10]
`dlsymdev`	development	8	+0.05	–	−2.09	−2.14
			(0/1) [8]		(8/0) [8]	(8/0) [8]
`dlheld4`	exposed to v1.2	12	−0.20	–	−1.90	−1.70
			(2/0) [12]		(12/0) [12]	(12/0) [12]
`dlheld5`	exposed to v1.2	12	+0.12	–	−2.63	−2.75
			(1/3) [11]		(11/0) [11]	(11/0) [11]
`dlheld6`	development	12	+0.06	–	−1.22	−1.28
			(3/3) [11]		(10/2) [11]	(11/1) [11]
`dlfamdev`	development	12	−0.19	–	no networks	–
			(3/1) [11]			
`dlheld7`	**held out**	12	−0.03	–	−1.30	−1.27
			(2/2) [11]		(9/1) [11]	(10/1) [11]
pooled		138	+0.07	1	−1.74	−1.88
			(28/26) [133]		(111/9) [122]	(117/5) [123]

	parities (4)	stair-cases (2)	multi-plexers (2)	half-spaces (13)	compo-sitions (13)	formulas (15)	random formulas (35)	sym-metric (25)	mixtures (29)
core vs DFG alone	+2.01	−0.06	+0.21	+0.32	+0.17	−0.08	−0.07	+0.06	−0.04
	(0/2) [3]	(0/0) [2]	(0/1) [2]	(3/7) [13]	(4/2) [13]	(4/2) [15]	(10/5) [35]	(0/2) [22]	(7/5) [28]
core vs `nets_best`	−1.97	−2.98	−0.66	−0.15	−2.92	−0.77	−1.89	−2.07	−1.85
	(2/2) [3]	(2/0) [2]	(2/0) [2]	(6/4) [12]	(12/0) [13]	(12/0) [12]	(34/1) [35]	(21/1) [21]	(20/1) [22]
DFG alone vs `nets_best`	−4.73	−2.92	−0.87	−0.49	−3.09	−0.67	−1.82	−2.13	−1.90
	(4/0) [4]	(2/0) [2]	(2/0) [2]	(8/2) [12]	(13/0) [13]	(12/0) [12]	(33/2) [35]	(21/1) [21]	(22/0) [22]

Table 20: The core DFG (post hoc). Top: per suite and pooled, the core against DFG alone on the same seeds (0 and 1) and training sets, and the core and DFG alone (same seeds) against `nets_best` [Numerical]. Cells as in Table 8: mean $\log_2$ ratio over the targets both learn, below it the targets clearly lower / clearly higher (here: lower = the first-named learner needs clearly less data) and in brackets the number both learn. "Newly never": learned by DFG alone on these seeds but never by the core. Bottom: the same by target family over all 138 targets (group sizes in the header; families post hoc). Sources: `study/part11/RESULTS.json` (`q5b_core`) and `study/part11/interp/ablations/ablations.json` (`core`).

par4, par5, rf17, thr30, thr20, h6_10, thrint, f7_01 and h6_00. DFG alone (same seeds) is clearly worse on thrint, f7_01, h6_00, h3rf0 and h3rf1; the core is better than the networks on the last two.

Compute [Numerical; post hoc]. On the same 3036 fits, the core's median FLOP count is 0.094 of DFG alone's (paired). Its largest fit uses 0.150 of one network run, against DFG alone's 0.500. Its total CPU time is 2302 s against DFG alone's 9058 s, with median 0.30 s against 2.01 s per fit. Both ran four workers on four CPUs, so CPU times are rough.

target	DFG alone	core	core vs DFG alone	degrees 4–6	compo-sition	linear, spectral	logistic readout	back-ground	weighted blocks	eigen. seeds	weighted + eigen.	backgr. + linear
the core needs more data												
par5	56	never	$+\infty$	**56**	never							
rf17	49	3550	+6.19	**49**	3550							
par4	48	3124	+6.02	**48**	3124							
poly4	161	3127	+4.28	**151**	3127							
thr30	158	623	+1.98			192	483	223				
thr20	134	455	+1.76			**140**	414	249				
h6_10	215	649	+1.60					668	494	649	**207**	
h5s2	367	678	+0.88					713	678	678	678	
sym8	54	100	+0.88			100		**54**	100	100		
maj15	122	206	+0.76			**122**	207	174				
h5t0	112	185	+0.72					146	185	185	187	
f7_09	222	355	+0.68					357	355	355	**234**	
maj21	109	174	+0.67			**109**	174	136				
maj9	84	110	+0.39			**84**	110	**94**				
f7_01	156	195	+0.32			**132**	195	**161**				
thr10	128	154	+0.27			**128**	**136**	**123**				
the core needs less data												
h3rf1	135	31	−2.13	31	31	31	31	52				52
h3rf0	120	30	−1.98	30	30	30	30	32				32
h4rf1	103	29	−1.84	29	29	29	29	59				**104**
and3	78	37	−1.09	37		**78**	37	44				
f7_11	63	30	−1.07	30		30	30	**54**				
rf06	31	16	−0.97	16		16	16	**27**				
fd_02	58	31	−0.93	31	31	31	31	42				42
dlist8	55	30	−0.86	30	30	30	30	31				31
h5rf0	293	164	−0.84	**292**	186	164	164	166				166
fd_11	109	61	−0.84	62		68	61	**125**				
rand0	27	16	−0.73	16		16	16	**26**				
rand1	105	63	−0.72	**88**		68	63	**102**				

Table 21: Leave one in (post hoc). For the core's largest losses and gains, n^* (seeds 0 and 1) of DFG alone, of the core, and of the core with one part put back: the degree-4 to degree-6 searches ($K = 4$, $K_{\text{hi}} = 6$), the extension and local classes, the linear and spectral blocks, the logistic readout, the kernel background, the weighted blocks, the eigenvector seeds. Empty: not run (the targets for each part were chosen after reading the core's results). Bold: putting this one part back returns n^* to within 0.25 of DFG alone's. Same training sets, column orders and folds throughout. Source: `study/part11/interp/ablations/leavein.py`, results in `study/part11/interp/ablations/results/leavein/`, summarized in `study/part11/interp/ablations/ablations.json` (`leavein`).

Where the core loses, and what it misses [Numerical; post hoc]. To attribute the core's largest differences, each removed part was put back on its own, on the targets where the core moved most (Table 21; the targets were chosen after reading the core's results). The losses fall into three groups.

- *Parities of degree 4 and 5 with no lower-degree footprint.* These are par4 (+6.02), rf17 (+6.19; its gates compute a 4-parity), poly4 (+4.28; its term $x_7x_8x_9x_{10}$) and par5 (never learned). Putting the degree-4 to degree-6 searches back restores all four: 48, 49, 151 and 56, against DFG alone's 48, 49, 161 and 56. Putting the extension and local classes back restores none, since these targets have no lower-degree character to extend. In the core they are found only when the block search hits the right inputs by chance, as par4 under `K3` (Section 5.3): par4 at $n = 512$ and from 2048 on in seed 0, and from 4096 on in seed 1. par5 is found only at $n = 16384$ in seed 0. In a refit of that cell, a product block whose two factors together cover

its five inputs, next to nine irrelevant ones, passes in round 0 at 6.7 times its threshold, and the exact block $(x_1 + x_2 - x_3 + x_4 + x_5)$ follows in round 1 (error 0). At $n = 512$ the round-0 searches' best candidates stand at 0.34–0.37 of their thresholds, and the model is the constant.

- *Dense halfspaces.* thr30 (+1.98), thr20 (+1.76), maj15 (+0.76), maj21 (+0.67), maj9 (+0.39), f7_01 (+0.32) and thr10 (+0.27). Without the linear block a degree-1 character must pass its own test, which most of them cannot do at $n = 128$–512 on 20 or 30 inputs. Putting the linear and spectral blocks back restores maj9, maj15, maj21, thr10, thr20 and f7_01 to within 0.25 of DFG alone, and most of thr30 (192 against 158). The background restores maj9, thr10 and f7_01 and part of the others; the logistic readout restores only thr10. In a refit of thr30 at $n = 512$ (seed 0), the core's model holds the 5 degree-1 characters that pass their own tests, a 12-input signed block and a modulated block, with error 0.108. DFG alone's holds the same 5 characters, the linear block's other 25 and modulated blocks, with error 0.034, and 0.188 without the linear block's characters.

- *Mixtures with an inner term, and a few single steps.* h6_10 (+1.60) and f7_09 (+0.68) come back only with the weighted blocks and the eigenvector seeds together (207 and 234, against 215 and 222), not with either alone (494 and 649; 355 and 355). In DFG alone, removing either one already cost h6_10 about a doubling (Table 18). In a refit of h6_10 at $n = 256$ (seed 0), DFG alone's model is the weighted folded block $|x_1 + \cdots + x_{12} + 3x_{13}|$ (share 0.96, error 0.017), and the core's is the two pair characters $\chi_{\{1,13\}}$ and $\chi_{\{3,13\}}$ (error 0.329). sym8 (+0.88) comes back with the background (54). At $n = 64$ (seed 0), DFG alone keeps the forced variant, whose forced block $|x_1 + \cdots + x_8|$ is exact (held-out errors 23 against 11; error 0). The core keeps the default variant (16 against 12) and predicts the constant (0.281): the background's noise had made the default variant look worse in the folds. sym8 is also one of the 8 targets the paired null moves (+0.69): its n^* hinges on that one choice at $n = 64$. h5t0 (+0.72) comes back in part with the background (146 against 112). h5s2 (+0.88; one grid step later on both seeds) comes back with none of the parts tried.

Where the core gains [Numerical; post hoc]. The core needs less data than DFG alone on 28 targets. The largest gains are on formulas learned at $n = 16$–64: h3rf1 (-2.13), h3rf0 (-1.98), h4rf1 (-1.84), and3, f7_11, rf06, fd_02, dlist8, h5rf0, fd_11, rand0 and rand1 (-1.09 to -0.72). Putting the background back removes the gain on rand0, rand1, rf06, f7_11 and fd_11, and part of it on h3rf1, h4rf1 and fd_02. With the linear block as well, it removes all of h4rf1's gain (104 against 103). The linear block alone removes the gain on and3, and $K = 4$ with the degree-4 to degree-6 searches removes it on h5rf0 (292 against 293). On h3rf0 and dlist8 no part tried removes the gain, alone or paired. In both learners their test error stays near 10% for several doublings (h3rf0: 0.07–0.12 at $n = 32$–256 in the core, 0.07–0.20 in DFG alone), so small differences move the 10% point by one or two doublings.

Reading [Heuristic]. On these suites, a learner with characters of degree at most 3, unweighted signed-sum blocks, residual-scaled tests, least squares and the small-n cross-validation keeps DFG alone's results against the networks, on the held-out suite too, at about a tenth of its FLOPs. What it gives up is what the removed parts are for. The degree-4 to degree-6 searches find parities with no lower-degree footprint, the linear block finds dense halfspaces, and the weighted blocks with the eigenvector seeds find sums with a heavy inner term. The single ablations could not show the first two, because the ablation suites contain one target of the first kind (par4) and no halfspace on more than 13 inputs. A target mix with more such targets would make the core worse than DFG alone. Conversely, the background and the linear block cost data at small n on formulas, and that is what the core gains. The core is a description of the frozen learner, not a tested design: it was defined

after every result had been read.

6 What it learns

This section describes what DFG v1.3, run alone, puts in its fitted models: across target families, on labels that carry no signal, where it fails, when structure enters as n grows, and in what order. It is descriptive. It says which features the models hold, not why DFG does well, and nothing in it is a causal test; the ablations of Section 5 are the interventions. Every fit analysed on a toy target is an instrumented rerun of an evaluation fit, with the same training rows, column permutation and random streams, and the same test error and FLOP count (Section 6.1); the fits on noise labels run the same frozen code. Each analysis was checked by an independent verifier, and the verifiers' corrections are used throughout. Measured quantities are [Numerical], with their source; readings are [Heuristic]; statements proved in `study/part11/interp/theory/THEORY.md` are [Proved]; Section 2.5 states them as P1–P5.

Findings.

1. *DFG holds each target in the form of its dictionary that is cheapest to detect.* Parities are one character from the first fit on, symmetric and signed-sum targets their exact signed block, and halfspaces degree-1 characters and then a block. Staircases, multiplexers and small formulas are their characters, after transient block encodings of a branch or a term. At $n = 16384$, 8 of the 37 analysed models are exactly the target's Fourier expansion (Section 6.2).

2. *Mixtures end in forms their definitions do not suggest.* A parity of two majorities is first two folded blocks over the union of their inputs. A parity, gate or selector partner becomes the inner term of a weighted block. At $n = 16384$ only 1 of the 7 signed-sum mixtures still holds its compact form, though all 7 are fitted without error (Section 6.2).

3. *Where DFG errs, features are missing, not weights.* In 133 of the 148 analysed cells with test error above 0.1, no weights on the learned features fit the target. In 9 they would, all on threshold-like targets just after the linear block entered (Section 6.2).

4. *The character tests keep their proved level on noise; the dense block tests and the forced variant do not.* On balanced labels independent of the inputs, none of 220 final models at $d = 30$ holds a feature that passed a test. The false admissions that do occur come from the linear and spectral blocks, the spectral block's growing with label bias and with d, and from the product of the forced variant's two guesses (Section 6.3).

5. *Spurious terms are mostly the linear block's irrelevant degree-1 characters*: 961 of the 1047 terms that read an irrelevant input on the 41 designed and random targets. They fade with n (Section 6.3).

6. *The residue classes mod 3 fail at the search, not at the test.* Their weight at Fourier degrees 1–2 is exactly $1.6 \cdot 10^{-6}$ to $2.1 \cdot 10^{-4}$, so the block search, whose seeds are degree-1 and degree-2 statistics, has no foothold. Their block would pass DFG's test from $n = 128$. h5s1 fails at its modulator, a pair with coefficient $-1/32$, and arbitrary accept sets are the last signed-sum targets to be learned (Section 6.4).

7. *On sum targets the search, not the test, sets the data cost.* Round 0 admits a character near its detection size n_{glob}, in which degree enters only through the logarithm of the union bound and coefficient size through $1/\hat{y}_S^2$, and later rounds admit the rest of a junta earlier. A block

is held only well after its test would admit it, and a stricter or more lenient test leaves n^* of the symmetric targets nearly in place (Section 6.5).

8. *DFG's order is set by detectability, the networks' by construction.* At the networks' sample sizes DFG admits a target's characters in one round. Where it has an order, it agrees with the networks' moderately, and it inverts where a large coefficient sits at high degree: mux's largest term is DFG's first character and the networks' last (Section 6.6).

9. *FLOPs follow the budgeted exhaustive searches, CPU time the sum-block search*, so the slowest fits are at $n = 64$–512, where the FLOP count is smallest (Section 6.7).

The per-target detail is in Appendix B.2.

6.1 How the models are read

The fits are the evaluation's [Numerical]. The analyses refit DFG through one wrapper (`study/part11/interp/dfg_hooks.py`). It loads the frozen file (md5 checked), builds the training rows, the fit's random stream and the column permutation exactly as the evaluation does, and records every call of DFG's fit routine without changing it: at $n \leq 512$ the $2F$ fold fits of the cross-validation, then the final fit on all n rows, whose model predicts. 517 seed-0 cells (47 targets of all 11 suites, every grid n) reproduce the test error and the FLOP count of the result files bit for bit, as do 66 cells of the runs at $d = 15$, 60 and 120 (`study/part11/interp/tables/identity.md`). The result files store no models, so the identity is of errors and FLOP counts, a strong fingerprint of the same code path. The other analyses pass the same check (Appendix B.2).

The atlas. The structure atlas (`study/part11/interp/ATLAS.md`) covers 37 targets from 9 of the 11 suites, chosen to cover the main families, at every grid n from 16 to 16384: 407 fits, all on seed 0, so one training set per (target, n). How much a seed can matter shows in h5s7: its seed-0 fits reach 10% error at $n_0^* = 1705$, the seed-mean error only at $n^* = 6186$.

Terms, shares and categories. A model is read as terms in the original input coordinates, with the column permutation undone: the constant, each character χ_S, each spectral feature, each sum block with all its level indicators as one term, and the kernel background. A term's *share* is the standard deviation of its contribution to the model's output over the 4096 test inputs; a group's share is that of its summed contribution, as a fraction of the output's. A term is a *target term* (T) if it is a character in the target's Fourier support or in one of its compact representations, or a block with a true class's input set and signs. It is *relevant* (R) if it reads only relevant inputs but is not a target term, and *spurious* (S) if it reads an irrelevant one. S-lin marks the irrelevant degree-1 characters that entered with the linear block, which adds every input not yet in the model.

The target's own representations. For each target the atlas writes down the compact representations it has in DFG's dictionary (`study/part11/interp/atlas_truth.py`). They are the sign of a few characters (parities, staircases, majorities of parities, halfspaces, DNFs, CNFs, decision lists), the exact Fourier expansion (multiplexers, random and read-once formulas), and sum blocks on the target's signed-symmetry classes, additive, modulated, weighted or as a product. Character representations are verified on all 2^k assignments of the k relevant inputs. Block representations are checked only as "f is a function of the block statistics"; their accept sets are not checked.

Complete, exact, and what the features can express. A representation is *complete* in a model when all its characters are present, or when every true block is a factor of a sum term with the true statistic (the true inner term or modulator, if any), or the factors of a product partition it. A model is *exactly* the target's Fourier expansion when it holds all the target's Fourier characters and nothing else but the constant and the background. "Complete" checks the dictionary, not the weights. What the learned features can express whatever DFG's weights is measured by the *oracle error*: the test error of a lightly regularized logistic fit of f on the model's features, fitted on 2048 test inputs and scored on the other 2048, without the kernel background. The categories depend on the representations written down. A model can be exact in a form not listed, and its terms then count as R. Five representations (the folded pair, the multiplexers' address blocks, the clause blocks, f7_06's and the auxiliary symmetry-class blocks) were added after the first fits had been looked at, and every cell was then reclassified.

6.2 What the models contain, family by family

Figure 5 shows every cell of the atlas and Table 22 summarizes each family at its first fit and at $n = 16384$. The paragraphs below give the families' models; "first fit" is the first grid n with test error at most 0.1.

Parities: one character [Numerical]. From the first fit on (par2 at $n = 16$, par4 and par5 at 64), every model is the constant plus the parity's character, with share 0.98–1.00, and nothing else up to $n = 16384$. Before it the models are the constant and the background. par2 at $n = 16$ needs the forced variant: no character passes a union-bound test on 16 rows, and the cross-validation's held-out errors are 7 (default) against 0 (forced). par5 is found at $n = 64$ by the final fit's subsampled degree-5 search, which the fold fits skip; their held-out errors are high and equal in both variants (21 and 21 of 64 at $n = 64$), so the cross-validation cannot see what the final fit will find. *Reading* [Heuristic]. For parities the dictionary is the target class, and DFG either has the character or nothing. The size at which it has it is set by the union bound (Proposition 2.2, Figure 3).

Staircases, multiplexers and formulas: characters, after block encodings [Numerical].

- *Staircases.* The prefix characters x_1, x_1x_2, ... are complete at the first fit, with error 0 (stair5 at $n = 128$, forced; stair6 at 256), next to one further Fourier character each. With more data the rest of the expansion fills in: at $n = 16384$ stair5's model is exactly its 16 Fourier characters, and stair6's holds 57 of its 63 and two relevant blocks. stair6 loses its degree-6 prefix character from $n = 512$ on, at error ≤ 0.0007.
- *Multiplexers.* The first fits mix characters and blocks. mux11 at $n = 512$ (error 0) holds 4 of its 8 modulated address blocks, such as $x_5 \cdot (-x_1 - x_2 + x_3)$, weighted blocks on address and data bits, and 17 of its 64 Fourier characters. mux6 at 256 (forced, error 0.002) holds 10 of its 16 characters and two additive blocks on two address bits and a data bit. From $n = 1024$ (mux6) and 4096 (mux11) on, the model is the complete Fourier expansion and nothing else.
- *DNF, CNF, decision lists, read-once formulas.* At the first fit the models are characters of degree 1–3 and blocks that encode a clause, a term or a rule. cnf3 at $n = 256$ (forced, error 0.004) holds one clause block exactly, $(-x_7 + x_8 - x_9)$; a second clause's block with two of the first clause's literals as inner term, $(-x_4 + x_5 + x_6 - 3\,\mathrm{AND}(\neg x_1, \neg x_2))$, share 0.49; and 10 of its 21 clause characters. f7_02 at 256 holds the block of its 4-literal term. The character representations are complete later: cnf3 at 1024, f7_02 and f7_06 at 2048, f7_04 at 4096. The decision lists start as their degree-1 characters, through the linear block.

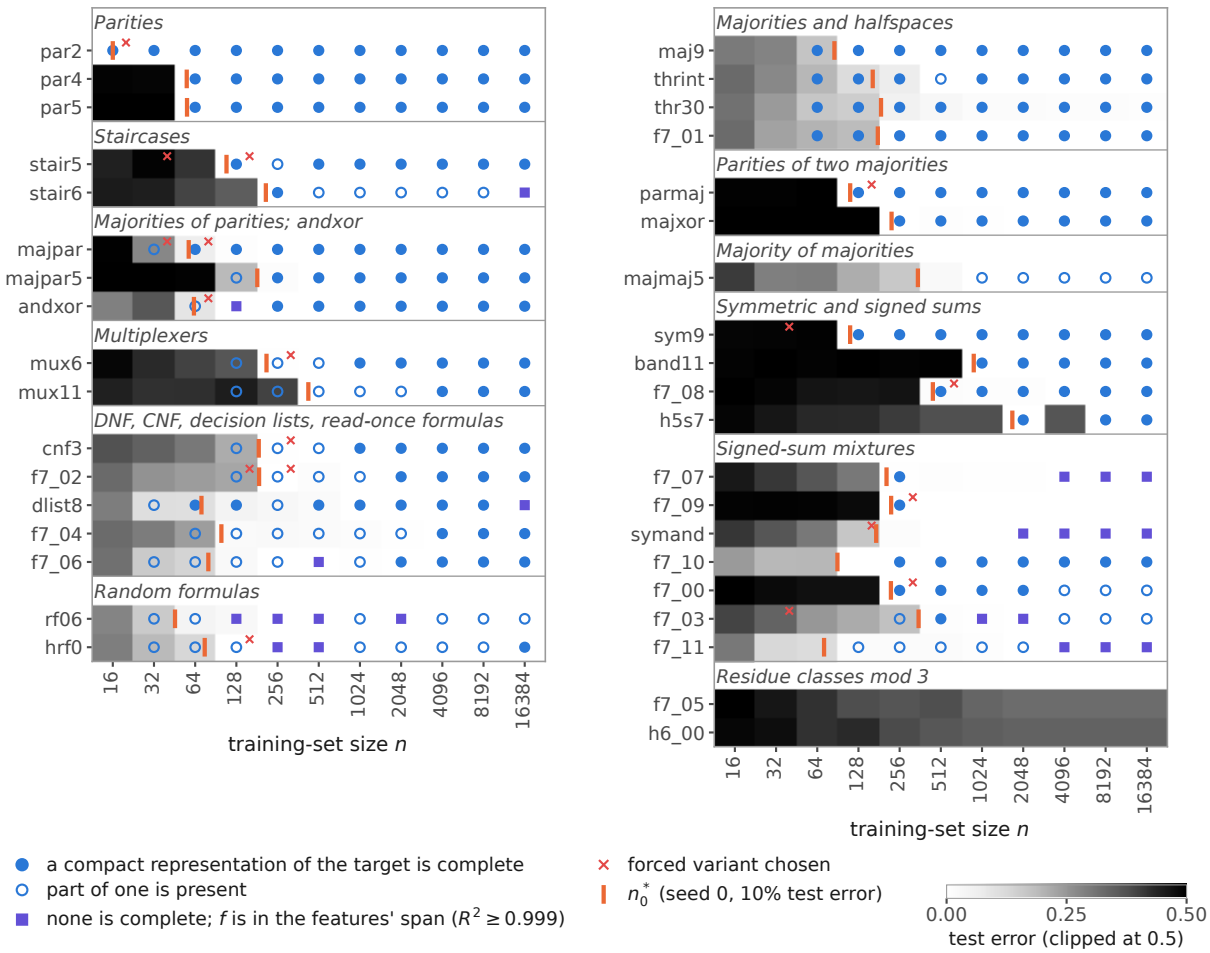


Figure 5: The structure atlas: DFG alone on 37 targets, seed 0, every grid n [Numerical]. Grey: test error (white is 0). Markers: a compact representation of the target is complete in the model (filled dot), part of one is present (open dot), or none is complete but f lies in the span of the model's features (square; held-out $R^2 \geq 0.999$ of f on them). Cross: DFG's own cross-validation chose the forced variant ($n \leq 512$). Bar: n_0^*, the seed-0 fit's 10% test-error point, interpolated in $\log n$. Families as in Table 22; the targets' suites are in Table 32. A filled dot on a grey cell means that the dictionary is complete and the weights are not yet right (for example the halfspaces at $n = 64$). Re-plotted by `paper_dfg/make_interp.py` from `study/part11/interp/tables/atlas_cells.csv` and `atlas_targets.csv`, with the markers of the analysis's `study/part11/interp/figures/fig_atlas_grid.pdf`; the plotted values are in `paper_dfg/tables/interp_numbers.json`.

	first n with test error ≤ 0.1			complete	$n = 16384$		
family (targets)	n	error	what the model holds	first / 16384	error	size	what the model holds
Parities (par2, par4, par5)	16–64	0	χ_S alone	3/3 ; 3/3	0	2	χ_S alone
Staircases (stair5, stair6)	128–256	0	the prefix characters x_1, x_1x_2, … and one more Fourier character	2/2 ; 1/2	0	17–81	Fourier characters: all of stair5's; most of stair6's, and relevant blocks
Majorities of parities; andxor (majpar, majpar5, andxor)	64–256	0.011–0.092	their pair characters	2/3 ; 3/3	0	4–29	their characters; one relevant block in majpar and majpar5
Multiplexers (mux6, mux11)	256–512	0–0.002	characters and blocks on address and data bits	0/2 ; 2/2	0	17–65	exactly the Fourier expansion
DNF, CNF, decision lists, read-once formulas (cnf3, f7_02, dlist8, f7_04, f7_06)	128–256	0–0.049	low-degree characters, blocks that encode a clause, term or rule, irrelevant degree-1 characters	1/5 ; 4/5	0	67–181	characters and a few blocks
Random formulas (rf06, hrf0)	64–128	0–0.045	the dominant input's character (hrf0: and a relevant block)	0/2 ; 1/2	0	32–66	Fourier characters (hrf0: its whole expansion)
Majorities and halfspaces (maj9, thrint, thr30, f7_01)	128–256	0–0.071	degree-1 characters from the linear block; the exact block (maj9, f7_01)	4/4 ; 4/4	0–0.009	43–149	degree-1 characters and a block
Parities of two majorities (parmaj, majxor)	128–256	0–0.0005	two folded blocks $\lvert t_A + t_B\rvert$, $\lvert t_A - t_B\rvert$	2/2 ; 2/2	0	19–137	the cross-pair characters; majxor also both folded blocks
Majority of majorities (majmaj5)	512	0.023	4-input pieces of a majority, modulated by its fifth input	0/1 ; 0/1	0	256	blocks and characters of odd degree 1–7: 256 features, DFG's cap
Symmetric and signed sums (sym9, band11, f7_08, h5s7)	128–2048	0–0.012	the exact signed block (h5s7: two factors that partition its class)	4/4 ; 4/4	0–0.0005	60–255	the block, with relatives (sym9: and a spurious product)
Signed-sum mixtures (f7_07, f7_09, symand, f7_10, f7_00, f7_03, f7_11)	128–512	0–0.016	the partner's character, and the compact form or a relative of it	4/7 ; 1/7	0	27–115	weighted blocks with the partner as inner term, and characters
Residue classes mod 3 (f7_05, h6_00)	never	–	never reached	0/2 ; 0/2	0.3245–0.3430	1	the constant and the kernel background, at every n

Table 22: What DFG alone's model holds, family by family, seed 0 [Numerical]. "First n": the first grid n at which the target's test error is at most 0.1; n and the error are ranges over the family's targets. "Complete": the number of targets whose model holds some compact representation of the target completely, at that first n and at $n = 16384$. "Size": the number of features at $n = 16384$, the constant included. The descriptions summarize `study/part11/interp/ATLAS.md`, §1; the numbers are computed by `paper_dfg/make_interp.py` from `study/part11/interp/tables/atlas_family.json` and `atlas_cells.csv`. Table 32 has every target.

- *Random formulas.* rf06 and hrf0 are dominated by one input (Fourier coefficients -0.94 and -0.81), and their first models are that character ($n = 32$–64). Error 0 follows at $n = 128$ for hrf0, with a relevant block on the other inputs, and at 256 for rf06. At $n = 16384$, hrf0's model is exactly its 31-character expansion.

At $n = 16384$, 8 of the 37 models are exactly the target's Fourier expansion: par2, par4, par5, stair5, andxor, mux6, mux11 and hrf0. *Reading* [Heuristic]. Characters enter in order of how detectable they are (Section 6.5). Blocks, often weighted with an AND of two literals as inner term, encode a whole clause, term or branch before its characters are all detectable, and give way to them as they become detectable.

Majorities and halfspaces: the linear block, then a block [Numerical]. In maj9, thrint, thr30 and f7_01 the linear block enters at $n = 64$ and puts all 30 degree-1 characters in the model. The degree-1 representation is then complete and the oracle error is at most 0.007, but DFG's own error is 0.17–0.20, and the 17–21 irrelevant characters carry an S-lin share of 0.45–0.56 (thr30 has no irrelevant input). The error falls when a block arrives: maj9's 9-input block at $n = 128$ (error 0, S-lin 0.01) and f7_01's 13-input block at 256 (error 0). thrint, which has no single exact block, reaches error 0 at 512 with a product of its two blocks and two modulated folded blocks, all relevant non-target terms. thr30, with Gaussian weights on all 30 inputs, is never fitted exactly (0.009 at 16384). At $n = 16384$ the degree-1 characters carry 0.76–0.91 of the output's standard deviation. *Reading* [Heuristic]. For a threshold function the dictionary is complete almost at once, and what remains is to learn the weights. The logistic readout of 30 noisy degree-1 coefficients learns them slowly; one block, whose levels encode the threshold, does it at once.

Parities of two majorities: two folded blocks [Numerical]. parmaj and majxor ($\mathrm{MAJ}_3 \oplus \mathrm{MAJ}_3$, $\mathrm{MAJ}_5 \oplus \mathrm{MAJ}_5$) are first fitted, at $n = 128$ (forced, error 0) and 256 (error 0.0005), by two folded blocks over the union of the two majorities' inputs, $|t_A + t_B|$ and $|t_A - t_B|$, with shares 0.56–0.77 and next to nothing else. They represent the target because $\mathrm{sign}(|t_A + t_B|^2 - |t_A - t_B|^2) = \mathrm{sign}(t_A t_B)$. With more data the cross-pair characters come in, all 9 of parmaj's at $n = 1024$ and all 25 of majxor's at 2048. The product block of the two majorities never appears, and the spectral pair once (majxor at 512). The majority of three majorities, majmaj5, is fitted without error from $n = 2048$ on without ever holding one of its compact forms, and at 16384 its model sits at DFG's cap of 256 features. *Reading* [Heuristic]. The folded pair is the block analogue of the spectral pair the quadratic feature was built for.

Symmetric and signed sums: the exact signed block [Numerical]. sym9 ($|s| \leq 1$ on 9 inputs) is its folded block alone at $n = 128$ (share 0.98, 4 features, error 0), and band11 at 1024 (share 0.99, 5 features). f7_08, an arbitrary accept set of an 11-input count, is its unfolded block at 512 (share 0.92), next to copies contaminated by irrelevant inputs. h5s7, an arbitrary set of a signed 13-input count, is constant to $n = 1024$; at 2048 it is a product of two factors that partition its class with its signs (share 0.72, error 0.0007), at 4096 the constant again (error 0.368), and from 8192 on the exact block. What varies across these targets is not the form but when the search finds it (Sections 6.4 and 6.5).

Mixtures: partner characters, then weighted blocks [Numerical]. In the seven signed-sum mixtures of the atlas, a sum combined with a parity, a gate, a selector or a second sum:

- *The partner comes first* in 5 of the 6 mixtures that have one: the characters x_{13} (f7_07), x_{11} (f7_03, f7_11), x_7 (symand) and x_6 (f7_10) enter one or more doublings before the block. In f7_09 the parity $x_{11}x_{12}$ enters at $n = 256$ together with its modulated block.
- *At the first fit* the compact form is complete in 4 of the 7: f7_07 and f7_09 (the block modulated by the parity, at 256), f7_00 (the product of its two sums, 256) and f7_03 (512).
- *With more data* the compact forms give way to weighted blocks with the partner as inner term, and to characters. At $n = 16384$ the models hold $\pm s - 3x_{13}$ (f7_07), $s \pm 3x_{11}x_{12}$ (f7_09), $s - 3\,\mathrm{AND}(\neg x_7, \neg x_8)$ and $s + 2\,\mathrm{AND}(\neg x_7, \neg x_8)$ (symand), and $s_1 - 3x_{11}$ and $s_2 - 3x_{11}$ (f7_03). Only f7_10 still holds its compact form there. All seven have test error 0, and their features span f with $R^2 \geq 0.998$ in five of them (0.96 and 0.89 for f7_03 and f7_00).

Reading [Heuristic]. DFG treats a partner variable as a heavy input of the block rather than as a modulator. A weighted block with the partner as its inner literal is easy to find, because the partner is itself a strong single input or pair. Two such blocks in opposite orientations, or one beside characters, represent the target as well as the modulated block does.

Dictionary or readout [Numerical]. Of the 148 cells with test error above 0.1, the learned features cannot express the target (oracle error above 0.1) in 133. In 9 they can (oracle error at most 0.02), and only DFG's own weights fall short: maj9, thrint, thr30 and f7_01 at $n = 64$ (thrint, thr30 and f7_01 also at 128), and dlist8 and f7_10 at 64, all just after the linear block entered. The other 6 lie in between. The oracle leaves out the kernel background, so "cannot express" concerns the learned features only: in 27 of the 133 cells DFG's own error is more than 0.05 below the oracle's, because the background carries the signal there. At the first fit, DFG's error is within 0.0005 of the oracle's for the median target. The largest gaps are andxor at $n = 64$ (0.092) and thrint at 256 (0.071), then majpar at 64 (0.067). andxor's and majpar's vanish without the background; thrint's is a readout gap, 0.071 with or without it, on a complete degree-1 dictionary (oracle error 0). *Reading* [Heuristic]. Outside threshold functions, DFG's data cost is the cost of getting the right features into the model. Once they are in, its least-squares and logistic readout is close to the best weights for them.

Spurious terms in the atlas [Numerical]. They are of three kinds.

- *The linear block's irrelevant degree-1 characters.* The block entered in 83 of the 407 cells. Its irrelevant characters carry an S-lin share of 0.40–0.78 at $n = 64$ in the 8 targets with irrelevant inputs where it entered then, and at most 0.035 at $n = 16384$.
- *Products of a true block with a factor of irrelevant inputs*, such as $|x_1 + \cdots + x_9| \times |x_{12} - x_{23}|$ in sym9: 9 cells of 4 targets, shares 0.07–0.41, test error at most 0.0083.
- *Blocks contaminated by irrelevant inputs* at the first n at which a block appears (f7_08 at 512, symand at 256, dlist8, f7_04).

At the first fit the spurious share is at most 0.1 for 22 of 35 targets (median 0.01, largest 0.36 for dlist8), and removing the spurious terms changes the test error by at most 0.035. At $n = 16384$ it is at most 0.1 for 36 of the 37 (the exception is sym9's spurious product, 0.21), and removing them leaves every error unchanged. In 5 cells the same block appears twice. *Reading* [Heuristic, from the code]: a block's key keeps the order of its inputs, so a block found again with its inputs in another order is not recognized as present.

DFG alone and v3's DFG member [Numerical]. Part 10 analysed the same code as a member of v3, on the same targets and training rows ([13]; `study/part10/interp/TOY_INTERP.md`, with

the corrections in `study/part10/interp/VERIFICATION.json`). Over the 255 seed-0 cells the two analyses share, the DFG test error is identical in 141: in all 15 cells at $n \geq 4096$, in 28 of 32 at $n = 1024$–2048, and in 30 of the 110 at $n \leq 64$. At $n \leq 64$, half of the 80 differing cells differ through the variant (v3's stacker kept the forced member, DFG's own paired test the default), and most of the rest are cells where the stacker gave DFG no weight, so that Part 10 described an out-of-fold fit. Part 10's first n with the exact signed blocks is the same for all 15 targets where both analyses have one. Part 10's findings on v3's member therefore carry over to DFG alone: the exact signed blocks of the symmetric targets, parities and small juntas as characters, halfspaces through the linear block, partners as characters and inner terms, spurious products, and the residue classes mod 3 never learned, a failure that Part 10 traced to the search, not the test. Part 11 adds the models of DFG alone itself; every target family, not only the 54 sum-structured targets, with each target's own representations and the oracle error; the calibration on noise; and exact low-degree weights for the failures (Section 6.4).

6.3 False admissions

How often does DFG admit a feature when there is nothing to find? Table 23 puts the proved guarantee, the null rate of each test's statistic and DFG's own fits on noise side by side.

What is proved [Proved] (P1, Proposition 2.1). Let the labels be independent of uniform inputs, with any law. In round 0 of any fit call, some character of degree k passes the global test with probability at most $\delta/(2K) = 0.0125$, for any d and n. The linear block's test over all d inputs passes with probability at most δ, so a false linear block in round 0 has probability at most $\delta + \delta/(2K)$. If round 0 admits nothing, the default variant's model is the constant plus the kernel background. The proof does not cover the spectral block, the sum blocks, the forced variant, the cross-validation's choice between the variants, or the rounds after an admission.

The character tests sit about ten times inside their bound [Numerical]. In round 0 of the final fits on noise, the global character classes passed in 2 of 2400 class searches at $d = 30$ and 1 of 840 at $d = 60$. The null rate of the degree-1 and degree-2 classes is about 0.1% per test, and all six classes are compatible with it. That is about a tenth of the bound: by THEORY.md the bound exceeds the independent-Gaussian rate by 10.9–14.3 at $d = 30$, close to the Mills-ratio factor of a Gaussian tail, so the union over characters is nearly tight and Hoeffding's inequality is what is loose [Heuristic]. On balanced labels none of the 220 final models at $d = 30$ holds a feature that passed a test (95% upper limit 0.017). In THEORY.md's 80 full fits on balanced labels at $n = 64$–4096, 78 final models are the constant plus the background. One holds a pair that passed its round-0 test at $z = 1.06$, and one is a forced model that the cross-validation preferred.

The false admissions are the dense block tests' [Numerical].

- *The linear block* passes at about 1%, close to the χ^2_d tail at its threshold and a tenth of its nominal $\delta = 0.1$. Its statistic, a sum of squared Rademacher sums, has a lighter tail than χ^2_d. When it passes on noise it adds all d inputs: the two false linear blocks on biased labels at $d = 30$ raise the test error to 0.336 and 0.283, against 0.195 for a constant prediction.
- *The spectral block* has no proved level. On balanced labels it never passed in the fits, and in at most 3 of 20000 null draws. On biased labels it passes often at $n = 4d$, the smallest n at which it runs: in 5.3% of null draws at $d = 30$ and 47% at $d = 60$ ($P(y{=}1) = 0.8$), 99% at $d = 120$, and 39% at $d = 30$ with $P(y{=}1) = 0.9$. The fits agree: 2 of 20 at $d = 30$, $n = 128$,

test	proved under the null (P1)	null pass rate of its statistic	admitted in DFG's own fits
global characters, degrees 1–6, round 0	at most $\delta/(2K) = 0.0125$ per degree, for any d, n and law of the labels	0.11% (degree 1) and 0.07% (degree 2) per test at $d = 30$	2 of 2400 round-0 class searches at $d = 30$, 1 of 840 at $d = 60$
linear block, round 0	at most $\delta = 0.1$ when it tests all inputs; a false linear block at most $\delta + \delta/(2K)$	0.84% ($d = 30$) and 1.04% ($d = 60$), close to the χ^2_d tail at its threshold (0.92%, 1.05%)	2 of 360 round-0 tests at $d = 30$, 4 of 160 at $d = 60$; each adds all d inputs
spectral block	nothing	balanced labels: at most 3 in 20000 draws. $P(y{=}1) = 0.8$, at $n = 4d$, the smallest n at which it runs: 5.3% ($d = 30$), 47% ($d = 60$), 99% ($d = 120$); $P(y{=}1) = 0.9$: 39% ($d = 30$)	biased labels: 2 of 20 fits ($d = 30$, $n = 128$), 5 of 10 ($d = 60$, $n = 256$); at $P(y{=}1) = 0.85$, $n = 128$: 7 of 30 fits; balanced: none
sum blocks	nothing	–	none in the final fits (largest statistic 0.93 of its threshold); in round 0 of 2 default-variant fold fits ($d = 30$, $P(y{=}1) = 0.8$); at $P(y{=}1) = 0.85$, $n = 128$: 1 of 30 fits
forced variant, then the local class	nothing: the variant adds a guess by design	–	a test-passed feature after a forced one in 229 of 900 forced fold fits ($d = 30$, $P(y{=}1) = 0.8$), mostly the product of the two forced characters; in fold-sized refits the local test passes after the two forced characters in 244 of 334, after two random ones in 0 of 334. The cross-validation chose the forced variant in 0 of 180 biased fits
final model	the constant and the kernel background, if round 0 admits nothing	–	a test-passed feature in 0 of 220 balanced and 6 of 220 biased fits at $d = 30$, 5 of 110 and 6 of 110 at $d = 60$

Table 23: False admissions on labels independent of uniform inputs [Proved, Numerical]. "Proved": P1, Proposition 2.1 (`study/part11/interp/theory/THEORY.md`, with the wording of its verifier, `VERIFY_THEORY.md`). "Null pass rate": the share of null draws in which the test's round-0 statistic, computed as DFG computes it, passes; 80000 draws for the characters and the linear block, 20000 for the spectral block (pooled with the verifier's own draws on biased labels). "Admitted": DFG alone's own fits on labels drawn independently of the inputs, with $P(y{=}1) = 0.5$ (balanced) or 0.8 (biased): 20 seeds per $(P(y{=}1), n)$ at $d = 30$ and 10 at $d = 60$, $n = 16, \ldots, 16384$, 660 fits in all; the fits at $P(y{=}1) = 0.85$ are THEORY.md's 30 at $n = 128$. "Test-passed": a feature admitted because it passed a test, not as a forced guess. Sources: `study/part11/interp/calib/calib.json` and `CALIB.md`, with the corrections of `study/part11/interp/calib/VERIFY_CALIB.md` (the spectral rates, the fold-fit counts, the scope of the sum-block row, the $d = 120$ and $P(y{=}1) = 0.9$ rates); `study/part11/interp/theory/check_theory.json`. Made by `paper_dfg/make_interp.py`.

and 5 of 10 at $d = 60$, $n = 256$; and in THEORY.md's fits with $P(y{=}1) = 0.85$ at $n = 128$, 7 of 30 final models end with spurious spectral features. *Reading* [Heuristic]. Its edge $2s\sqrt{d/n}$ is the semicircle edge of a matrix with many comparable weights. On biased labels the minority rows carry residual weights of about -1.6 against $+0.4$, and the Marchenko–Pastur edge of their own Gram matrix reaches 0.99 and 1.08 of DFG's edge at $d = 30$, $n = 128$ and $d = 60$, $n = 256$. At $n = 4d$ that edge is fixed while DFG's margin factor $1 + \sqrt{2\ln(1/\delta)/d}$ shrinks with d, so the false admissions grow with d. The low rates measured at larger n therefore need not carry over to strongly biased targets, such as and3 with $P(y{=}1) = 1/8$, or to the designed suite at $d = 120$; the 15 spurious spectral features that DFG admits on the designed and random targets at $n = 128$–256 are plausibly the same effect.

- *The sum blocks* have no proved level either. They never passed in the 660 final fits on noise (the largest statistic was 0.93 of its threshold), but they did in round 0 of 2 default-variant fold fits at $d = 30$ with $P(y{=}1) = 0.8$, and in 1 of THEORY.md's 30 fits at $P(y{=}1) = 0.85$.

The forced variant is not calibrated after its guesses [Numerical]. When nothing passes, the forced variant adds the best candidate as a guess, at most twice. After two forced characters χ_A and χ_B, the local class, which tests every subset of the inputs in use, can admit their product $\chi_A\chi_B$ by its own test. On biased noise at $d = 30$, 229 of the 900 forced fold fits hold a test-passed feature after a forced one, mostly such a product. In fold-sized refits the local test passes after the two forced characters in 244 of 334 cases, and after two random characters of the same degrees in none of 334. *Reading* [Heuristic]. The selection does it: the characters that best match a biased residual are nearly constant on its minority rows, and so is their product, which least squares on $\{1, \chi_A, \chi_B\}$ leaves in the residual. DFG's paired cross-validation protects the final fit: it chose the forced variant in none of the 180 biased fits. On balanced labels it chose it in 5 of 120 fits at $d = 30$, with test errors 0.491–0.500, and in 2 of 60 at $d = 60$, one of which then admitted the product.

On noise the background, not the features, costs error [Numerical]. With $P(y{=}1) = 0.8$ the test error falls from 0.339 at $n = 16$ to the test set's minority rate, 0.195, from $n = 1024$ on, although no fit at $n \le 32$ holds a feature. The excess is the kernel background's random direction: with q the share of test points predicted -1 and a the minority rate, $a + q(1 - 2a)$ reproduces the mean error at every n to within 0.001. On such data the background carries no information and costs up to 0.144.

Spurious features on the designed and random targets [Numerical]. On the 21 designed and 20 random targets (seed 0, every grid n; 451 fits, Table 33), 1047 of the 4499 admitted terms read an irrelevant input. 961 of them are the linear block's degree-1 characters, which come in with the relevant ones whenever the block passes; the other spurious terms are 86 of 3364 (0.026). The share of spurious terms falls from 0.69 at $n = 64$ to 0.08 at 16384 (without the linear block's characters, from 0.044 to 0.005), and their share of the output from at most 0.86 at $n = 64$ to at most 0.086 from $n = 2048$ on, with median 0 at every n. Removing them lowers the mean test error by 0.020 at $n = 64$ and changes it by at most 0.0002 from $n = 128$ on. Only 2 characters admitted by their own round-0 test read an irrelevant input, both wholly irrelevant, against 2.2 expected under an independent-Gaussian approximation and at most 29.8 by the argument of P1 applied to the irrelevant characters. Once a spurious term is in the model, the residual correlates with characters on its irrelevant inputs, and later rounds can admit them (rand2 at $n = 1024$). From $n = 2048$ on, 11 of the 14 spurious terms other than the linear block's characters are products of two blocks, one of which reads irrelevant inputs, as in the atlas.

					true block: test		search's seeds		DFG alone
target	definition	k	$W_1 + W_2$	peak	passes from	over threshold at 16384	signal/noise at 16384	eigenvector needs n	error at 16384
f7_05	$[B \bmod 3 \neq 0]$	14	$1.6 \cdot 10^{-6}$	11	128	134	0.017	$1.1 \cdot 10^{7}$	0.3245
h6_00	$[B \bmod 3 \neq 1]$	13	$5.4 \cdot 10^{-6}$	9	128	139	0.033	$3.1 \cdot 10^{6}$	0.3430
fd_10	$[B \bmod 3 = 0]$	10	$2.1 \cdot 10^{-4}$	7	128	155	0.265	$8.6 \cdot 10^{4}$	0.3240

Table 24: The three residue classes mod 3 [Proved, Numerical, Heuristic]. B counts the block's k signed inputs that equal $+1$. $W_1 + W_2$: the target's exact Fourier weight at degrees 1 and 2 [Proved] (Proposition 2.5); "peak": the degree with the most weight. True block: the first grid n at which DFG's test statistic of the true (unfolded) block exceeds its threshold on the seed-0 training rows, and the statistic over the threshold at $n = 16384$, as Part 10 computed them for the same code [Numerical]. Search's seeds: the expected signal-to-noise ratio of one entry of the residual pair matrix inside the block at $n = 16384$, and the n from which the top eigenvector of that matrix would point at the block, by the BBP threshold for a rank-one perturbation [Heuristic]. DFG alone: the test error at $n = 16384$, the same in its three seeds and equal to the test set's minority rate. Sources: `study/part11/interp/theory/check_theory.json` (`P4.targets`), `study/part10/interp/results/toy_final/power/` and the Part 11 result files, via `paper_dfg/make_interp.py`.

Pooled over the 62 targets of the ablation suites that both runs learn, a stricter ($\delta = 10^{-4}$) or more lenient ($\delta = 0.9$) level of every test moves n^* by $+0.06$ and $+0.07$ in mean $\log_2$ ratio (median 0.00). Section 6.5 shows that it moves the targets whose n^* one character test sets, and barely the symmetric ones.

6.4 Where it fails, and why

The residue classes mod 3: detectable, but out of the search's reach [Numerical]. Let B count the k signed inputs of a block that equal $+1$. f7_05 $= [B \bmod 3 \neq 0]$ on 14 inputs, h6_00 $= [B \bmod 3 \neq 1]$ on 13 and fd_10 $= [B \bmod 3 = 0]$ on 10 are three of the four sum-structured targets that DFG alone never learns; the fourth is h5s1, below. v3 does not learn them either (Section 4.2). DFG alone's model is the constant plus the kernel background at every n (f7_05 and h6_00 in the atlas, fd_10 in the identity sample), and at $n = 16384$ its test error is the test set's minority rate in every seed. v3's member was not constant at small n: it held one or two characters, and for h6_00 a sum block at $n = 256$–512. Table 24 sets the test against the search, and Proposition 2.5 proves the exact statements.

- *The test would accept the block.* One block, with one group per value of its signed sum, represents each target with at most $k+1$ parameters, where the target's Fourier expansion has all $2^k - 1$ non-constant characters [Proved]. DFG's grouping merges the rarest values, which costs 0.1–0.2% error [Numerical]. Part 10 found the true block passing DFG's test from $n = 128$ on the same training rows, and at 134–155 times the threshold at $n = 16384$ [Numerical].
- *The search's seeds see nothing.* The targets' Fourier weight at degrees 1 and 2 is exactly $1.6 \cdot 10^{-6}$, $5.4 \cdot 10^{-6}$ and $2.1 \cdot 10^{-4}$, and it peaks at degrees 11, 9 and 7. Every starting point of the block search is computed from statistics of one or two inputs, whose expectations in round 0 are the target's degree-1 and degree-2 coefficients: here 2^{-13}, 2^{-12} and 2^{-9} in absolute value [Proved].
- *How far out of reach* [Heuristic]. At $n = 16384$ one entry of the pair matrix inside the block has a signal-to-noise ratio of 0.017, 0.033 and 0.27. Its top eigenvector would point at the block only from $n \approx 1.1 \cdot 10^7$, $3.1 \cdot 10^6$ and $8.6 \cdot 10^4$, and a ranking of the inputs by their degree-1 coefficients only from $n \approx 4.1 \cdot 10^8$, $1.0 \cdot 10^8$ and $1.6 \cdot 10^6$. Part 10 measured the reach on v3's

member: at $n = 16384$ the eigenvector orders put 0.50–0.57 of the block's inputs among their first k, against 0.40–0.52 for the better of two random orders [Numerical].

Reading [Heuristic]. A residue class mod 3 is detectable once proposed and invisible from every foothold the search has; more data within any practical range would not help the seeds. Only the most regularized network, `nets_cewd1.0`, learns h6_00 ($n^* = 12304$), the one sum-structured target on which a network beats DFG alone (Section 4.2). How it gets there was not analysed.

h5s1: a modulator below its test's reach [Numerical]. h5s1 is an arbitrary accept set of a signed 9-input sum s, multiplied by the parity $x_{10}x_{11}$, and no learner in the study reaches 10% error on it. Its Fourier weight is 92% on the single degree-11 character of all its inputs, and the plain block carries no signal: at $n = 16384$ its statistic is 0.04 of the threshold (Part 10). The block modulated by $x_{10}x_{11}$ would pass DFG's test from $n = 128$, but DFG builds a modulated block only on a character already in the model. The pair's coefficient is exactly $-1/32$, the mean of the sum's condition, so its round-0 test admits it in expectation only from $n_{\text{glob}} = 2.3 \cdot 10^4$ (Proposition 2.4), above the grid [Heuristic]. Refits of DFG alone at $n = 8192$ and 16384, made for this paper, show the mechanism (`paper_dfg/check_h5s1.json`; all six refitted test errors equal the result file's):

- at $n = 16384$ the pair passes its round-0 test in seeds 0 and 2 ($z = 1.14$ and 1.23), the modulated block $x_{10}x_{11} \cdot s$ enters in round 1, and the test error is 0. In seed 1 the pair misses ($z = 0.98$), and the model is the constant;
- at $n = 8192$ the pair passes only in seed 2 ($z = 1.04$), where the block does not follow and the error stays at 0.48;
- at every $n \leq 8192$ every seed's test error is 0.48–0.51.

Reading [Heuristic]. h5s1's bottleneck is a low-degree statistic like the residue classes', but one that becomes visible within the grid: the target is learned when the pair's test fluctuates above its threshold.

Arbitrary accept sets come last [Numerical]. The six signed-sum targets that DFG alone learns last are all pure arbitrary accept sets: h5s6, h5s2, f7_08, band11, h5s3 and h5s7, with n^* from 597 to 6186 (the median over the 50 sum-structured targets it learns is 186). Their share of variance at Fourier degrees 1–2, $(W_1 + W_2)/(1 - W_0)$, is 0.008–0.081, against a median of 0.34 over the 54 (Part 10's `study/part10/interp/tables/toy_timing.csv`). The models hold the exact block once it arrives (Section 6.2), and Table 25 shows that for f7_08, band11 and h5s7 it arrives two to six doublings after the test would admit it. h5s7 also shows how much one training set matters: its seed-0 model is exact at $n = 2048$, empty at 4096 and exact again from 8192 on.

6.5 When the structure appears, and when its test would admit it

DFG adds a feature to its model only when the feature passes a test, apart from the forced variant's guesses (Section 2.4). So each piece of a target's structure has two sizes: the training-set size from which DFG's test would admit it if the search offered it, and the size from which the fitted model holds it. For characters the search offers every candidate of a class, so the two sizes can differ only through the class and the round in which a character is tested. For sum blocks the search offers only blocks grown from starting points that it computes from statistics of one or two inputs (Proposition 2.5), so the two sizes can be far apart. This subsection measures both sizes for characters, for parities and for sum blocks. It then moves the tests' threshold to see which of the two sizes sets the data cost.

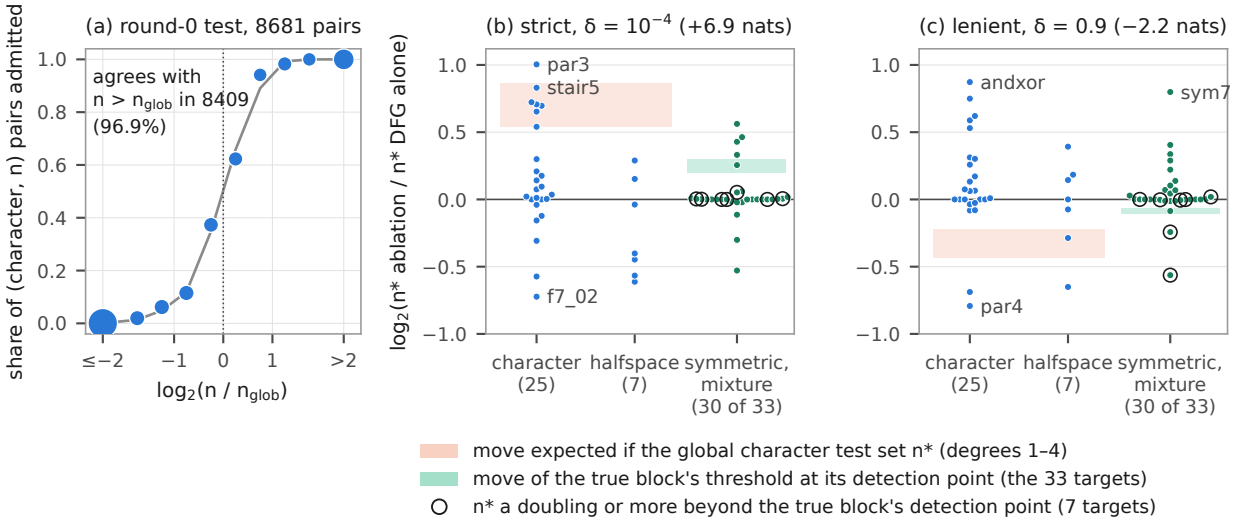


Figure 6: The test and the data cost. (a) DFG's round-0 decisions on the target characters of degree ≤ 4 against the detection size n_{glob}, on the 41 targets of `designed` and `random` (seed 0, every grid size; 8681 (character, n) pairs) [Numerical]. Points: the share admitted in each band of $\log_2(n/n_{\text{glob}})$ (area $\propto \sqrt{\text{pairs}}$; the two end bands are open). Line: a normal approximation for v_S. (b, c) The threshold ablations `strict` and `lenient`: $\log_2(n^*_{\text{ablation}}/n^*_{\text{DFG alone}})$ per target (seeds 0 and 1, paired: the same training sets, column orders and folds), by the class the ablation analysis assigns to each target (character targets: parities, staircases, multiplexers, compositions, DNF, CNF, decision lists and read-once or random formulas; halfspaces; symmetric and mixture targets: functions of one signed sum, alone or with a parity, a gate, a second sum or an inner term) [Numerical]. The three residue classes mod 3, which neither run learns, are not drawn. Orange: the move of n_{glob} for one character of degree 1–4, which is the move of n^* if a round-0 character test set it [Heuristic]. Green: the move of the true block's threshold at its detection point, from the smallest to the largest over the 33 symmetric and mixture targets (Part 10's block statistics re-thresholded) [Numerical]. Rings: the 7 targets whose n^* lies a doubling or more beyond that detection point. Sources: `study/part11/interp/theory/check_theory.json` (`P3.test0`), `study/part11/RESULTS.json` (`q5_ablations`), `study/part11/interp/ablations/families.py` and `study/part11/interp/ablations/ablations.json` (`block_thr_shift`, `block_gap`), the numbers of `study/part11/interp/ablations/ABLATIONS.md`.

Characters: round 0 admits by detectability [Proved, then Numerical]. In round 0 the residual is the labels minus their mean. A character χ_S of degree k searched on all n rows passes its test exactly when

$$n > 2c^2 L_k \, s^2/v_S^2, \qquad L_k = \ln(4K\tbinom{d}{k}/\delta),$$

where v_S is the mean of the residual times χ_S and s^2 the residual's mean square (Proposition 2.4). At $d = 30$ degrees 1 and 2 are searched on all rows at every grid size, and degrees 3 and 4 from $n = 32$ on. Replacing v_S by the target's Fourier coefficient $\hat{y}_S = \hat{f}(S)$ and s^2 by $1 - \hat{y}_\emptyset^2$ gives the *detection size* of Equation (7),

$$n_{\text{glob}}(S) = 2c^2 L_{|S|} \, (1 - \hat{y}_\emptyset^2)/\hat{y}_S^2,$$

a first-order prediction [Heuristic]. Degree enters only through the logarithm. With $K = 4$, $\delta = 0.1$ and $c = 1$, L_k is 8.5, 11.2, 13.4 and 15.3 nats for $k = 1$–4 at $d = 30$. Size enters through $1/\hat{y}_S^2$, so a coefficient $\sqrt{15.3/8.5} = 1.34$ times larger at degree 4 than at degree 1 already has the smaller n_{glob}.

Figure 6a shows how closely DFG's round-0 decisions follow n_{glob} on the 41 targets of `designed` and `random`, seed 0, at every grid size [Numerical]. These are 451 fits, each equal to the evaluation's in test error and FLOP count; in each, every target character's test was replayed with DFG's own residual (`study/part11/interp/theory/THEORY.md`, checked by an independent verifier).

- *The decisions.* Over the 8681 (character, n) pairs with $|S| \le 4$, DFG's round-0 decision agrees with the rule "admit when $n > n_{\text{glob}}$" in 8409 (96.9%). The disagreements lie near n_{glob}: 100 of the 104 pairs above it that fail lie within one doubling of it, and 139 of the 168 pairs below it that pass. The share admitted rises from 6% one doubling below n_{glob} to 94% one doubling above. A normal approximation for v_S gives the share to within 0.051 in every band.
- *First entries.* The 249 characters that first enter the model through this test do so at a median $2^{0.26}\, n_{\text{glob}}$. The order analysis finds the same +0.26 doublings over 151 character-seeds of the Part 5 targets (Section 6.6).

Later rounds make the rest of a junta cheaper [Proved identity, then Numerical]. In round j a class with N_{cls} candidates admits χ_S when $n > 2c^2 \ln(4KN_{\text{cls}}/\delta)\, s_j^2/v_{S,j}^2$. So n_{glob} divided by this size is the product of three factors (Proposition 2.4c): a smaller union bound, $L_{\text{glob}}/L_{\text{cls}}$; a smaller residual, $(1-\hat{y}_\emptyset^2)/s_j^2$; and the residual's coefficient on S, $v_{S,j}^2/\hat{y}_S^2$. On the same 41 targets [Numerical]:

- *Against the prediction.* For 427 characters the predicted grid size (the first at which the rows searched exceed n_{glob}) lies inside the grid. 105 of them first appear at the predicted grid size, 266 earlier, 54 one doubling later and 2 never. Each of the 54 had $z < 1$ at the predicted size (at most 0.994), a fluctuation of the test. Another 321 characters appear although no grid size was predicted for them, 262 of them through the local class.
- *By class* (Table 25a). The classes of the later rounds admit characters a median 1.4 (the global test in rounds ≥ 1) to 3.7 doublings (the local class, every subset of the inputs in use) below n_{glob}. In every class the residual is the main factor: its median is 3.2 to 19 times smaller than the labels' variance. The smaller union bound gains a median factor of 1.0 to 1.7 per class (at most 0.8 doublings), though single characters of the local class gain up to 3.3.

Reading [Heuristic]. Round 0 ranks characters by how detectable they are against the labels' variance. Once the large terms are fitted, the residual is small, and characters on the inputs already in use pass the tests of small classes. This resembles two of Part 5's findings for the networks [12], junta completion and the faster learning of products of learned features. A class label says which test admitted a character, not what would have happened without that class; the ablations ask that (Section 5).

Parities: no gap [Proved, then Numerical]. For a k-parity χ_S passes the round-0 test exactly when $n(1-\bar{y}^2) > n_{\text{thr}}(k,d)$, with $\bar{y}$ the mean label (Proposition 2.2). In all 250 parity cells at $n \ge 32$ ($d = 15$–120) the test error is 0 exactly where the criterion holds; at $n = 16$, where no test passes, the forced variant can put the parity into the model earlier. For a parity no search stands between the test and the model.

Sum blocks: detection is cheap, discovery lags [Numerical]. Part 10 measured both sizes for v3's DFG member on the 54 sum-structured targets [13].

- *Definitions.* n_{detect} is the first grid size at which DFG's block test would admit the target's *true* block if it were proposed in round 0 on the seed-0 rows; it is an oracle. n_{struct} is the first grid size at which the model holds every signed-symmetry class of size ≥ 3 exactly, as a block, a factor of a product block or a modulator.
- *Part 10's result* (`study/part10/interp/TOY_INTERP.md`). Part 10's verifier corrected the detection points of the gate and multiplexer mixtures, testing the modulated block on a residual already fitted on its modulator, as DFG tests it. With these points, 39 of the 54

(a) Characters by the class that first admits them: 41 targets, seed 0, every grid n

class	characters	below n_{glob} (doublings)	class gain (doublings)	union bound $L_{\text{glob}}/L_{\text{cls}}$	residual $(1-\hat{y}_\emptyset^2)/s_j^2$	coefficient $v_{S,j}^2/\hat{y}_S^2$
global test, round 0	249	−0.26	–	1	1	1
global test, rounds ≥ 1	37	+1.4	1.7	1.00	3.3	0.85
extension by one input	33	+1.7	2.3	1.32	3.2	0.86
extension by two inputs	19	+1.9	2.6	1.20	5.0	1.06
local (subsets of the inputs in use)	332 (277)	+3.7	4.5	1.71	19.2	0.80

(b) The 13 sum-structured targets of the structure atlas: DFG alone, seed 0

target	form	k	low-degree share	n_{detect}	n_{struct} (alone = v3)	delay (doublings)	n_0^*
f7_01	threshold	13	0.688	128	256	+1	177
f7_08	arbitrary	11	0.081	128	512	+2	448
band11	arbitrary	11	0.047	128	1024	+3	891
h5s7	arbitrary	13	0.017	128	8192	+6	1705
f7_05	mod	14	0.000	128	never	–	never
h6_00	mod	13	0.000	128	never	–	never
f7_10	threshold; weighted	5	0.703	128	128	0	90
f7_00	band/interval; two sums	7	0.267	256	256	0	221
f7_07	interval; parity	12	0.658	128	256	+1	205
f7_09	threshold; parity	10	0.061	128	256	+1	222
symand	band; gate	6	0.777	256	256	0	173
f7_03	mod/band; mux	6	0.540	512†	512	0	352
f7_11	arbitrary/band; mux	6	0.828	1024†	never	–	72

Table 25: When characters and blocks enter (seed 0) [Numerical]. (a) Characters of the 41 `designed` and `random` targets, by the class whose test first admitted them into the model: the round-0 global test (249 characters), and the classes of later rounds (the characters that entered through them before their n_{glob}, or without a predicted grid size; every character that first entered through a later class except 3). "Below n_{glob}": the median of $\log_2(n_{\text{glob}}/n_{\text{first}})$ over the characters of degree ≤ 6 (their number in brackets where it is smaller); negative in the first row, where characters enter just after n_{glob}. "Class gain": the median of $\log_2(n_{\text{glob}}/n_{\text{cls}})$ at the round of entry; it exceeds the entry gain because a character enters at the first grid size at which its class passes. The last three columns are the medians of its three factors (Proposition 2.4c); single characters spread widely around them. (b) The 13 sum-structured targets of the structure atlas. Form: the accept set of the signed sum (mod: a residue class), then the mixture; k: the inputs of the tested true block; low-degree share: $(W_1+W_2)/(1-W_0)$, which for a mixture includes the partner's own weight. n_{detect}: Part 10's first grid size at which DFG's block test admits the true block in round 0 on the seed-0 rows; †: the verifier's correction for the multiplexers of two sums. n_{struct}: the first grid size at which the model matches every signed-symmetry class of size ≥ 3 exactly (Part 10's rule); DFG alone's equals v3's member's on all 13. Delay: $\log_2(n_{\text{struct}}/n_{\text{detect}})$. n_0^*: DFG alone's 10% point on seed 0. Sources: `study/part11/interp/theory/check_theory.json` (`P3.factors`, `P3.entry_vs_nglob`, `P3.entry_vs_nglob_all`), `study/part11/interp/tables/atlas_targets.csv`, `study/part10/interp/tables/toy_timing.csv` and `study/part10/interp/VERIFICATION.json`.

targets are detectable at $n = 128$. Over the 49 found, n_{struct} is earlier than n_{detect} for 6, equal for 20, one doubling later for 9 and two or more doublings later for 14. On the 25 pure symmetric targets every true block passes at $n = 128$; 22 are found, 9 of them two or more doublings later. The delay grows as the target's share of variance at Fourier degrees 1–2 shrinks (Spearman -0.86).

- *Part 10's reading.* Part 10 read this as the search, not the test, setting the data cost. It labelled the reading Heuristic: nothing isolated the search, and the stacker's choice between DFG's two variants could also have contributed.

What Part 11 adds is threefold: the same timing for DFG alone, without the stacker; an intervention on the test (the next paragraph); and, for the residue classes mod 3, exact low-degree weights that put numbers on the search's missing foothold (Section 6.4). For DFG alone (Table 25b; seed 0): on the 15 atlas targets where both runs have an n_{struct}, DFG alone's equals v3's member's. On all 13 sum-structured targets of the atlas the two runs agree; three of them never reach it. n_{detect} depends only on the training rows and the test, so Part 10's detection points apply unchanged. On these 13 targets the delay is 0 for 4, one doubling for 3, and two to six doublings for f7_08, band11 and h5s7. The structure never appears for the two residue classes mod 3, nor for f7_11, whose two branch classes are never matched in one model. When the block does enter, its statistic is well above the threshold: in Part 10's runs, the median over the 22 pure symmetric targets found is 2.3 times the threshold at n_{struct} (range 0.89–74). The residue classes mod 3 are the extreme case, with a true block that passes the test from $n = 128$ but no foothold for the search.

Moving the threshold [Numerical, then Heuristic]. Two of the pre-registered ablations change only δ (Section 5): `strict` ($\delta = 10^{-4}$) adds 6.9 nats to the log term of every test, and `lenient` ($\delta = 0.9$) subtracts 2.2. They ran on the 65 targets of five suites, seeds 0 and 1, each fit paired with DFG alone on the same training set, column order and folds. The ablation analysis (`study/part11/interp/ablations/ABLATIONS.md`, post hoc and not yet checked by an independent verifier) reads them against a paired null: DFG alone refitted on the same training sets with another column order and other folds. That null moves 3 of the 65 targets by more than 0.25 (mean $|\log_2 \text{ratio}|$ 0.038); the unpaired seed-against-seed spread overstates the noise of a paired comparison about tenfold.

- *What the tests would predict* [Heuristic]. n_{glob} is proportional to L_k, so `strict` multiplies it by $(L_k + 6.9)/L_k$: $+0.54$ (degree 4) to $+0.86$ doublings (degree 1). `lenient` moves it by -0.22 to -0.43. The block test of Equation (5) counts every block of a given size and sign pattern, so its log term is larger, 33.7 nats for an unmodulated 11-input block in round 0, and the same change of δ moves it less. At the true blocks' detection points of the 33 symmetric and mixture targets, `strict` moves the threshold by $+0.20$ to $+0.30$ doublings of n (median $+0.26$) and `lenient` by -0.07 to -0.11 (Part 10's block statistics, re-thresholded).

Figure 6b,c shows what happened, by the classes of the ablation analysis (`study/part11/interp/ablations/families.py`): 25 character targets, 7 halfspaces, and 33 symmetric and mixture targets [Numerical].

- *Parities: the test sets the cost exactly.* Proposition 2.2 with δ moved, applied to each training set's own $\bar{y}$, predicts the measured changes to within 0.03. `strict` moves $n_{\text{thr}}(3, 30)$ from 26.8 to 40.6, above the grid point 32, and par3's n^* doubles ($+0.98$ predicted, $+1.00$ measured). par4 moves by $+0.20$ against $+0.21$ under `strict`, and by -0.79 against -0.79 under `lenient`, which moves $n_{\text{thr}}(4, 30)$ from 30.6 to 26.2, below 32. par2 does not move: the forced variant fits it at $n = 16$.

- *Other character targets: one grid step where the test binds.* `strict` raises n^* by more than 0.25 on 8 of the 25 (class mean +0.17; 3 lower), mostly by about the predicted amount: par3 +1.00, stair5 +0.83, rand3 +0.72, rand2 +0.71, majpar +0.70, sparsepoly +0.65, h6_03 +0.54, h6_05 +0.30. The first fit moves one grid step: stair5's test error at $n = 128$ is 0 on both seeds with $\delta = 0.1$, and 0.309 and 0.184 with $\delta = 10^{-4}$. `lenient` is not the mirror image. It lowers n^* on 2 (par4, rand2 −0.69) and raises it on 8 (andxor +0.87, majpar +0.75, mux +0.59 and 5 more; mean +0.12). These are first fitted at $n = 32$–64, often by the forced variant. With $\delta = 0.9$ something passes in the default variant at $n = 64$, and the first fit is lost: majpar's error at 64 is 0.341 and 0.277 on seeds 0 and 1, against 0.067 and 0.099.
- *Halfspaces: a stricter test helps* (−0.23; 4 lower of 7). In a refit of f7_01 (seed 1, $n = 128$), with $\delta = 0.1$ the model is $\chi[x_7]$, admitted by its own test, plus the linear block (error 0.091). With $\delta = 10^{-4}$ no character passes on its own in round 0, and the exact 13-input block enters next to the linear block (error 0). A block must also pass on the residual projected off the round's admitted characters, so a degree-1 character that passes first takes part of the block's signal.
- *Symmetric and mixture targets: n^* stays.* Both runs learn 30 of the 33; the three residue classes are learned by neither. The class means are +0.04 under `strict` and +0.05 under `lenient` (5 and 4 higher, 2 and 1 lower). On the 7 targets whose n^* lies a doubling or more beyond n_{detect} (band11, f7_08, h6_01, h6_02, h6_06, f7_03, h6_08; ringed in Figure 6), `strict` changes n^* by at most 0.05, although it moves their blocks' threshold by about a quarter of a doubling. On the other 23, n^* lies less than a doubling beyond n_{detect}, and for 18 of them below it, so the grid cannot separate the test from the search. Their 10% point is often reached through characters (fd_08: $n^* = 51$, against $n_{\text{detect}} = 512$) or through the forced variant's block (sym7). The movers under `strict` among them (f7_09 +0.56, fd_11 +0.46, fd_08 +0.43, f7_10 +0.26; h6_09 −0.53, fd_05 −0.30) are all targets whose true block's detection point does not move. h6_07 (+0.33) is the exception, and the paired null moves it too (+0.40).

Reading [Heuristic]. Where n^* lies beyond n_{detect}, a test about a quarter of a doubling stricter changes nothing, because once the search proposes the block, the test admits it far above its threshold (in Part 10's runs a median 2.3 times the threshold at n_{struct}): *the search, not the test, sets the data cost on sum targets.* Where the first fit comes from a character test, the test binds and n^* moves with the threshold: exactly for the parities, within a grid step for the staircases and small formulas. Where the first fit comes from the forced variant, δ acts through the cross-validation instead, and the sign of its effect does not follow from the test. Part 10's reading rested on correlations; the ablations add an intervention on the test. They still do not isolate the search, for example by seeding it with the true block, and they say nothing about the residue classes, which neither run learns.

6.6 The order of acquisition, against the Part 5 networks

Part 5 traced the order in which small ReLU networks acquire the Fourier characters of 11 `designed` targets at $d = 30$ [12]. It found four things: a linear stage first; bundles of characters that one new hidden-unit direction carries together; an order that follows how the target is built (low degree first, products of learned features after their factors) more than coefficient size; and, at finite n, later stages that stop partway. This subsection compares DFG alone's order with the networks' on the same 11 targets and the same training sets (`study/part11/interp/order/ORDER.md`, corrected after an independent verification).

- *The fits.* There are 363 fits: 11 targets, 11 grid sizes, 3 seeds. Each equals the evaluation's fit in test error and FLOP count.
- *The replay.* Every decision DFG made about a target character agrees with a replay of its test statistic, apart from 13 forced picks and 17 characters cut by one round's limit of 64 admissions.

The orders, and what is like for like. A model's *learned fraction* of a character, φ_S, is the coefficient on χ_S of the model's output averaged over the irrelevant inputs, divided by the target's coefficient $\hat{y}_S$.

- *The networks' orders.* The *time order* ranks characters by the training step at which the seed-mean φ_S first reaches $1/2$ and stays above 0.4, online. The *n order* ranks them by the smallest grid size n from which the network after 20000 steps has $\varphi_S \geq 1/2$, with $\varphi_S \geq 0.4$ at every larger n. It comes from a fixed-n study with the same networks and training sets, which recorded characters of degree ≤ 4 only.
- *DFG's orders.* DFG has three: the *round order* within one fit; the *round-0 priority*, the order of the characters' best round-0 z-scores; and the *n order*, by the networks' rule applied to the model's least-squares function.
- *Like for like.* Only the two n orders are like for like. They share the training sets, the grid and the rule, but not the precision of the estimate. The networks' φ_S on the n axis is a mean over the 4096 test inputs, whose sampling standard deviation on φ_S is about $0.0156/|\hat{y}_S|$. That is 0.06 for $|\hat{y}| = 0.25$, 0.25 for 0.0625 (rmaj9, dlist5), and 0.57 and 4.0 for dnf3's characters of size 0.027 and 0.0039. A network equal to the target would score φ from -9.9 to $+9.6$ on the latter, so the networks' n onsets of dnf3's small characters are mostly noise.
- *Ranks or analogies only.* The other comparisons are not like for like. A round is a test-and-refit step on a fixed sample, not time, so the round order and the time order are compared by ranks only. The round-0 priority and the n order are compared with the time order only as analogies.

At Part 5's sample sizes DFG's rounds collapse [Numerical] (Table 26a). Part 5 trained at $n = 512$, 2048 and 16384. At these sizes DFG admits every target character it will ever carry in a single round in 24, 30 and 32 of the 33 fits. All 99 fits then stop by the sign rule, because every training sign is right; the test error is 0 in 97 of them. DFG's round axis is empty where the networks' time axis is richest. Fits with two or more entry rounds are most common at $n = 128$ (24 of 33).

At small n the round order agrees moderately [Numerical]. Over the fits with two or more entry rounds, the mean Kendall τ between the networks' online onset order and DFG's round order is 0.77 at $n = 32$ (2 fits), 0.44 at $n = 64$ (11 fits, 9 positive) and 0.44 at $n = 128$ (24 fits, all positive); it is 0.30 at $n = 256$ and 0.10–0.25 above. Per fit it is negative in 3 of the 69 multi-round fits: mux at $n = 64$ in seeds 2 and 1 (-0.48 and -0.05), and rmaj9 at $n = 512$ in seed 0 (-0.01); it is 0.00 for sparsepoly at $n = 256$ in seed 1. At $n = 128$, where DFG's round order is richest, its own cross-validation picks the forced variant in 15 of the 33 fits.

Inside round 0 the priority agrees on every target [Numerical]. DFG's round-0 priority is positively rank-correlated with the networks' onset order on all 11 targets, in each of four conditions:

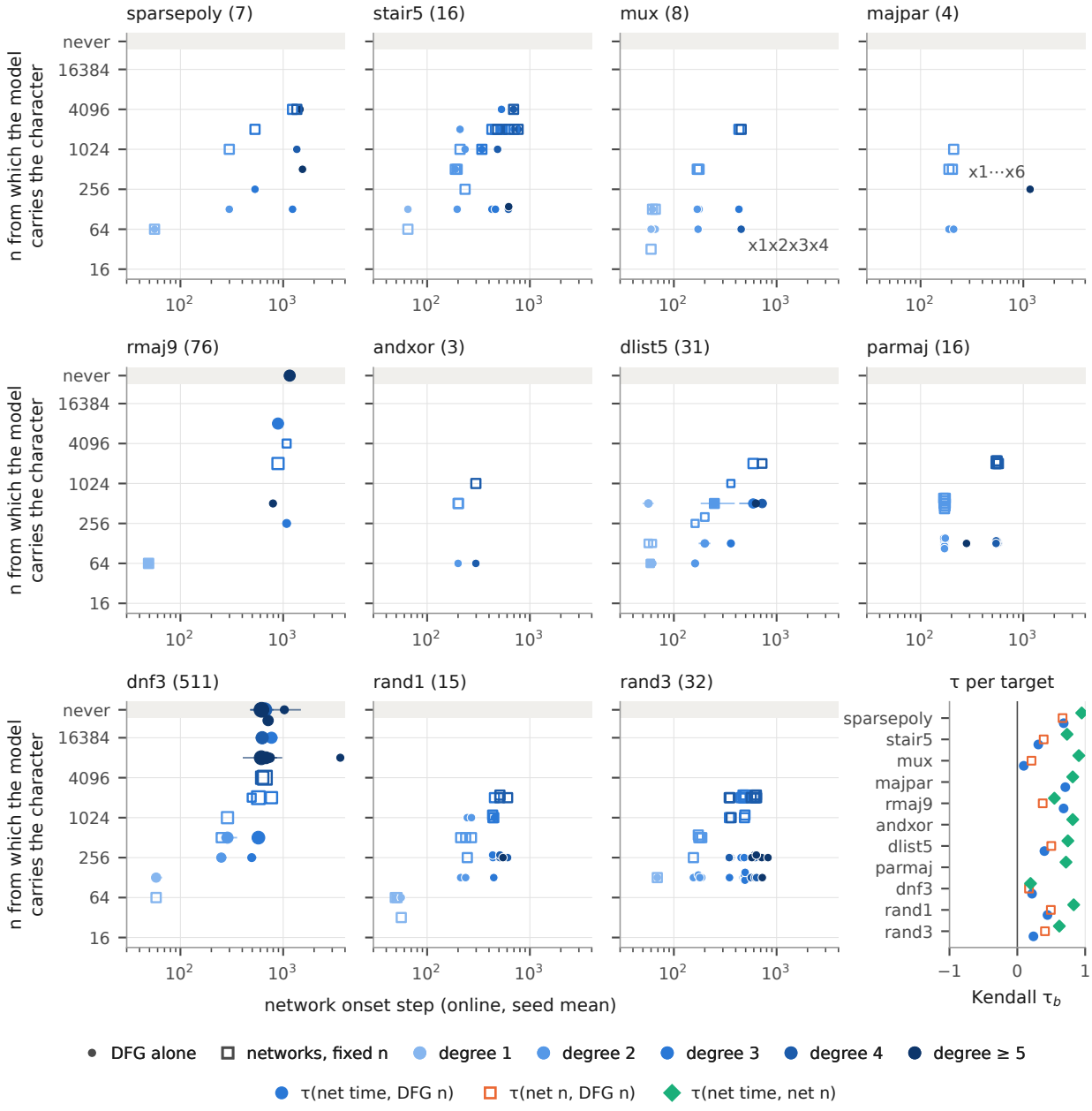


Figure 7: The order of acquisition, DFG alone against the Part 5 networks [Numerical]. One panel per target, with its number of non-constant Fourier characters on the relevant inputs; each point is a character. x: the networks' onset step online (the seed-mean learned fraction reaches 1/2 for good, log-interpolated). y: the smallest grid size from which the final model carries the character (learned fraction $\geq 1/2$, and ≥ 0.4 at every larger n). Filled: DFG alone's least-squares function, median of 3 seeds. Open: the networks of the fixed-n study, median of 3 seeds, $|S| \leq 4$ only. Grey band: never within $n \leq 16384$. Colour: degree. For rmaj9, dlist5 and dnf3 there is one marker per (degree, $|\hat{y}|$) group, at the group's median, with area $\propto \sqrt{\text{count}}$ and a bar over the 10–90% range of the network onsets. Coinciding points are offset vertically by up to ± 0.26 doublings. dnf3's open squares for $|\hat{y}| \leq 0.027$ are dominated by the noise of the networks' estimate. Last panel: the three Kendall τ_b of Table 26b per target; DFG's n order of andxor and of parmaj is a single tie, so these have no τ against it. Re-plotted from `study/part11/interp/order/tables/order_chars.csv` and `order_numbers.json`; the analysis' own figure is `study/part11/interp/order/figures/fig_order.pdf`.

at $n = 16384$, 2048 and 512, network seed s against DFG seed s on the same training set; and online, against DFG at $n = 16384$. All 44 target-condition means are positive, and 131 of the 132 per-seed values; the other is 0.00 (mux, $n = 2048$, seed 2). At $n = 16384$ the means range from 0.22 (dnf3) to 0.81 (sparsepoly), median 0.54 (Table 26b). The online networks see fresh data at every step and have no DFG counterpart, so that row is not seed-matched; seed-matched, all its 33 per-seed values are positive.

On the n axis the agreement is moderate [Numerical] (Figure 7, Table 26b).

- *Against the networks' time order.* τ is 0.09 (mux) to 0.71 (majpar), median 0.40, on the 9 targets whose DFG n order is not a single tie. On andxor and parmaj every character enters DFG at the same n.
- *Like for like*, against the networks' own n order ($|S| \leq 4$). τ is 0.17 to 0.67, median 0.40, on the 8 targets where DFG's order is not a single tie.
- *The networks against themselves.* The networks' time and n orders agree at 0.19 to 0.95, median 0.75: they agree with each other more than either agrees with DFG.
- *dnf3.* Its two n-axis values (0.17 and 0.19) rest on characters the networks' estimate cannot resolve. On its 21 characters of size 0.19 they are 0.56 and 0.87. With those values the two ranges become 0.21–0.67 (median 0.45) and 0.55–0.95 (median 0.82).

The networks follow construction, DFG follows detectability [Numerical, then Heuristic].

- *Degree or size.* On the 8 targets whose coefficient sizes differ, the networks' time order follows degree more than coefficient size on 7. dnf3 is a near-tie that slightly favours size (0.25 against 0.27). DFG's n order follows $1/|\hat{y}|$ more than degree on 5 of the 8 (stair5, mux, dlist5, dnf3, rand1) and degree on the other 3 (sparsepoly, rmaj9, rand3) (Table 26b).
- *Mux, the clearest case.* Its largest term $x_1x_2x_3x_4$ ($\hat{y} = 0.75$, degree 4) is the networks' last character: online at 452 steps, against 60–66 for the linear terms, 169–176 for the pairs x_ix_4 and 430 for $x_1x_2x_3$; in the fixed-n study at $n = 2048$. It is DFG's first character. It has the highest round-0 z-score of mux's 8 characters in all 27 fits with $n \geq 64$, and in 2 of the 3 at $n = 32$. It is in the model from $n = 64$ in all 3 seeds: admitted by the global degree-4 test (seed 1), carried by a block whose levels span it (seed 0), or picked by the forced variant (seed 2). Its n_{glob} is 54, against 271 for the linear terms. The coefficients differ by a factor 3 and the z-scores by 2.2.
- *Data cost, like for like.* For the 64 characters of degree ≤ 4 with $|\hat{y}| \geq 0.25$, DFG's median n onset is 4 times smaller than the networks' (median $\log_2$ ratio -2.0). DFG needs less data for 46, the same for 11 and more for 7. For the 349 smaller characters the median ratio is 1 (147 less, 64 equal, 138 more or never). 255 of these are dnf3's, whose network onsets are mostly estimator noise; without dnf3 the split is 47, 15 and 32 of 94, and the median ratio is still 1.
- *Bundles, cut by detectability.* DFG also admits characters in bundles, but they are sets of near-equal detectability. It merges a whole target whose coefficients are equal: parmaj's 16 characters all enter at $n = 128$ (median seed), carried mostly by two folded blocks. It splits network bundles that mix sizes: the networks learn stair5's x_1x_3, x_1x_4, x_1x_5 ($\hat{y} = -0.125$) with x_1x_2 (0.375) at 188–234 steps, while DFG takes them at $n = 512$–2048 against 128 for x_1x_2.

(a) DFG's rounds: 11 targets × 3 seeds = 33 fits per n

n	16	32	64	128	256	512	1024	2048	4096	8192	16384
fits whose characters enter in ≥ 2 rounds	0	2	11	24	12	9	2	3	4	1	1
mean τ(network time, DFG round)	–	0.77	0.44	0.44	0.30	0.25	0.18	0.15	0.22	0.12	0.10
of which positive	–	2	9	24	11	8	2	3	4	1	1

(b) Per target: Kendall τ_b over its Fourier characters (positive: same order)

		order against order					against degree / $1/\lvert\hat{y}\rvert$	
target	chars	net time, DFG z_0	net time, DFG n	pairs c / t / i	net n, DFG n ($\lvert S\rvert \leq 4$)	net time, net n	net time	DFG n
sparsepoly	7	+0.81	+0.68	13 / 1 / 1	+0.67 (5)	+0.95	+0.88 / +0.53	+0.85 / +0.55
stair5	16	+0.52	+0.31	45 / 18 / 18	+0.39 (15)	+0.73	+0.73 / +0.16	+0.13 / +0.79
mux	8	+0.31	+0.09	7 / 10 / 4	+0.21 (8)	+0.91	+0.89 / −0.50	+0.11 / +0.38
majpar	4	+0.67	+0.71	3 / 0 / 0	tied (3)	+0.82	+0.71 / tied	+1.00 / tied
rmaj9	76	+0.56	+0.68	764 / 0 / 2	+0.37 (39)	+0.55	+0.67 / +0.41	+0.80 / +0.61
andxor	3	+0.33	tied	0 / 2 / 0	tied (3)	+0.82	+0.82 / tied	tied / tied
dlist5	31	+0.54	+0.40	180 / 117 / 45	+0.50 (30)	+0.75	+0.77 / +0.47	+0.35 / +0.77
parmaj	16	+0.54	tied	0 / 69 / 0	tied (15)	+0.72	+0.63 / tied	tied / tied
dnf3	511	+0.22	+0.22	34171 / 8989 / 13265	+0.17 (255)	+0.19	+0.25 / +0.27	+0.22 / +0.47
rand1	15	+0.65	+0.44	51 / 6 / 14	+0.49 (14)	+0.83	+0.89 / +0.61	+0.43 / +0.82
rand3	32	+0.58	+0.24	107 / 136 / 35	+0.41 (26)	+0.62	+0.69 / +0.39	+0.52 / +0.33
dnf3*	21				+0.56 (21)	+0.87		

Table 26: DFG alone's order against the Part 5 networks' [Numerical]. (a) Per grid size: the fits (of 33) in which the target's characters enter DFG's model over two or more rounds; the mean Kendall τ between the networks' online onset order and DFG's round order over those fits (a character carried only by a sum block counts half a round later); how many of them are positive. (b) Per target, Kendall τ_b over its Fourier characters ("tied": one of the two orders is a single tie). The columns are: the networks' onset order in Part 5's run on the same 16384 training rows against DFG's round-0 priority there (mean of 3 seeds, seed s against seed s); the networks' online onset order against DFG's n order (median seed); the pairs the networks order (onsets more than 1.35 times apart) that DFG's n order keeps, ties or inverts; the networks' n order against DFG's ($\lvert S\rvert \leq 4$; the characters in brackets); the networks' time order against their n order. The last two columns give the networks' time order and DFG's n order against degree and against $1/\lvert\hat{y}\rvert$; positive means that higher degree, or a smaller coefficient, comes later. dnf3*: dnf3's 21 characters of size 0.19, which the networks' n-axis estimate can resolve. Source: `study/part11/interp/order/tables/order_numbers.json`.

- *Above degree 4* DFG's global search reaches only subsamples at degrees 5–6, and there DFG does build. Of the 490 character-seeds of degree ≥ 5 that enter for good, 473 are carried by sum blocks, 11 enter through the local class, 3 by extension and 3 through the degree-5 subsample search. They come after the lower-degree characters they build on: majpar's degree-6 term follows its pairs, and stair5's top term enters by extending its lower terms.
- *Linear stage and stopping.* On all 8 targets with linear terms the networks learn every linear term before any other. In DFG's n order the linear terms come strictly first in 6 of the 24 target-seed cases, tie in 12 and do not come first in 6. At $n = 512$ the networks have learned 36% of a target's characters ($\varphi \geq 1/2$; mean over the 33 target-seed cells, targets weighted equally) and hold 30% partially ($0.25 \leq \varphi < 0.75$, a range that overlaps the first); DFG has learned 80% and holds 7% partially. At $n = 16384$ the networks have learned all of them and DFG 92%. DFG stops once every training sign is right, so its model is sign-sufficient rather than an estimate of the labels' conditional mean. On the n axis it drops characters it carried at a smaller n: 1139 of 2157 character-seeds, 945 of them on dnf3.

Reading [Heuristic]. The two cost laws are close to opposite. For DFG a character costs data roughly logarithmically in its degree, through the union bound, and quadratically in $1/\hat{y}_S$. Part 5's model of the networks charges steeply for the number of inputs a new feature coordinates, and only as $\rho^{-1/2}$ in its share ρ of the residual. The orders therefore agree where a target's large coefficients sit at low degree (sparsepoly, majpar, rmaj9: $\tau = 0.68$–0.71 against the time order). They invert where a large coefficient sits at high degree (mux), and they disagree where a stage mixes sizes (stair5, dnf3, rand3). DFG's foothold is chosen by size. The composition steps that resemble the networks' come after it, through the later rounds' small classes (Section 6.5) and through sum blocks. *Caveats.* Three seeds, one network architecture and $d = 30$ only. The 11 targets come from a few designed families, so no p-values are quoted. DFG's φ is that of its least-squares function; the logistic refit can distort single coefficients.

6.7 Where the compute goes

An instrumented copy of `dfg13.py` cuts each fit into regions; its 64 inserted lines only read DFG's FLOP counter, the process clock and statistics DFG has just computed. The profile covers 35 cells, seed 0. Most are par4, maj9, mux, stair5, f7_07, f7_08, f7_05 and labels independent of the inputs, at $n = 64$, 512, 4096 and 16384; the other three are the largest-FLOP fits of three suites at $n = 16384$. Every profiled fit equals the frozen learner's in test error and FLOP count (`study/part11/interp/calib/CALIB.md`, part B, and its independent verification `study/part11/interp/calib/VERIFY_CALIB.md`, whose corrections are used here). Section 4.7 gives the totals.

- *FLOPs are fixed by the budgeted searches* [Proved, then Numerical]. At $d = 30$ one round of the global search over degrees 1–4 on all n rows costs exactly 768,390 FLOPs per training row (Proposition 2.6). Degrees 1–3 (53,970 per row) run again in every later round, and the budget shares would bind only above 29,627 rows. Round 0 adds the degree-5 and degree-6 searches on $\min(n, 1374)$ and $\min(n, 147)$ rows, each 0.030 of a network run from $n = 1374$ on. So before anything else a fit costs 0.057 of a network run at $n = 1024$, 0.079 at 4096 and 0.137 at 16384, whatever the target. At $n \le 512$ the final fit has half the budget (0.015 of a run at $n = 64$, 0.028 at 512), and the cross-validation's fold fits share the other half, of which they spend little. The formula matches the recorded search FLOPs of every profiled call exactly.

- *By component* [Numerical], over the 8 profiled cells at each n. At $n = 64$ the degree-6 subsample search is 50–64% of the FLOPs and the fold fits 27–43%. At $n = 16384$ the degree-4 search is 43–52%, and degrees 5 and 6 are 18–22% each. The largest fits are those with many features. The three largest profiled fits cost 0.32–0.47 of a network run, 41–47% of it in the logistic readout. The largest fit in the result files (h5s5, $n = 16384$, seed 2; 0.519 of a run, 250 features) spends 37% in the readout, by the verifier's refit.

- *CPU time follows the sum-block search instead* [Numerical]. In the final fits the block search runs at 0.75 GFLOP/s, against 16.5–17.6 for the character searches of degrees 3–5 and 32–33 for least squares and the readout; inside the fold fits it runs at 0.27. At $n = 64$ it takes 94–95% of the fold fits' CPU time for 42–46% of their FLOPs; par4 is left out here, because its parity passes in round 0 and no block search runs. In the final fit at $n \ge 4096$ it takes up to 67% of the CPU time (f7_08 at 4096) for at most 5.7% of the FLOPs. So the slowest fits in the result files are at $n = 64$–512 (suite medians 2.2–9.2 seconds at $d \ge 30$, the longest 17.4 seconds), where the FLOP count is smallest: there every one of the cross-validation's small fits runs its own block search.

- *The budget never binds* [Numerical]. DFG's cap stopped none of the 6786 recorded fit calls; a

round starts only while a call has spent less than 0.75 of its budget. The largest call spent 0.471 of its budget (a final fit of fd_08), and the largest fold fit 0.313. In the result files no fit exceeds 0.519 of a network run.

The CPU times come from a 4-core machine shared with 3–4 evaluation workers, and repeats of a cell vary by up to 2.6 times (median 1.44). The CPU shares are medians of three repeats and are rough; the FLOP shares are exact.

7 Discussion

7.1 What kind of learner DFG turned out to be

An Occam learner over a Boolean dictionary, made greedy by a residual [Heuristic, from Sections 2 and 2.5]. DFG's character test admits a feature when the residual sum of squares it would remove, in units of the residual's variance, exceeds $2c^2 \ln(4KN/\delta)$ for a class of N candidates (for a feature orthogonal on the sample to those already present): a code-length criterion, with the union bound over the class as the price of naming the feature (Section 2.6). In round 0 a character therefore passes, to first order, once n exceeds its detection size $n_{\text{glob}} = 2c^2 \ln(4K\binom{d}{|S|}/\delta)\,(1 - \hat{f}(\emptyset)^2)/\hat{f}(S)^2$ (Proposition 2.4). The degree of the feature enters only through the logarithm, and its coefficient through its inverse square. The candidates come from the low-degree algorithm's enumeration, at $d = 30$ for characters up to degree 4 on all rows and degrees 5–6 on a subsample, and from a heuristic search for signed-sum blocks seeded by the residual's degree-1 and degree-2 statistics. The readout is least squares. The residual makes the loop greedy: each round's tests are calibrated to the noise left by the previous round, so weaker features pass once stronger ones are fitted (Section 5.5).

Two regimes. The analyses sort DFG's targets by what sets their data cost.

- *Test-limited* [Proved, Numerical]. Where the target's structure lies in an enumerated class (parities, juntas, the characters of staircases, multiplexers and small formulas), the search offers every candidate, and the first fit is set by the union bound. For parities this is exact: Proposition 2.2's criterion predicts the outcome in all 250 parity cells with $n \geq 32$, and moving the test's level moves n^* by the predicted amount, to within 0.03 (Section 5.2). For the other character targets a stricter test delays the first fit by about one grid step. Later rounds admit the rest of a junta below its detection size, mainly because the residual is smaller (class medians 1.4 to 3.7 doublings below n_{glob}; Section 6.5).
- *Search-limited* [Numerical, then Heuristic]. Where the structure is a signed-sum block, the test would admit the true block well before the model holds it. Part 10's analysis of the same code inside v3 found every true block of the 25 pure symmetric targets passing at $n = 128$, and 9 of the 22 found were found two or more doublings later [13]. For DFG alone, a stricter test leaves n^* in place where it lies a doubling or more beyond the detection point (7 targets, at most 0.05; Section 6.5). The data cost is then set by when the block search first proposes the block, and the less of the target's variance sits at Fourier degrees 1–2, the later that is (Spearman -0.86 in Part 10). At the extreme, the residue classes mod 3 give the search no foothold at all (Section 6.4).

Halfspaces sit between the two. In the four analysed halfspaces the linear block makes the dictionary complete at $n = 64$, where the oracle error is at most 0.007 but DFG's own error is 0.17–0.20; the error falls when one block, whose levels encode the threshold, enters (Section 6.2).

Its relation to the networks' greedy feature growth [Heuristic]. Part 6 built DFG from what Parts 1–5 had seen the networks do: grow features greedily and cheapest first, at a cost set by a feature's correlation with the residual against the noise of the competing directions, with features on inputs already in use cheaper and a near-kernel component from the start [11, 12]. DFG kept the loop and replaced the search: enumeration and explicit tests take the place of noisy gradient steps in weight space (Section 2.6). What survived of the picture is the loop: residual-driven growth, later rounds that make a junta's remaining characters cheaper, as in the networks' junta completion, and a Hebbian background (Sections 5.4 and 6.5). What changed is the price of a feature, and with it the order in which features come (Section 7.4).

7.2 Why it needs less data than the networks here

The logarithm of the dictionary against a power of d. For a k-parity the comparison is explicit. DFG's round-0 test passes χ_S exactly when $n(1-\bar{y}^2) > n_{\text{thr}}(k,d) = 2c^2 \ln(4K\binom{d}{k}/\delta)$, which grows like $2k \ln d$ [Proved; Proposition 2.2]. Over $d = 15$–120, n_{thr} is 19.5–42.0 for $k = 2$–4, and DFG's n^* on the parities moves by at most one grid step [Numerical; Figure 3]. The networks' n^* grows as $d^{1.03}$, $d^{1.96}$ and $d^{2.40}$ for $k = 2$, 3 and 4 [Numerical; Table 15], about d^{k-1} for $k \le 3$, the rate of the leap and correlational-SQ analyses [1, 2, 4]. Part 3's model of the networks explains the gap [12]. There a move that needs k new inputs and has correlation ρ with the residual is accepted once $n\rho^2 \gtrsim c\, d \max(1, d/a)^{k-2}$, about d^{k-1}: the noise level of a continuous search over directions, not the best-of-N level of a union bound. With a noise-free search, that model's n^* fell to 16–87 on every target, as far from the networks as an Occam learner. DFG's test is the union bound: in round 0 it accepts once $n\rho^2$ exceeds about $2\ln(4KN/\delta)$ (Proposition 2.4) [Heuristic]. In those terms DFG is the networks' greedy loop with the search's sampling noise removed and paid for in enumeration, and its advantage grows with d: the median data factor against `nets` is 6.2× at $d = 15$ and 42.5× at $d = 120$ (Table 15) [Numerical].

Enumeration pays in FLOPs, not samples [Proved, by the code; Numerical]. An exhaustive degree-k search costs about $n\binom{d}{k}$ FLOPs times a constant, 12 for degree 3 and 24 for degree 4 (Proposition 2.6), while one network run costs FLOPs linear in d. On 16384 rows a full degree-4 search costs 0.072, 1.04 and 6.5 network runs at $d = 30$, 60 and 120. DFG's budget therefore subsamples the search: at $d = 120$ its degree-4 test still sees 364 rows, far above a 4-parity's $n_{\text{thr}} = 42.0$, and its largest fit at $d = 120$ used 0.18 of a network run (Sections 2.4 and 4.5). Proposition 2.3 says where this ends. At $n > 512$, the budgeted global search can pass no degree-4 character from $d = 179$ on, and no degree-3 character from $d = 630$ on, because its rows fall below the floor at which any character of that degree can pass. Beyond that point a degree-4 parity with no lower-degree footprint is found only by the block search's chance hits, as in the `K3` and core runs (Sections 5.3 and 5.6). At $d = 1000$ an exhaustive degree-4 search on just 64 rows would cost 55.9 network runs (Section 2.5). So within one network run of compute, DFG's Occam rate on degree-4 structure holds up to d of about 120–180. Further out it needs more compute than the networks use, unless a search replaces the enumeration [Heuristic].

Signed sums: the same foothold, rounded and tested [Heuristic]. DFG's block search starts from the residual's degree-1 and degree-2 statistics (Proposition 2.5(f)), the statistics a network's early gradient also sees. For a function of one signed sum the residual pair matrix is about $a_2(ss^\top - I)$ on the block, so its top eigenvector gives the support and the signs. DFG needs fewer samples because it rounds a coarse estimate of that direction to an exact signed subset and then tests the whole block under one union bound, while a network has to learn the direction to useful precision

by noisy local steps, as in the information-exponent picture of single-index learning [5, 14]. On the five sum-structured wins of `dlheld7`, DFG alone needs about 4.3× less data than the best network variant (Section 4.1) [Numerical].

Where it has no advantage. On halfspaces and small formulas, the learners that search no features already beat the networks: on five of the ten `dlheld7` wins, logistic regression or default gradient boosting is also clearly better than `nets_best`, and against the better of the two DFG alone is clearly worse on two of them (Section 4.1) [Numerical]. There the networks are slow, and DFG's search earns nothing.

7.3 Where it fails, and why

No low-degree foothold [Proved, Numerical, Heuristic as marked]. The residue classes mod 3 of 10–14-input counts (f7_05, h6_00, fd_10) put $1.6 \cdot 10^{-6}$ to $2.1 \cdot 10^{-4}$ of their Fourier weight at degrees 1–2 [Proved; Proposition 2.5(e)]. One block represents each of them with at most $k+1$ weights, and the true block passes DFG's test from $n = 128$ on the same training rows [Numerical]. But every starting point of the block search is built from statistics of one or two inputs, which see these targets only through that weight. By the BBP threshold, the eigenvector seed would point at the block only from $8.6 \cdot 10^4$ to $1.1 \cdot 10^7$ rows [Heuristic; Section 6.4]. Exhaustive search does not help within the budget: Part 10 counted $1.5 \cdot 10^{10}$ to $1.2 \cdot 10^{12}$ blocks of the true size, each costing at least n operations to score, against $1.62 \cdot 10^{11}$ FLOPs per fit [13]. The limit is shared: no learner in the study learns f7_05, and only the most regularized network learns h6_00, at $n^* = 12304$. h5s1 is a milder case of the same kind. Its block is modulated by a pair whose coefficient is $-1/32$, so the pair's test passes in expectation only from $n_{\text{glob}} = 2.3 \cdot 10^4$, and DFG learns it at $n = 16384$ only in the two seeds where the pair's statistic fluctuates above its threshold (Section 6.4).

Small n [Numerical, then Heuristic]. DFG alone's weakest range is $n \leq 128$, before its tests admit features. There v3 is ahead: its error is lower by more than 0.02 on 63–67 of the 138 targets at $n = 16$–64, and on most of the 42 targets where DFG alone needs clearly more data than v3, v3's other members (trees, nearest neighbours, logistic regression) carry the fit (Section 4.3). DFG's own small-n machinery is conservative: its paired cross-validation keeps the forced variant's right guesses less often than a stacker does, and its kernel background costs data on balance on the ablation suites (Section 5.4). This is the one place where the portfolio around DFG added something on Boolean targets.

Dense real-valued signal [Numerical, then Heuristic]. DFG v1.3 does not run on real-valued data. Part 7's three adaptations of DFG v1.1, frozen there, replaced its dictionary with thermometer bits, threshold gates or thresholded directions. On 11 held-out tabular datasets they were about level with a plain MLP (+0.04 to +0.10 in mean $\log_2$ error ratio) and behind an L2-regularized one at $n \leq 3000$ (+0.16 to +0.23); on MNIST and Fashion-MNIST each was worse than a plain MLP in 7 of 14 cells [11] (Section 4.6). Part 7's reading was that the useful features of dense real-valued signal are not in any small enumerable dictionary, and that union-bound tests stop growth early on many weak, correlated effects. The Boolean results here contain a small echo: thr30, a halfspace with Gaussian weights on all 30 inputs, is never fitted exactly (test error 0.009 at $n = 16384$; Section 6.2).

Calibration at larger d [Numerical]. The character tests keep their proved level on noise, about ten times inside the bound (Section 6.3). The spectral test has no proved level, and on biased labels

near $n = 4d$ its null pass rate grows with d: 5.3% at $d = 30$, 47% at $d = 60$ and 99% at $d = 120$ for $P(y{=}1) = 0.8$. At larger d, or on strongly biased targets, DFG can therefore be expected to admit more spurious spectral features than on these suites [Heuristic].

7.4 DFG as a model of gradient descent

What the order comparison supports [Numerical] (Section 6.6).

- *Inside one round the priorities agree.* At a fixed training set, DFG's round-0 test statistics order a target's characters much as the Part 5 networks acquire them: Kendall's τ is positive on all 11 targets in all four conditions (median 0.54 at $n = 16384$).
- *At small n the round order agrees moderately*, with mean τ 0.44 at $n = 64$ and $n = 128$ over the fits with two or more entry rounds.
- *Both loops make the rest of a junta cheaper* once its first characters are in (Section 7.1), and both carry a linear, Hebbian component.

What it does not support [Numerical, then Heuristic].

- *A correspondence of time.* At Part 5's sample sizes DFG's rounds collapse: every target character it will carry enters in a single round in 24, 30 and 32 of the 33 fits at $n = 512$, 2048 and 16384. A round is a test-and-refit step on a fixed sample, not a stretch of training, so DFG has no counterpart of the networks' time axis where that axis is richest.
- *The same price of a feature.* On 7 of the 8 targets whose coefficients differ in size, the networks' order follows degree more than coefficient size; DFG's follows coefficient size on 5. The orders invert where a large coefficient sits at high degree: mux's $x_1x_2x_3x_4$ ($\hat{y} = 0.75$) is DFG's first character and the networks' last. On the n axis the networks' two orders agree with each other (median τ 0.75) more than either agrees with DFG's (0.40).
- *A linear stage.* On all 8 targets with linear terms the networks learn every linear term before any other; in DFG's n order the linear terms come strictly first in only 6 of the 24 target-seed cases.
- *The same sample complexity.* The networks' n^* grows as about d^{k-1} on parities and DFG's as $\ln d$ (Section 7.2). DFG is not a model of how much data gradient descent needs. The race model of Parts 2–3, which keeps the search's sampling noise, is such a model [12].

Reading [Heuristic]. DFG models the outcome of greedy feature growth, the features a greedy residual loop ends with, at one extreme of the trade between data and compute that networks also make [16]. It does not model the dynamics. The two orders agree where a target's large coefficients sit at low degree, so that detectability and construction coincide, and they come apart where they do not.

7.5 The core DFG as the object to study next

Everything here is post hoc (Section 5.6). The core keeps five things: the global characters of degree at most 3; the unweighted signed-sum blocks, searched from pairs and single inputs, with their modulated, product, split, beam and forced moves; the union-bound tests scaled to the residual; least squares; and the small-n cross-validation between the default and the forced variant. Pooled, it is at +0.07 against DFG alone on the same seeds (over the 133 of 138 targets both learn), and at −1.74 against `nets_best` (over 122 of 126) where DFG alone is at −1.88; on `dlheld7` it is at −1.30. Its median fit costs 0.094 of DFG alone's FLOPs [Numerical]. Its largest losses are where a

removed part has no substitute: parities of degree 4–5 with no lower-degree footprint, which need the degree-4 to degree-6 searches; dense halfspaces, which need the linear block; and sums with a heavy inner term, which need the weighted blocks and the eigenvector seeds together (Table 21). One loss, h5s2's, comes back with none of the parts tried.

Why it is the better object [Heuristic]. Every test in the core is a union bound over an explicit class, its only heuristic is the block search, and its readout is plain least squares, with no logistic refinement, spectral directions or background. The questions it poses are therefore sharper than DFG's:

- *When does the block search find a block?* The BBP argument for the eigenvector seed is heuristic (Section 6.4), and the core does not even use that seed: it grows blocks from the strongest pairs and single inputs. A statement of the form "a function of a k-input signed sum with low-degree weight $W_1 + W_2 \geq w$ is proposed from $n \geq f(k, d, w)$" would turn the search-limited regime into a sample bound.
- *Does the block test hold its level?* Its groups are a data-dependent partition of the rows, so the Gaussian calibration of Equation (5) is not proved (Section 2.5); on noise it never passed in a final fit, but it did in fold fits on biased labels (Section 6.3).
- *What does the forced variant buy?* Its guesses are the first fit of 12 of the 37 atlas targets (Appendix B.2), and the paired test that chooses it is conservative (Section 5.4). A calibrated replacement would make the small-n behaviour analysable.

The core was defined after every result had been read, `dlheld7`'s included, so any claim about it beyond a description needs fresh data.

7.6 What would test it further

1. *Fresh suites from other families.* The held-out evidence is one draw of 12 targets from one generator whose design was known. A family written independently of both the blind family and DFG's dictionary, for example higher-degree sparse polynomials, intersections of several halfspaces, functions of two or three interacting sums, or residues mod q, would test DFG and the core on structure their designers did not see.
2. *Larger d with the networks.* Targets with degree-3 and degree-4 structure at $d = 60$–250, where the budget first subsamples and then skips the degree-4 search (Section 7.2), with the networks trained at the same d.
3. *The core, pre-registered,* against DFG alone and the networks on fresh suites, with a target mix that includes the kinds it is predicted to lose (Section 7.5).
4. *Real-valued data with dictionaries derived from the data.* Keep the residual loop and the union-bound test, and let the classes be thresholds, quantile bits or residual-derived directions, with their sizes counted honestly in the union bound. Compare with gradient boosting and regularized MLPs, with Part 7's adaptations as the baseline to beat.
5. *Theory for the search.* A success probability for the block search as a function of the low-degree weights and n; levels for the block and spectral tests; and a bound on the test error of the fitted model, which none of Section 2.5's statements gives.
6. *Stronger references.* Networks tuned per target (width, learning rate, steps, regularization), more seeds, and CPU time measured under a controlled load.

7.7 Limits

- **The held-out set is small.** `dlheld7` has 12 targets from one family and one draw, and on 5 of DFG's 10 wins there a learner without feature search also beats the networks. Its standard error of 0.27 treats the targets as independent draws from the family.
- **The other suites are in-sample** (Table 5). Pooled numbers summarize the record; they are not held-out estimates. The margin against `nets_wd` is smallest on the three suites held out for some version of DFG, which is what selection on the others would produce, though the suites also differ in kind (Section 4.2).
- **The references are untuned.** One architecture, width, learning rate and step count per variant, not tuned per target; `nets_best` is a hindsight minimum over 2 to 5 such variants, which favours the networks; and the d-scaling uses `nets`, the weakest variant at $d = 30$ (Sections 3.2 and 4.5). Stronger networks would plausibly narrow the margins.
- **One input law and one domain.** Uniform ± 1 inputs, mostly at $d = 30$, and two tabular datasets whose 9 cells resolve little. Nothing here is about real-valued data or images.
- **Post-hoc analyses.** The core, the paired null, the target families, the leave-one-in runs and the refits were defined after the pre-registered results had been read (Section 5). The ablation analysis was being checked by an independent verifier while this paper was completed (Appendix C).
- **Descriptive interpretability.** Most analyses use seed 0, one training set per target and n. Their readings are labelled Heuristic, and the ablations, one part at a time, are the only interventions (Section 6).
- **Compute.** FLOPs are DFG's own count, prediction is not charged, and CPU times were measured on a shared machine (Section 4.7).
- **AI-written.** The code, the runs, the analyses and this paper were produced and checked by AI agents (Claude), with independent AI verifiers for the analyses. No human has reviewed any of it (Appendix C).

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A Names used in this paper

Tables 27 and 28 list the names of learners, parts of DFG, references, sets, targets and metrics that the paper uses. DFG v1.3 is `study/part10/hyb_portfolio3/dfg13.py`, every toy target is defined in `study/boolfuncs.py`, and the blind family in `study/part9/heldout6/FAMILY.md`.

name	meaning
Learners	
DFG; v1, v1.1, v1.2, v1.3	discrete feature growth (Section 2): v1, v1.1 (Part 6), v1.2 (Part 9), v1.3 (Part 10)
DFG alone, `dfg13-perm`	this paper's object: the frozen DFG v1.3 run by itself, inputs permuted per fit (Section 3.4)
v1, v2, v3	the portfolios of Parts 8, 9, 10, which stack DFG v1.1, v1.2, v1.3 with classical learners
`dfg`, `dfg_forced`	DFG's default and forced variants as two members of a portfolio
`p9_sum`, `p10_sum`	the Part 9 and Part 10 toy builders' candidates: v2's and v3's code on toy inputs
`lowdeg`	the thresholded low-degree algorithm (Section 3.2)
ablation tags	DFG alone with one constructor argument changed (Section 5): `nosums`, `strict`, `lenient`, `noeig`, `noblocks`, `nocomp`, `noweighted`, `nosmalln`, `nobg`, `lsread`, `K3`, `labelnoise`
core DFG, `dfg13-core-perm`	post hoc: DFG alone with nine parts removed by constructor arguments, keeping global characters of degree ≤ 3, unweighted signed-sum blocks, residual-scaled tests, least squares and the small-n cross-validation (Section 5.6)
leave one in	post hoc: the core with one removed part (or two) put back, on the targets where the core moved most (Table 21)
paired null, `dfg13-reperm`	post hoc: DFG alone refitted on the same training and test sets with another fit stream, so with another column order and other folds and nothing else; what an ablation is compared with (Section 5)
unpaired null	the protocol's null: DFG alone's seed 1 (or 2) against its seed 0, so other training sets
Part 7's adaptations	`dfg_bits`, `dfg_dirs`, `dfg_gates`: DFG adapted to real-valued data (context for Q6)
DFG's search (Section 2)	
character χ_S; degree	$\chi_S(x) = \prod_{i \in S} x_i$; $\lvert S \rvert$
residual r; s	$r = y - \hat{y}$, with $\hat{y}$ the least-squares fit on the current features; $s = \mathrm{rms}(r)$
round	a pass that searches every class on r, admits what passes its test and refits; at most 8
class; union bound N	candidates tested together; their number N enters the threshold as $\ln(4KN/\delta)$; $K = 4$, $\delta = 0.1$
global, extension, local	every character of one degree $\leq K$ on all d inputs (degrees 5–6 on a subsample); $S \cup \{j\}$ and $S \cup \{j, l\}$ for admitted S; every subset of the ≤ 12 inputs in use
linear block	all remaining degree-1 characters at once, admitted by a χ^2 test on their coefficients
spectral feature	$(u \cdot x)^2 - \lvert u \rvert^2$, u an eigenvector of the residual pair matrix whose eigenvalue clears the noise edge
sum block; statistic t	signs s_i on a subset S, $t = \sum_{i \in S} s_i x_i$; its features indicate groups of values of t
folded, modulated block	$\lvert t \rvert$ in place of t; the block's indicators times an admitted character χ_T
product, split block	the joint cells of two disjoint blocks; a mixed-sign block taken as the product of its + and − parts
weighted block	$u = t + a\tau$ in place of t, with $a \in \{\pm 2, \pm 3\}$ and τ a literal, a 2-parity or an AND of two literals
block search	how blocks are proposed: from the residual's degree-1 and pair statistics
forced variant	at $n \leq 512$, takes the best candidate if none passes; chosen by a paired cross-validation test
kernel background	one Hebbian step on the final residual, $\sum_i g_i x_i$ with $g_i = \frac{1}{n} \sum_t r_t x_{t,i}$, added to the output
readout	least squares on the admitted features, refined by 15 logistic Newton steps when $n \geq 8p$

Table 27: Names used in this paper, 1 of 2: learners and DFG's search.

name	meaning
References (Section 3.2)	
`nets`, `nets_wd`	the toy networks with square loss, without and with weight decay 0.1
`nets_cewd`, `nets_cewd0.3`, `nets_cewd1.0`	the toy networks with logistic loss and weight decay 0.1, 0.3, 1.0
`nets_best`	per target, the network variant with the smallest n^* among those run on its suite (hindsight)
`lin`, `hgb`	cross-validated logistic regression on the inputs; default histogram gradient boosting
`mlp`, `mlp_reg`, `mlp_strong`	the baseline MLP (256-256); the same with an L2 penalty; an ensemble of three regularized MLPs
`mlp_best`, `mlp_best+`	per cell, the best MLP variant run there (hindsight); `mlp_best+` adds Part 9's wider ones
Sets and targets (Section 3.1)	
the 11 toy suites	`designed`, `random`, `dlheld`, `dlheld2`–`dlheld7`, `dlsymdev`, `dlfamdev`, at $d = 30$ (Table 5)
Q4 targets	the `designed` targets at $d = 15$, 60 and 120: 7, 7 and 4 of them
the blind family	the random generator of `dlheld6`, `dlfamdev` and `dlheld7`, written blind to the sum-block design
sum-structured target	one of the 54 targets of `dlsymdev`, `dlheld4`–`dlheld7`, `dlfamdev` that is not one of their 14 controls
accept set	the values of a sum on which a target, or a function of the sum in it, is $+1$
ablation suites	`designed`, `dlsymdev`, `dlfamdev`, `dlheld6`, `dlheld7`: the 65 targets of Q5
target classes	post hoc, by hand: character targets, halfspaces, symmetric and mixture targets (Section 5)
kr_vs_kp, mushroom	the two PMLB datasets with only binary or categorical columns and two classes
target names	f7_, h6_, fd_, h5: `dlheld7`, `dlheld6`, `dlfamdev`, `dlheld5`; rf, rand, hrf, h$k$rf: random formulas; others: par$k$ (k-bit parity), maj$k$, thr$k$ (halfspace), stair$k$, mux, dnf, cnf, dlist, sym, ...
statuses	development, exposed, held out for v1.1 only, held out: a set's status for DFG v1.3 (Table 5)
Metrics (Section 3.3)	
n^*	the training-set size at which the seed-mean test error reaches 10%
$\log_2$ ratio	$\log_2(n^*_A/n^*_B)$, negative when A needs less data
clearly better, worse	the ratio is below -0.25 (above $+0.25$), or only A (only B) reaches 10% error
suite mean; pooled	the mean ratio over the targets both learn; per-target ratios pooled over suites
cell; better, tie, worse	a (dataset, n) pair on tabular data; the verdicts of the cell rule
budget	one toy network run in FLOPs (1.62×10^{11} at $d = 30$); for real data, three training runs of `mlp`
$n_{\text{thr}}(k, d)$	$2\ln(4K\binom{d}{k}/\delta)$: a k-parity passes round 0's test exactly when $n(1-\bar{y}^2) > n_{\text{thr}}$ (Proposition 2.2)
n_{glob}	$2\ln(4K\binom{d}{\lvert S\rvert}/\delta)(1-\hat{f}(\emptyset)^2)/\hat{f}(S)^2$, the n at which χ_S passes round 0's global test in expectation
n_{detect}	the first n at which DFG's test would admit the target's true block if proposed (Section 6)
n_{struct}	the first analysed n at which DFG's model holds the target's exact signed blocks (Section 6)
Other	
Q1–Q7	the questions fixed by the Part 11 protocol (Section 3.4)
builder	an agent that designed a component on development data and froze it (Parts 8–10)
Part k	the k-th part of the study: Parts 1–5 in [12], 6–9 in [11], 10 in [13]; 11 is this paper

Table 28: Names used in this paper, 2 of 2: references, sets and targets, metrics and other names.

B Further results

This appendix holds the per-target detail behind Sections 4 to 6: every toy target's n^* for every learner, the post-hoc core included (Appendix B.1), and what the fitted models contain, target by target (Appendix B.2). Apart from the rows of `dlheld7`, none of it is held out for DFG v1.3 (Table 5).

B.1 Every toy target

Tables 29 to 31 give, for each of the 138 targets of the 11 toy suites, n^* of DFG alone and of every reference, and the post-hoc core's n^* next to DFG alone's on the same seeds [Numerical]. The suite summaries of Tables 8, 9, 14 and 20 are means over these rows, and the targets' definitions are in `study/boolfuncs.py` (for `dlheld7`, in Table 6).

B.2 What the models contain: the detail

This subsection holds the per-target detail behind Sections 6.2 and 6.3: every atlas target (Table 32), the spurious terms by n (Table 33), the identity checks of the analyses, DFG's choice of variant, the kernel background and the cells where a model loses structure as n grows. All of it is seed 0, from `study/part11/interp/ATLAS.md` and `study/part11/interp/calib/CALIB.md` with their verifications.

The analyses' fits are the evaluation's [Numerical]. Every analysis refits DFG through the evaluation's own code path and checks the refits against the result files, which store test errors and FLOP counts but no models.

- *The atlas*: 517 cells at $d = 30$ (47 targets of all 11 suites, every n, seed 0) and 66 cells at $d = 15$, 60 and 120, equal in test error and FLOP count (`study/part11/interp/tables/identity.md`).
- *The order analysis*: 363 fits (11 `designed` targets, every n, seeds 0–2), equal in both; the residual rebuilt from the recorded weights matches DFG's own round log in all of them (`study/part11/interp/order/ORDER.md`).
- *The theory checks*: 330 new seed-0 cells of `designed` and `random`, equal in both, and the 110 fits on null labels, whose round-0 decisions an independent replica reproduces in all 110 (`study/part11/interp/theory/THEORY.md`).
- *The false-admission and compute analysis* runs an instrumented copy of the frozen file with 64 inserted lines that only read. Its verifier rebuilt the copy byte for byte and found that removing the inserted statements leaves a syntax tree identical to the frozen file's. Every fit checked against the frozen module or the result files is equal (88 on noise, 451 on the designed and random targets, 35 profiled cells), and the unchecked fits are equal by that construction (`study/part11/interp/calib/VERIFY_CALIB.md`).
- *This paper's check* of h5s1 (Section 6.4): 6 refits, equal in test error (`paper_dfg/check_h5s1.json`).

DFG's own choice of variant [Numerical]. At $n \leq 512$ the paired cross-validation (Section 2.4) kept the default variant in 204 of the atlas's 222 cells and chose the forced one in 18; in 12 of these 18 the forced model is the target's first fit (par2 at 16, andxor and majpar at 64, stair5, parmaj and

target	DFG alone	v3	lowdeg	lin	hgb	nets_best (variant)	DFG alone vs nets_best	DFG alone vs v3	core (seeds 0, 1)	core vs DFG alone (seeds 0, 1)
designed: 21 targets; development (v1.1, v1.2, v1.3); networks: nets, nets_wd										
dict	22	16	16	27	–	40 (wd)	−0.85	+0.48	16	−0.65
and3	60	25	158	68	–	70 (wd)	−0.21	+1.30	37	−1.09
maj3	56	31	85	56	–	91 (wd)	−0.70	+0.82	49	−0.48
maj9	81	80	323	80	–	149 (wd)	−0.88	+0.02	110	+0.39
thr10	146	121	730	113	–	307 (wd)	−1.07	+0.27	154	+0.27
par2	16	16	28	never	–	109 (wd)	−2.77	+0.00	16	+0.00
par3	28	55	28	never	–	443 (wd)	−3.99	−0.99	28	+0.01
par4	52	56	28	never	–	1726 (wd)	−5.06	−0.10	3124	+6.02
stair5	111	109	225	never	–	871 (wd)	−2.97	+0.02	103	−0.08
majpar	61	60	97	never	–	1026 (wd)	−4.08	+0.00	49	−0.34
parmaj	111	110	418	never	–	434 (wd)	−1.96	+0.02	111	−0.00
mux	63	70	148	never	–	729 (wd)	−3.53	−0.14	58	−0.28
dnf3	178	118	532	never	–	353 (wd)	−0.99	+0.60	195	−0.02
dlist5	61	32	173	85	–	85 (wd)	−0.49	+0.93	52	−0.16
rmaj9	167	163	403	never	–	1273 (wd)	−2.93	+0.03	209	+0.32
andxor	52	40	83	never	–	204 (wd)	−1.98	+0.37	44	−0.15
sparsepoly	102	103	185	never	–	1238 (wd)	−3.61	−0.03	82	−0.45
rand0	25	16	29	never	–	125 (wd)	−2.35	+0.62	16	−0.73
rand1	104	95	130	never	–	316 (wd)	−1.60	+0.13	63	−0.72
rand2	28	16	16	50	–	59 (wd)	−1.06	+0.83	28	+0.13
rand3	116	114	995	never	–	692 (wd)	−2.58	+0.01	136	+0.26
random: 20 targets; development (v1.1); networks: nets, nets_wd										
rf00	32	40	16	83	60	62 (wd)	−0.95	−0.33	30	−0.10
rf01	52	48	85	never	120	256 (wd)	−2.31	+0.11	74	+0.51
rf02	51	49	98	never	228	191 (wd)	−1.90	+0.07	49	−0.07
rf03	91	62	294	never	139	276 (wd)	−1.60	+0.54	108	+0.24
rf04	54	48	97	never	111	203 (wd)	−1.91	+0.16	56	+0.04
rf05	75	62	145	never	250	493 (wd)	−2.72	+0.26	72	−0.04
rf06	31	16	16	37	58	50 (wd)	−0.67	+0.97	16	−0.97
rf07	104	88	231	never	168	281 (wd)	−1.44	+0.23	97	−0.10
rf08	50	21	146	69	53	62 (wd)	−0.32	+1.22	49	−0.02
rf09	51	45	37	never	292	182 (wd)	−1.83	+0.18	37	−0.46
rf10	43	36	86	66	53	79 (wd)	−0.89	+0.26	37	−0.21
rf11	28	56	28	never	416	445 (wd)	−4.00	−1.00	28	−0.00
rf12	97	87	289	never	524	610 (wd)	−2.65	+0.16	87	−0.16
rf13	54	48	150	never	213	245 (wd)	−2.18	+0.15	59	+0.13
rf14	53	53	97	never	378	404 (wd)	−2.92	−0.00	55	+0.06
rf15	51	22	108	67	53	75 (wd)	−0.55	+1.23	37	−0.48
rf16	44	36	74	never	214	200 (wd)	−2.19	+0.27	49	+0.16
rf17	49	56	28	never	never	1715 (wd)	−5.14	−0.19	3550	+6.19
rf18	51	46	74	never	93	112 (wd)	−1.13	+0.16	56	+0.12
rf19	57	59	296	never	366	654 (wd)	−3.53	−0.06	74	+0.39

Table 29: Every toy target, part 1 of 3: `designed` and `random` [Numerical]. n^*: the training-set size at which the seed-mean test error reaches 10% (3 seeds; 2 on `random`), interpolated in $\log n$; "never": not by $n = 16384$; "–": not run on the suite. DFG alone, v3 and `lowdeg` see the inputs permuted per fit, `lin` and `hgb` in the natural order. `nets_best`: the per-target minimum over the network variants run on the suite (named in each suite's line), with the variant attaining it in brackets: wd, cewd, cewd0.3, cewd1.0 for `nets_wd`, `nets_cewd`, `nets_cewd0.3`, `nets_cewd1.0`. Ratios are $\log_2$ of n^*, negative when the first-named learner needs less data; $\pm\infty$: only one of the two reaches 10% error; n/a: neither. Core (post hoc): seeds 0 and 1, against DFG alone on the same seeds and training sets (Section 5.6). Each suite's line gives its status for DFG v1.3 as in `study/part11/PROTOCOL.md`. Source: `study/part11/RESULTS.json` (`q2_suites`, `q3_lowdeg`, `q5b_core`) via `paper_dfg/make_tables.py` (`tab_targets`).

target	DFG alone	v3	lowdeg	lin	hgb	nets_best (variant)	DFG alone vs nets_best	DFG alone vs v3	core (seeds 0, 1)	core vs DFG alone (seeds 0, 1)
dlheld: 12 targets; development (v1.1); networks: nets, nets_wd										
maj15	130	105	622	110	1278	210 (wd)	−0.69	+0.31	206	+0.76
thr30	158	149	1820	148	1645	311 (wd)	−0.97	+0.09	623	+1.98
majxor	225	224	1724	never	4076	792 (wd)	−1.82	+0.01	225	+0.02
majmaj5	351	360	5997	never	1638	2355 (wd)	−2.75	−0.04	352	−0.00
par5	56	839	never	never	never	7096 (wd)	−6.99	−3.91	never	+∞
mux6	197	123	339	never	188	358 (wd)	−0.86	+0.67	210	+0.01
stair6	207	190	1075	never	695	1495 (wd)	−2.85	+0.13	205	−0.04
inter2	441	346	2204	never	416	454 (wd)	−0.04	+0.35	540	+0.06
majpar5	128	127	198	never	4127	4648 (wd)	−5.19	+0.01	108	−0.62
hrf0	80	62	32	191	63	133 (wd)	−0.74	+0.36	46	−0.75
hrf1	119	116	121	never	461	789 (wd)	−2.73	+0.04	123	+0.09
hrf2	46	54	26	93	60	90 (wd)	−0.98	−0.23	56	+0.16
dlheld2: 7 targets; held out for an earlier version only; networks: nets, nets_wd										
thr20	138	124	1054	117	1262	304 (wd)	−1.14	+0.16	455	+1.76
maj21	112	109	693	116	1956	216 (wd)	−0.94	+0.04	174	+0.67
xmaj	61	66	168	never	1538	871 (wd)	−3.83	−0.10	56	−0.15
dnf4x2	235	112	467	never	98	306 (wd)	−0.38	+1.07	252	+0.14
sym8	54	53	1553	never	2689	377 (wd)	−2.81	+0.03	100	+0.88
h2rf0	119	103	118	never	220	389 (wd)	−1.71	+0.21	112	−0.06
h2rf1	82	31	179	103	58	103 (wd)	−0.33	+1.40	80	+0.00
dlheld3: 10 targets; held out for an earlier version only; networks: nets, nets_cewd, nets_wd										
cnf3	189	126	600	never	164	361 (cewd)	−0.93	+0.58	164	−0.26
sym9	111	111	3267	never	7845	350 (cewd)	−1.65	+0.00	111	−0.00
dlist8	39	24	29	91	61	90 (wd)	−1.21	+0.69	30	−0.86
thrint	186	139	659	137	243	190 (cewd)	−0.03	+0.42	287	+0.14
par3or	71	63	177	never	2009	725 (wd)	−3.35	+0.18	60	−0.21
poly4	176	177	195	never	never	2745 (cewd)	−3.96	−0.01	3127	+4.28
mixmaj	114	96	289	never	368	326 (cewd)	−1.52	+0.24	89	−0.22
mux11	509	466	1486	never	573	1245 (wd)	−1.29	+0.13	850	+0.42
h3rf0	108	46	16	112	60	96 (wd)	+0.17	+1.22	30	−1.98
h3rf1	133	59	31	101	63	101 (wd)	+0.39	+1.17	31	−2.13
dlsymdev: 8 targets; development (v1.2, v1.3); networks: nets, nets_cewd, nets_wd										
sym7	56	55	1356	never	–	329 (cewd)	−2.56	+0.02	55	−0.02
ex3of8	123	118	6509	never	–	664 (cewd)	−2.43	+0.06	94	−0.14
band11	828	829	never	never	–	1836 (cewd)	−1.15	−0.00	889	+0.20
far9	98	71	1433	never	–	296 (wd)	−1.60	+0.47	86	−0.13
ssym9	111	104	2992	never	–	324 (cewd)	−1.55	+0.10	111	−0.00
symxpar	112	111	806	never	–	1410 (cewd)	−3.66	+0.00	112	−0.00
twosym	415	415	5326	never	–	1817 (cewd)	−2.13	−0.00	449	+0.21
thrsum	79	80	406	never	–	330 (cewd)	−2.07	−0.02	140	+0.32
dlheld4: 12 targets; exposed (v1.2 built after exposure); networks: nets, nets_cewd, nets_wd										
sym11	167	160	7778	never	never	374 (cewd)	−1.16	+0.06	187	−0.04
ex4of9	205	194	6580	never	never	411 (cewd)	−1.01	+0.08	193	−0.07
tail13	134	100	3992	never	never	420 (cewd)	−1.65	+0.42	148	−0.09
band10	440	432	never	never	never	3440 (cewd)	−2.97	+0.02	432	−0.02
ssym8	148	177	3873	never	8072	396 (cewd)	−1.42	−0.27	100	−0.07
symxor	406	405	2569	never	6909	946 (wd)	−1.22	+0.00	461	+0.07
symand	149	151	761	never	1307	370 (cewd)	−1.31	−0.02	100	−0.33
sumpar	181	164	316	never	never	1837 (cewd)	−3.34	+0.14	146	−0.17
majsym	102	90	978	never	2840	595 (cewd)	−2.54	+0.17	96	−0.03
dnf3x3	193	115	538	never	148	373 (wd)	−0.95	+0.74	212	+0.15
h4rf0	55	47	98	never	160	206 (wd)	−1.90	+0.22	55	+0.01
h4rf1	96	54	114	158	61	143 (wd)	−0.58	+0.82	29	−1.84

Table 30: Every toy target, part 2 of 3: dlheld, dlheld2, dlheld3, dlsymdev and dlheld4 [Numerical]. Columns as in Table 29.

target	DFG alone	v3	lowdeg	lin	hgb	nets_best (variant)	DFG alone vs nets_best	DFG alone vs v3	core (seeds 0, 1)	core vs DFG alone (seeds 0, 1)
dlheld5: 12 targets; exposed (v1.2 built after exposure); networks: nets, nets_cewd, nets_wd										
h5s0	201	205	never	never	never	803 (wd)	−2.00	−0.03	216	+0.11
h5s1	never	never	never	never	never	never	n/a	n/a	never	n/a
h5s2	598	1154	6456	never	10860	1850 (cewd)	−1.63	−0.95	678	+0.88
h5s3	1642	1757	never	never	never	6454 (cewd)	−1.97	−0.10	1752	−0.01
h5s4	222	184	1766	never	1465	1801 (wd)	−3.02	+0.27	178	−0.06
h5s5	420	421	never	never	never	12254 (cewd)	−4.87	−0.00	449	+0.10
h5s6	597	573	never	never	never	3963 (cewd)	−2.73	+0.06	354	−0.07
h5s7	6186	6760	never	never	never	7641 (cewd)	−0.30	−0.13	1698	−0.01
h5t0	109	105	402	never	2886	352 (cewd)	−1.68	+0.06	185	+0.72
h5t1	100	95	223	never	993	325 (cewd)	−1.70	+0.07	100	−0.06
h5rf0	215	203	318	never	2658	1547 (cewd)	−2.85	+0.08	164	−0.84
h5rf1	98	97	476	never	9281	2512 (cewd)	−4.68	+0.01	149	+0.60
dlheld6: 12 targets; development (v1.3); networks: nets, nets_cewd, nets_cewd0.3, nets_cewd1.0, nets_wd										
h6_00	never	never	never	never	never	12304 (cewd1.0)	+∞	n/a	never	n/a
h6_01	431	430	7470	never	never	1655 (cewd1.0)	−1.94	+0.00	428	−0.01
h6_02	384	389	666	never	never	996 (cewd0.3)	−1.37	−0.02	414	−0.01
h6_03	129	98	270	never	103	231 (cewd1.0)	−0.84	+0.40	188	+0.59
h6_04	79	29	259	96	59	96 (wd)	−0.27	+1.44	65	−0.29
h6_05	42	33	16	48	59	56 (wd)	−0.39	+0.38	27	−0.55
h6_06	347	349	never	never	never	1872 (cewd0.3)	−2.43	−0.01	387	−0.03
h6_07	448	447	5684	never	4492	1812 (cewd1.0)	−2.02	+0.00	495	+0.35
h6_08	479	449	2542	never	5940	1720 (cewd0.3)	−1.84	+0.09	522	−0.16
h6_09	391	380	7850	never	2706	744 (cewd0.3)	−0.93	+0.04	227	−0.69
h6_10	213	206	never	never	never	451 (cewd1.0)	−1.08	+0.05	649	+1.60
h6_11	192	184	2993	never	2264	357 (cewd1.0)	−0.90	+0.06	156	−0.14
dlfamdev: 12 targets; development (v1.3); networks: none run										
fd_00	138	75	239	–	–	–	–	+0.87	139	+0.29
fd_01	82	79	189	–	–	–	–	+0.05	86	+0.09
fd_02	61	66	31	–	–	–	–	−0.11	31	−0.93
fd_03	120	81	205	–	–	–	–	+0.56	116	−0.04
fd_04	170	171	never	–	–	–	–	−0.00	111	+0.00
fd_05	158	147	2555	–	–	–	–	+0.11	167	+0.11
fd_06	355	318	9543	–	–	–	–	+0.16	223	−0.46
fd_07	134	98	344	–	–	–	–	+0.45	105	−0.23
fd_08	51	29	89	–	–	–	–	+0.81	53	+0.04
fd_09	104	101	3210	–	–	–	–	+0.04	99	−0.08
fd_10	never	never	never	–	–	–	–	n/a	never	n/a
fd_11	114	123	61	–	–	–	–	−0.10	61	−0.84
dlheld7: 12 targets; held out (the only suite); networks: nets_cewd0.3, nets_cewd1.0, nets_wd										
f7_00	211	203	7061	never	6208	1545 (cewd0.3)	−2.87	+0.06	211	−0.01
f7_01	119	106	595	111	684	119 (cewd1.0)	−0.00	+0.17	195	+0.32
f7_02	190	106	435	never	120	337 (wd)	−0.83	+0.85	186	−0.01
f7_03	352	352	4205	never	2199	1599 (cewd1.0)	−2.18	−0.00	344	−0.04
f7_04	97	70	121	never	84	164 (wd)	−0.76	+0.47	99	+0.01
f7_05	never	never	never	never	never	never	n/a	n/a	never	n/a
f7_06	82	44	179	127	59	108 (wd)	−0.39	+0.90	82	+0.02
f7_07	167	174	1123	never	1373	415 (cewd1.0)	−1.32	−0.06	121	−0.44
f7_08	644	634	never	never	never	2932 (cewd1.0)	−2.19	+0.02	454	+0.03
f7_09	222	215	591	never	never	883 (cewd0.3)	−1.99	+0.05	355	+0.68
f7_10	84	82	336	87	114	119 (cewd1.0)	−0.50	+0.03	104	+0.23
f7_11	51	55	29	119	63	114 (wd)	−1.16	−0.12	30	−1.07

Table 31: Every toy target, part 3 of 3: dlheld5, dlheld6, dlfamdev and the held-out dlheld7 [Numerical]. Columns as in Table 29. dlfamdev has no network, lin or hgb runs.

				first n with test error ≤ 0.1					$n = 16384$		
target	suite	n_0^* (n_3^*)	complete	n	error	var.	size	T / R / S	error	size	T / R / S
Parities											
par2	designed	16 (16)	16	16	0	f	2	1.00 / 0.00 / 0.00	0	2	1.00 / 0.00 / 0.00
par4	designed	56 (52)	64	64	0	d	2	1.00 / 0.00 / 0.00	0	2	1.00 / 0.00 / 0.00
par5	dlheld	56 (56)	64	64	0	d	2	1.00 / 0.00 / 0.00	0	2	1.00 / 0.00 / 0.00
Staircases											
stair5	designed	108 (111)	128 (512)	128	0	f	7	0.99 / 0.00 / 0.00	0	17	1.00 / 0.00 / 0.00
stair6	dlheld	210 (207)	256 (–)	256	0	d	8	0.97 / 0.00 / 0.00	0	81	0.96 / 0.21 / 0.00
Majorities of parities; andxor											
majpar	designed	57 (61)	64	64	0.067	f	4	0.92 / 0.00 / 0.00	0	11	0.92 / 0.22 / 0.00
majpar5	dlheld	182 (128)	256	256	0.011	d	6	0.98 / 0.00 / 0.00	0	29	0.99 / 0.11 / 0.00
andxor	designed	63 (52)	256	64	0.092	f	3	0.85 / 0.00 / 0.00	0	4	1.00 / 0.00 / 0.00
Multiplexers											
mux6	dlheld	213 (197)	1024	256	0.002	f	43	0.89 / 0.63 / 0.16	0	17	1.00 / 0.00 / 0.00
mux11	dlheld3	430 (509)	4096	512	0	d	76	0.83 / 0.56 / 0.02	0	65	1.00 / 0.00 / 0.00
DNF, CNF, decision lists, read-once formulas											
cnf3	dlheld3	187 (189)	1024	256	0.004	f	48	0.85 / 0.48 / 0.09	0	67	0.92 / 0.10 / 0.07
f7_02	dlheld7	188 (190)	2048	256	0	f	43	0.91 / 0.30 / 0.12	0	69	0.98 / 0.04 / 0.00
dlist8	dlheld3	71 (39)	64 (–)	128	0.049	d	40	0.89 / 0.02 / 0.36	0	149	0.91 / 0.04 / 0.10
f7_04	dlheld7	99 (97)	4096	128	0.021	d	33	0.96 / 0.00 / 0.24	0	181	0.92 / 0.09 / 0.05
f7_06	dlheld7	80 (82)	2048	128	0.024	d	42	0.87 / 0.50 / 0.28	0	84	0.93 / 0.10 / 0.09
Random formulas											
rf06	random	46 (31)	–	64	0.045	d	2	0.96 / 0.00 / 0.00	0	66	0.98 / 0.00 / 0.03
hrf0	dlheld	75 (80)	16384	128	0	f	8	0.97 / 0.37 / 0.00	0	32	1.00 / 0.00 / 0.00
Majorities and halfspaces											
maj9	designed	85 (81)	64	128	0	d	37	1.00 / 0.00 / 0.01	0	54	1.00 / 0.00 / 0.00
thrint	dlheld3	163 (186)	64 (1024)	256	0.071	d	34	1.08 / 0.39 / 0.22	0	43	0.85 / 0.28 / 0.00
thr30	dlheld	187 (158)	64	256	0.058	d	31	1.00 / 0.00 / 0.00	0.009	149	0.91 / 0.15 / 0.00
f7_01	dlheld7	177 (119)	64	256	0	d	56	0.75 / 0.00 / 0.27	0	57	0.92 / 0.30 / 0.00
Parities of two majorities											
parmaj	designed	111 (111)	128	128	0	f	37	1.00 / 0.00 / 0.00	0	19	1.00 / 0.00 / 0.00
majxor	dlheld	223 (225)	256	256	0.0005	d	9	0.96 / 0.00 / 0.10	0	137	1.00 / 0.00 / 0.00
Majority of majorities											
majmaj5	dlheld	348 (351)	–	512	0.023	d	49	0.91 / 0.69 / 0.11	0	256	0.90 / 0.20 / 0.00
Symmetric and signed sums											
sym9	dlheld3	111 (111)	128	128	0	d	4	1.00 / 0.00 / 0.00	0	75	0.88 / 0.00 / 0.21
band11	dlsymdev	891 (828)	1024	1024	0	d	5	1.00 / 0.00 / 0.00	0	62	0.98 / 0.12 / 0.00
f7_08	dlheld7	448 (644)	512	512	0.012	f	52	0.96 / 0.00 / 0.20	0	60	1.05 / 0.18 / 0.01
h5s7	dlheld5	1705 (6186)	2048 (8192)	2048	0.0007	d	164	0.79 / 0.43 / 0.07	0.0005	255	0.59 / 0.45 / 0.05
Signed-sum mixtures											
f7_07	dlheld7	205 (167)	256 (–)	256	0.006	d	21	1.00 / 0.00 / 0.00	0	87	1.00 / 0.00 / 0.00
f7_09	dlheld7	222 (222)	256 (–)	256	0.0005	f	15	1.00 / 0.00 / 0.00	0	35	1.00 / 0.00 / 0.00
symand	dlheld4	173 (149)	–	256	0.012	d	36	1.00 / 0.00 / 0.19	0	80	1.00 / 0.00 / 0.00
f7_10	dlheld7	90 (84)	256	128	0	d	33	1.18 / 0.64 / 0.12	0	27	1.00 / 0.00 / 0.00
f7_00	dlheld7	221 (211)	256 (–)	256	0	f	21	1.00 / 0.00 / 0.00	0	101	0.98 / 0.04 / 0.00
f7_03	dlheld7	352 (352)	512 (–)	512	0.008	d	68	0.85 / 0.45 / 0.01	0	110	1.00 / 0.00 / 0.00
f7_11	dlheld7	72 (51)	–	128	0.016	d	8	0.97 / 0.00 / 0.17	0	115	0.97 / 0.11 / 0.06
Residue classes mod 3											
f7_05	dlheld7	never (never)	–				never		0.3245	1	0.00 / 0.00 / 0.00
h6_00	dlheld6	never (never)	–				never		0.3430	1	0.00 / 0.00 / 0.00

Table 32: Every target of the structure atlas: DFG alone, seed 0 [Numerical]. n_0^* (n_3^*): the 10% test-error point of the seed-0 fits (of the seed mean in the result file; 2 seeds for rf06), interpolated in $\log n$. "Complete": the first grid n at which some compact representation of the target is complete in the model and, in brackets where it differs, the n from which it stays complete (–: it does not stay complete to $n = 16384$). Variant: d = default, f = forced, as DFG's cross-validation chose it. Size: features, the constant included. T / R / S: the shares of target, relevant non-target and spurious terms in the model's output, each the standard deviation of the terms' summed contribution over that of the output; parts that partly cancel can exceed 1. Families as in Table 22. Source: `study/part11/interp/tables/atlas_family.json` and `atlas_targets.csv`, via `paper_dfg/make_interp.py`; the leading terms of every model are in `study/part11/interp/tables/atlas_summary.md`.

			terms reading an irrelevant input			spurious share of the output		test error
n	fits with error ≤ 0.1	admitted terms	all (share)	linear-block characters	others (share of the other terms)	mean	max	with / without them (mean)
16	2	5	1 (0.200)	0	1/5 (0.200)	0.024	1.000	0.3713 / 0.3654
32	7	18	3 (0.167)	0	3/18 (0.167)	0.033	0.752	0.2818 / 0.2800
64	22	362	250 (0.691)	247	3/68 (0.044)	0.136	0.863	0.1158 / 0.0958
128	38	399	204 (0.511)	185	19/164 (0.116)	0.063	0.493	0.0190 / 0.0191
256	41	405	174 (0.430)	159	15/212 (0.071)	0.025	0.170	0.0027 / 0.0028
512	41	403	103 (0.256)	92	11/300 (0.037)	0.016	0.111	0.0007 / 0.0006
1024	41	408	96 (0.235)	76	20/323 (0.062)	0.012	0.148	0.0002 / 0.0002
2048	41	519	53 (0.102)	50	3/464 (0.006)	0.003	0.086	0 / 0
4096	41	681	56 (0.082)	50	6/625 (0.010)	0.006	0.067	0 / 0
8192	41	608	52 (0.086)	50	2/553 (0.004)	0.003	0.055	0 / 0
16384	41	691	55 (0.080)	52	3/632 (0.005)	0.003	0.060	0 / 0
all		4499	1047 (0.233)	961	86/3364 (0.026)			

Table 33: Spurious terms on the 21 `designed` and 20 `random` targets, seed 0, 41 fits per n [Numerical]. A term is spurious if it reads an input on which the target does not depend; a sum block counts as one term. "Linear-block characters": irrelevant degree-1 characters that entered with the linear block. Spurious share: the standard deviation of the spurious terms' summed contribution over that of the model's output. Test error with and without the spurious terms' contributions, mean over the 41 fits; from $n = 128$ on the two differ by at most 0.0002 (`study/part11/interp/calib/VERIFY_CALIB.md`). Source: `study/part11/interp/calib/calib.json` (`A2.by_n`), via `paper_dfg/make_interp.py`.

hrf0 at 128, and six more at 256–512). Refitting the variant DFG did not choose on the same rows (`study/part11/interp/cv_counterfactual.py`), the two differ in test error in 140 cells, and the choice made is the better one in 89 of them. The forced variant would have been better in 48 of the 204 default cells, most of all par4 at $n = 32$ (0 against 0.488): its forced character is right on all 32 rows, but the 10-fold fits on 28–29 rows do not search degree 4. The mean test error over the 222 cells is 0.2257 for DFG's choice, 0.2319 for the default always, 0.2256 for the forced variant always and 0.2111 for the better of the two in each cell. *Reading* [Heuristic]. The paired test is conservative where the folds are too small for the forced guess to be right, which is where the guess helps most. There v3's stacker, which kept both variants, did better (Section 6.2).

Below the structure [Numerical]. In the 198 cells (36 targets) below the first n at which a compact representation is complete, 187 models are the default variant and 11 the forced one. 108 are the constant alone, with the background; 87 hold at least one true character; the median test error is 0.31 (`study/part11/interp/tables/atlas_below.csv`).

The kernel background [Numerical]. The background is added after the weights are fitted, so removing it from a model gives exactly DFG's prediction without it. Removed, the test error is lower in 85 of the 407 cells, higher in 54 and unchanged in 268. It helps most at $n = 16$–32, where the model holds little else and the target has a strong degree-1 part (maj9, thrint, thr30, majmaj5, symand, f7_04, f7_10, rf06): without it the error would be 0.19–0.27 higher (maj9 at $n = 32$: 0.491 against 0.283). It hurts where the residual's degree-1 correlations are noise: on the residue classes at $n = 16$–32 (h6_00 at 16: 0.488 with it, 0.343 without), and at the first fits of andxor and majpar at $n = 64$ (0.092 and 0.067 against 0).

Structure lost as n grows [Numerical]. On one training set per n the models are not monotone in n. A compact representation present at one n is missing at a larger one, with the test error at or near 0, in stair5 (its degree-5 prefix character at $n = 256$ only), stair6 (its degree-6 prefix

character from 512 on), thrint (at 512), dlist8 (at 256 and 16384), f7_07 and f7_09 (complete only at 256), f7_03 (only at 512) and f7_00 (from 4096 on). h5s7 loses its whole block at 4096 (error 0.368) and recovers it at 8192. After the first fit the test error rises with n in nine targets, by at most 0.013 (f7_02: 0 at $n = 256$, 0.013 at 512). In the order analysis, DFG drops 1139 of the 2157 character-seeds it carried at a smaller n, because each fit stops once every training sign is right (Section 6.6).

C How Part 11 was run

Part 11 and this paper were produced by an AI system, Claude (Anthropic), on 2 and 3 November 2026, in one orchestrating session that delegated parts of the work to separate agents. Kaarel Hänni posed the request (Appendix C.1). No human has reviewed the code, the results or this text. Times are UTC, and the commits named are on the working branch `claude/nifty-mccarthy-j0lw2a`. The records are:

- the protocol `study/part11/PROTOCOL.md`, with its post-hoc addendum;
- the runners `study/part11/eval_dfg.py` (toy suites) and `study/part11/dfg_real.py` (tabular data), the shard merger `study/part11/merge.py`, and the comparison script `study/part11/compare_dfg.py` with its outputs `study/part11/RESULTS.md` and `study/part11/RESULTS.json`;
- the five analyses in `study/part11/interp/`, each with its scripts, tables and result files, and the verification records of four of them (Table 34);
- this paper's generators and check scripts in `paper_dfg/`.

Organisation. The orchestrating session chose the object, wrote the protocol and the comparison script, and ran the queue of pre-registered runs and the core. It delegated three kinds of work to separate agents, each with its own instructions: five analysts (the structure atlas, the order of acquisition, the theory notes, false admissions and compute, and the ablations); one independent verifier for each analysis; and writing agents for the sections of this paper. Each writing agent drafted from the records it names. The abstract, the introduction, the discussion and this appendix were written last, from the drafted sections and the records.

C.1 The request and the choice of object

After the Part 10 paper [13], the request was: "can you please suggest a discrete method that is more like 'one thing', less an ensemble of different things. eg maybe one of the components of your current proposal would do? the aim behind this last question is to get a specific interesting object worth studying. then write an analogous paper but specifically about that". The orchestrating session chose DFG v1.3 run alone, for the three reasons of Section 1, and set aside v3's image boosters, the portfolio itself and a new DFG for real-valued data. It wrote the choice, its reasons and its domain limit into the protocol before any Part 11 run. The choice was made after every Part 10 result had been read, v3's on `dlheld7` included, and the protocol says so (Section 3.4).

C.2 The protocol and the runs

The protocol was written on 2 November and committed at 22:15:39 together with the runners and the merge script (`aca1f3c8`), before the first run. It fixed the object, the seven questions

with their runs, suites and seeds, the rules of Parts 8–10, the rule for crashes, and what is held out (Section 3.4). Its pre-registered part has not changed since. The addendum that defines the core run was committed at 02:25:54 on 3 November (`12eae4e0`; its heading says 02:30), after every pre-registered run had finished and its results had been read, and two seconds before the core's first job started.

The queue. The pre-registered runs were 282 jobs in one file:

- 48 shards of DFG alone on the 11 toy suites, 3 to 7 per suite;
- 11 `lowdeg` jobs, one per suite;
- 18 jobs at $d = 15$, 60 and 120, one per target and d;
- one job for the two tabular datasets;
- 204 ablation shards, 17 for each of the 12 ablations over the five ablation suites.

Four worker processes started at 22:15:41. Each worker took the next unclaimed job in file order (claiming it by creating a directory, which is atomic), ran it at nice 10 with single-threaded BLAS, and marked it done or failed. A merger joined the shards of each run and suite with `merge.py` once all of them were done, and the orchestrator committed the merged files. No job failed. The first results, DFG alone on the held-out `dlheld7`, were committed at 22:23 (`5bd788c4`). DFG alone had finished on all 11 suites by 23:31, and `lowdeg`, the runs at other d and the tabular runs by 23:41. The ablations' results were committed from 23:42 on, and at 02:24 all 282 jobs, 87 result files, were complete (`28d64a85`).

The container restart at about 01:09 on 3 November stopped every process. Four ablation shards were running: `noblocks` on `dlheld6` (shards 2 and 3) and on `dlheld7` (shards 1 and 2). A shard writes its result file only when it finishes, so these four had written nothing. Their claims were removed, three workers were restarted at 01:10:54, and the four shards ran again from the start, with the same code, seeds and random streams. The 158 jobs already finished kept their files, and the merger skips any run and suite merged before a restart. So no finished result was lost or computed twice, and no result file mixes two runs; the worker logs show 286 starts for the 282 jobs. After the restart the queue ran on three workers instead of four. The calibration analysis's record writer was also stopped, at 8 of the 220 noise fits at $d = 60$ and 32 of the 105 profile records. It was resumed from its records, which skip every key already written, so no cell was fitted twice. The commit `ba81503c` (01:10:57) kept the analysts' partial edits.

The post-hoc runs.

- *The core* (Section 5.6): 36 shards on four workers, from 02:25:56 to 02:38:32; its 11 result files were committed by 02:39 (`22e45319`). With them, the 98 result files of Part 11 are all complete (`study/part11/RESULTS.md`, "Run status").
- *The paired null* (`study/part11/interp/ablations/perm_null.py`) on all 11 suites, seeds 0 and 1, after the core had finished and until 03:36; *the leave-one-in runs* (`study/part11/interp/ablations/leavein.py`), until 03:24; and 36 refits of single cells for the ablation analysis (`study/part11/interp/ablations/refit.py`). Every refit reproduces its run's result file in test error and FLOP count. The leave-one-in targets were chosen after reading the core's results, and five of its configurations after reading the first leave-one-in results (they are marked in `leavein.py`).

- *This paper's checks*: `paper_dfg/check_parity_variant.py` (43 refits of the parity cells at small n) and `paper_dfg/check_h5s1.py` (6 refits of h5s1). Every refitted test error equals the result file's.

All of these ran single-threaded, as separate processes outside the queue.

C.3 The analyses and their verification

The analyses ran from about 23:00 on 2 November, in parallel with the queue, on the result files as they landed (Section 6). Each fit they analyse is an instrumented rerun of an evaluation fit, and each analysis checks its reruns against the result files in test error and FLOP count (Appendix B.2). The theory notes' proofs are about the frozen code, line by line. The protocol fixed Q7's questions, not the analyses' methods: each analysis chose its targets, definitions and groupings itself, and the atlas revised some definitions after its first fits had been looked at (Section 6.1). The paired null, the leave-one-in runs, the refits and the target families of the ablation analysis are post hoc (Section 5).

report	verification	what it covers	claims checked	confirmed	corrected	omissions listed
`ATLAS.md`	`VERIFY_ATLAS`	what the models contain: 37 targets, 407 fits	61	39	22	9
`ORDER.md`	`VERIFY_ORDER`	the order of acquisition, against the Part 5 networks	60	41	19	9
`THEORY.md`	`VERIFY_THEORY`	the exact statements P1–P5 and their checks	54	47	7	4
`CALIB.md`	`VERIFY_CALIB`	false admissions; where the compute goes	28	16	12	8
`ABLATIONS.md`	–	the ablations and the core DFG	not verified when this table was made			

Table 34: The independent verifications of the Part 11 analyses, counted from the verifiers' own records [Numerical]. "Corrected": the verifier changed a number, narrowed a statement or rejected a reading; "omissions": further problems it listed that were not tied to one claim. The reports are in `study/part11/interp/`, each next to its verification record (`.md` and `.json`). Generated by `paper_dfg/make_tables.py` (`tab_verif`) from the `VERIFY_*.json` files.

The verifications. Each verifier was a separate agent with its own instructions. It read the report, the code and the stored files, recomputed the numbers with its own scripts, refitted cells through its own code path, and listed every claim as confirmed or corrected, with the evidence. The analyst then checked each correction against the stored files, applied it or gave its reason for declining, and listed every change in the report's Verification section. Table 34 counts the verdicts. The paper uses the corrected numbers and wording throughout.

- *The structure atlas* (`ATLAS.md`; verified at about 00:36). The verifier refitted 11 cells and 6 counterfactual variants and recomputed every count. All 22 corrections were applied and none was declined. Among them: the 37 targets come from 9 of the 11 suites; block representations are checked only as functions of the block statistics; thrint is first fitted without error at $n = 512$ by a product block, not at 1024 by weighted blocks; and DFG alone's model of the residue classes is constant at every n, whereas v3's member was not at small n.
- *The order of acquisition* (`ORDER.md`; about 00:34). The verifier made 12 refits through a separate code path and recomputed all 23,727 round-0 z-scores. All its corrections and omissions were applied, two of them only in part, with the stored evidence for the rest, and

the analyst's own re-check found eight further errors (for example 27 fits with $n \geq 64$, not 30). Among them: DFG's round order is negative in 3 of the 69 multi-round fits; the n-axis comparison is like for like in data and rule, but not in the precision of the networks' estimate for small coefficients; and DFG's bundles are cut by detectability, not by coefficient size alone.

- *The theory notes* (`THEORY.md`; about 02:59). The verifier re-derived every proof against the cited lines of the code, refitted 615 cells, and gave the verdict "sound". All 7 corrections and 4 omissions were applied. No proof or measured value changed. The corrections narrowed wording that overstated what is proved (passing a test against being admitted to the model; n_{thr}'s exact decomposition) and fixed one count, and the omissions added P1a′, the bound for other input laws (Section 2.5).
- *False admissions and compute* (`CALIB.md`; about 02:43). The verifier rebuilt the instrumented copy byte for byte, showed that its syntax tree minus the hooks equals the frozen file's, and made 17 independent refits. It corrected 12 claims, among them the identity claim (only the checked fits are checked; the others equal the frozen learner by construction), the linear block's null rate (close to the χ^2_d tail, not at it), sum-block admissions in the fold fits, a count (229, not 238) and one step of a proof. Its important omission is the spectral block's null pass rate on biased labels, which grows with d to 99% at $d = 120$. Section 6.3 reports the corrected rates.
- *The ablations* (`ABLATIONS.md`) were being checked by an independent verifier while this paper was completed. Section 5 reports the analysis as written.

C.4 Disclosures

- *The timing check.* Before the protocol was written, DFG alone was fitted on two `dlheld7` targets, f7_03 and f7_08 (seed 0; $n = 64$, 512, 4096 and 16384), to time it, and the 8 test errors were printed. The protocol records them, and the queue's runs reproduce all 8 to the three decimals recorded (`study/part11/RESULTS.md`, "Consistency checks"). The code had been frozen in Part 10 and nothing is tuned in Part 11.
- *Smoke tests* of the ablations and of `lowdeg` used three `designed` targets: maj9, stair5 and par4.
- *A replaced setting.* The first choice for `lenient`, $\delta = 10$, crashed: the linear block's threshold takes the square root of a multiple of $\ln(1/\delta)$, which is negative for $\delta > 1$. It was replaced by $\delta = 0.9$ before any run, and the protocol records both.
- *The object was chosen after v3's results* on every suite had been read, `dlheld7`'s included (Appendix C.1).
- *Post-hoc choices.* The core was defined after every pre-registered result had been read, and the leave-one-in targets after the core's (Section 5.6). The target families and classes were assigned by hand after the results were in.
- *Load.* The queue ran four workers at nice 10 on a four-CPU machine, three after the restart, next to the analyses, so the CPU times are rough (Section 4.7).
- *Written and checked by AI agents.* The analyses, their verifications and this paper were produced by AI agents (Claude), each with its own instructions. No human has reviewed them.

C.5 Reproducing the numbers

Which files stand for which learner.

- *DFG alone, its ablations, the core and* `lowdeg` on the toy suites: `toy__dfg13-perm__<suite>.json`, `toy__dfg13-<tag>-perm__<suite>.json`, `toy__dfg13-core-perm__<suite>.json` and `toy__lowdeg-perm__<suite>.json`, all in `study/part11/results/`. The paired null's files are in `study/part11/interp/ablations/results/`.
- *DFG alone on tabular data*: `dfg13__kr_vs_kp.json` and `dfg13__mushroom.json` in `study/standard_tasks/results/`.
- *v3 on toy inputs*: `p10_sum`'s files in `study/hybrid/results/`, which is v3's code there (Section 3.2). The networks and the other references are read from their own result files, as the docstring of `study/part11/compare_dfg.py` lists them.

Commands. The commands are run from the folder `gd-circuit-growth/`, with single-threaded BLAS (the environment variables `OMP_NUM_THREADS` and `OPENBLAS_NUM_THREADS` set to 1):

- the runs: `cd study/part11`, then `python3 eval_dfg.py dfg13 <suite>` (an ablation adds `--kw=<json> --name=<tag>`, with the arguments of Section 3.4 and, for the core, those of the protocol's addendum; shards add `--fns=` and `--part=`, and `python3 merge.py <run> <suite>` joins them), `python3 eval_dfg.py lowdeg <suite>`, `python3 eval_dfg.py dfg13 designed_d<d>`, and `python3 dfg_real.py kr_vs_kp,mushroom`;
- the comparisons: `python3 compare_dfg.py` (writes `RESULTS.md` and `RESULTS.json`);
- the analyses: the scripts that each report names in `interp/` (for example `atlas_run.py` and `atlas_make.py`; `order/order_fit.py` and `order_analysis.py`; `theory/theory_fits.py` and `check_theory.py`; `calib/calib.py` and `calib_summary.py`; `ablations/perm_null.py`, `leavein.py`, `refit.py` and `ablations.py`);
- this paper: `cd ../../paper_dfg`, then `python3 make_tables.py`, `python3 make_figures.py`, `python3 make_interp.py`, `python3 check_parity_variant.py` and `python3 check_h5s1.py`, and `latexmk -pdf main.tex`; `dfg-paper.pdf` is a copy of `main.pdf`.

Every table and figure is generated from these files; derived numbers that the text quotes and no table shows are in `paper_dfg/tables/numbers.json`, `paper_dfg/tables/interp_numbers.json` and the figures' `.json` files.

What is not in the repository, and what stays on the working branch. The job queue, the worker and merger scripts and their logs stayed in the session scratch directory and are in neither branch; they only wrap the committed runners, whose commands are above. The networks' raw outputs also live in a scratch directory, as in Parts 1–10. On the working branch, the LaTeX build output in `paper_dfg/` (`main.pdf` and the auxiliary files) and the run logs of the post-hoc runs (`study/part11/interp/ablations/logs/`) stay there; `dfg-paper.pdf` is the published copy of the paper.